Chapter 6

The Numerical Physics of Micron-Length and Submicron-Length Semiconductor Devices

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I. INTRODUCTION

A turning point in the study of semiconductor devices occurred with the publication of the special January 1966 issue of the *IEEE Transactions on Electron Devices*. Here papers dealing with the numerical simulation of the space- and time-dependent behavior of the charge distribution within solid-state devices implied, either explicitly or implicitly, that these types of calculations could be used to develop the intuition needed to explain device behavior.

Today, numerical simulations are regarded as part of a device physicist's tools and are routinely used (1) when the device transport is nonlinear and the device differential equations do not admit to exact solutions, (2) as surrogates for laboratory measurements that are costly and/or not feasible, and (3) in computer-aided device design. Specifically, the kinds of devices most often treated numerically either require multidimensional concepts for their operation or are dominantly hot-carrier dependent. Transferred electron devices are examples of hot-carrier devices, but an MOS or MESFET requires two dimensions for conceptual operation and is often modified by hot-carrier contributions. In any case, through a heirarchy of differential equations, we are becoming increasingly able to simulate the operation of devices such as these and to numerically represent the effects of

(i) material properties (e.g., doping variations),
(ii) physical boundaries (e.g., contacts, surface states), and

(iii) the environment (e.g., adjacent devices, circuits) on device operation.

This numerical ability, coupled with heightened interest in device physics and generated in large measure by the current VLSI and VHSIC programs, is also forcing a long-needed reassessment of the assumptions used in device simulations. For example, most modeling of single-species transport represents the current response by the equation

$$\mathbf{J} = Ne\mu\mathbf{F} + eD\,\mathbf{grad}\,N + \epsilon\,\partial F/\partial t, \tag{1}$$

where N is the carrier concentration, e the electronic charge, F a field, μ the mobility, D the diffusivity, ϵ the dielectric constant, and t the time. This equation is based on the Boltzmann transport equation (BTE) for a distribution function only slightly modified by a self-consistent field [1]. For high-field, nonlinear, hot-carrier effects, the distribution function is strongly dependent on F and the usual approach replaces μF by a nonlinear $v(F)$ curve, and D by a field-dependent diffusivity [2]. This approach, although useful, avoids the problem of finding solutions to nonlocal transport equations, which include spatial and temporal field-dependent relaxation. Such solutions are especially important in GaAs, where recent history has shown that concepts like velocity overshoot[1] [3–5] profoundly affect transport in short-channel devices and the more conventional devices operated at high frequencies. To account for these effects, the BTE, rather than Eq. (1), must be solved.

The BTE is the cornerstone of semiclassical transport. Its central concept is the idea of a *single* carrier-distribution function $f(r, p, t)$, which may be used to compute expectation values for macroscopic current flow. When we examine this idea from a quantum-mechanical viewpoint, we see some necessary averaging if f is to be regarded as a simultaneous function of position and momentum. From a device perspective, care must therefore be exercised if f and/or the critical device dimension L are smaller than an electron wavelength Λ. For a quasi-particle with an effective mass m^*,[2]

$$\Lambda^* = h/m^* v_{\text{therm}}, \tag{2}$$

[1] The effects of overshoot are implicit in the results of [4].

[2] Note that on an ultra, short time scale, electrons accelerate with the free, rather than the effective, mass [6]. Also, although the constraint $L > \Lambda^*$ is a safe one, it does not imply that the BTE is invalid in a range

$$\Lambda < L < \Lambda^*, \tag{3}$$

for the range of validity of the BTE has not yet been delineated. Here $\Lambda = h/mv_{\text{therm}}$.

where Λ^* is the thermal de Broglie wavelength, h Planck's constant, and v the thermal velocity. For GaAs central valley electrons, $\Lambda^* \sim 270$ Å (see Table I).

In addition to spatial considerations, from a temporal perspective, the BTE describes irreversible phenomena and it is assumed that collisions occur on a scale short compared with an observation time. Thus, for example, relaxation times should be short compared to transit times. This assumption will be routinely violated in submicron devices (see Table I); and again, caution must be exercised in using the BTE.

It is clear that from a device-modeling viewpoint we are faced with serious problems. On the one hand, the BTE overcomes many objections to the use of equations like (1). On the other hand, the preceding arguments suggest abandoning the BTE for ultrasmall devices and replacing it with a quantum transport formulation. While the latter may be necessary, the current approach is to retain the BTE as long as useful quantum effects can be incorporated, even when first-order quantum effects occur, as, for example, the transport in quantized inversion layers [7].

The critical choices that must be made in examining the physics of semiconductor devices is the principal reason for this chapter. In the discussion that follows we take a detailed look at some of the physical assumptions underlying transport in semiconductor devices. In Section II we discuss the semiconductor equations and the picture they provide for the numerical simulation of GaAs and Si devices. We illustrate the discussion with simulations of the transient behavior of two- and three-terminal

TABLE I

Critical Boltzmann Transport Parameters for GaAs

Parameter	Variable	Value
Active region length	L	10^{-5} cm, 10^{-4} cm
de Broglie wavelength	$\Lambda^* = h/m^*v$	2.7×10^{-6} cm at $v = 4.5 \times 10^7$ cm/sec, $m^* = 0.067\, m_e$
Transit time	L/v	5×10^{-13} sec at $L = 10^{-5}$ cm, $v = 2 \times 10^7$ cm/sec 5×10^{-12} sec at $L = 10^{-4}$ cm, $v = 2 \times 10^7$ cm/sec
Mean time between collisions (momentum relaxation)	τ_m	3×10^{-13} sec at 300 K, 5×10^{-14} sec at 3000 K
Collision duration	τ_c	2×10^{-14} sec

GaAs devices. For silicon we consider the role of ion implantation in altering the charge distribution within an n-channel MOSFET. In Section III, we examine the Boltzmann transport equation. Two aspects are considered: (1) What are the assumptions, limitations, and methods of solution? When the limits are exceeded, what quantum modifications are possible? (2) Where does the device physics using the BTE show significant departures from that using the equations of Section II. In the first instance, modifications are introduced to account, for example, for finite collision-duration effects. Here, on a very short time scale, collisions cannot be regarded as instantaneous. Instead, during collisions, the carrier may absorb or lose energy to the self-consistent field. In the second case, transient transport calculations emphasizing velocity overshoot show significant BTE departures.

The discussion of velocity transients naturally leads into problems associated with collisionless transport in submicron devices. These problems are, at first glance, trivial. However, this notion should be dispelled by the unusually long list of questions asked by others in connection with vacuum-tube transport, such as: (1) How do contact properties affect the current–voltage relation? (2) How will the distribution of injected carriers and subsequent carrier–carrier interaction, for example, affect device transport? (3) How is one to interpret diffusionlike terms, such as grad NT (where T is an electron temperature) when diffusion in the sense of an Einstein relation is not a viable concept? The presence of these questions for collisionless transport does not lessen their importance for transport with collisions. In both cases the answers have not been found and represent current work in progress.

The discussion of Section III is primarily concerned with solutions to the BTE. The numerous corrections to it serve as a reminder of the extent to which new physics forces a close scrutiny. Most of this reexamination is done by comparing approximate evaluations of the one-particle quantum-density matrix with its classical analog, the distribution function. The connection between the two is reviewed in Section IV, where we formally introduce von Neumann's density matrix and show its relation to measurements of statistical averages. From the density matrix, a fully quantum-mechanical distribution function, the so-called Wigner distribution is introduced. The Wigner function is particularly intriguing insofar as its equation of motion is very close to the BTE, subject to the constraints of the uncertainty principle. Approximate solutions to the equation of motion and its connection to the results of Section III are given.

Analysis of semiconductor devices relies on the assumption that current flow is by drift and diffusion. We show in Section III that the formulation of drift on a submicron scale is not at all obvious and that proper use of diffusion currents is even less certain. Diffusion is related to the spatial spreading of an ensemble of carriers with time, as the ensemble responds to both applied drift forces and random forces such as are generated by collisions. For examining transient diffusion processes, a Fokker–Plank equation can be generated whose solution is the transistion probability for a particle at X_0, t_0 to transition to X, t. On the short time scale over which relaxation processes occur, this equation does not reduce to the normal diffusion equation. We review diffusion processes on a short time scale by an ensemble Monte Carlo method, highlighting differences with the semi-classical description.

In summary, semiconductor device physics separates into three overlapping categories, identified by a particular type of equation:

(1) the semiconductor mobility equation,
(2) the Boltzmann transport equation, and
(3) the quantum transport equation

We shall discuss the modeling of transport in nonlinear semiconductor devices by solving, when available, or at least by examining each of these equations.

II. THE SEMICONDUCTOR EQUATIONS

A. Introduction

It has long been recognized that a proper understanding of today's devices requires detailed computer modeling of the space- and time-dependent charge distribution within the device. Simple, two-terminal configurations are usually represented by one-dimensional equations,[3] but the more important designs require a full account of at least two directions—one along the channel length and a second normal either to the metal–semiconductor interface (MESFET, for example) or the

[3] For one-dimensional devices, see Scharfetter and Gummel [8a]. This paper deals with the large-signal simulation of a silicon Read diode oscillator. Simulation of a TRAPATT oscillator is discussed by DeLoach and Scharfetter [8b]. For transferred electron devices, see Shaw *et al.* [8c]. For two-dimensional simulations, see Barnes and Lomax [8d]. This paper deals with finite-element methods. Nonisothermal carrier flow in a two-dimensional bipolar transistor under reactive circuit conditions is discussed by Turgeon and Navon [8e]. The phenomenon of avalanche breakdown in MOSFETs is discussed by Toyabe *et al.* [8f]. Yamaguchi *et al.* [8g] discuss the two-dimensional simulation of GaAs FET. See also Grubin and McHugh [8h].

oxide–semiconductor interface (MOSFET, for example). Many two-dimensional computer codes have been developed for this purpose [8a–8h]. Each of these codes involves obtaining self-consistent solutions of

(i) Poisson's equation,

$$\nabla^2\phi = -\rho/\epsilon, \tag{4}$$

where ϕ is the scalar potential and ρ the charge density;
(ii) the continuity equation,

$$\text{div} \cdot \mathbf{J}_c + (\partial\rho/\partial t) = G - R, \tag{5}$$

where J_c is the carrier current and the quantities on the right-hand side represent the possibility of local generation (G)–recombination (R) events; and
(iii) the "semiconductor equation,"

$$\mathbf{J}_c = qN\mathbf{v} - qD\,\mathbf{grad}\,N, \tag{6}$$

where q is the electric charge.

Equation (5) is for single-species transport, and each of these equations is constrained by the external circuit.

As indicated in the introduction, the transport picture associated with Eq. (6) is based on the approximation that the carrier velocity v responds instantaneously to changes in electric field. For electrons:

$$\mathbf{v}(F) = -\mu(F)\mathbf{F}, \tag{7}$$

where

$$\mathbf{F} = -\mathbf{grad}\,\phi. \tag{8}$$

Thus any spatial or temporal dependence in velocity arises because of such dependencies on field. This extremely important assumption breaks down at high frequencies [9]. In gallium arsenide, serious discrepancies appear at 10 to 20 GHz. The significance of this becomes clear with a simple example.

Consider solutions to Newton's equation of motion for a single carrier subjected to instantaneous change in field and scattering centers:

$$m^*\frac{dv}{dt} + \frac{m^*v}{\tau_p} = -eF. \tag{9}$$

For $v(t = 0) = 0$,

$$v(t) = (-e\tau_p/m^*)F[1 - \exp(-t/\tau_p)]. \tag{10}$$

Thus, on a temporal scale, instantaneous response implies that the observation or measurement takes place over a time interval that is long compared to the momentum relaxation time τ_p. For GaAs with scattering by intravalley–central-valley phonons, $\tau_p \simeq 0.3 \times 10^{-12}$ sec, and for a measurement over a time interval $\Delta t \simeq 3\tau_p$, the velocity is within 95% of the steady-state v ($t = \infty$) value. Note that $(3\tau_p)^{-1} \simeq 1100$ GHz, but in a solid containing a sufficiently high density of carriers (and here the solid must also be large enough to contain a sufficient number of carriers), the response of the devices is more closely related to the time required for the collection of carriers, as represented by their distribution function, to relax. For gallium arsenide, the distribution of carriers responds sluggishly to changes in field, so well before the 1100-GHz limit the "instantaneous" mobility model breaks down. The question arises of just how serious the breakdown is at lower frequencies. Although this question is a subject of current research, the indication (see Section III) is that, at least for gallium arsenide and frequencies below 20 GHz, qualitative device behavior will not be altered by the nonlocal contributions.

For the remainder of this chapter, we shall discuss, through the use of examples, the development of a numerical physics using the semiconductor equations. Three examples will be discussed:

(i) one-dimensional analysis of GaAs two-terminal devices,
(ii) two-dimensional analysis of GaAs FETs, and
(iii) two-dimensional analysis of a silicon MOSFET.

The GaAs devices are chosen because they are the most thoroughly studied transferred-electron-effect devices and exhibit the most dramatic spatial and temporal behavior. The family of GaAs devices is also among the leading candidates for future VLSI applications. The silicon devices are chosen because they are the most widely used and studied devices.

B. One-Dimensional Analysis of GaAs Two-Terminal Devices

1. *Introduction*

The standard one-dimensional simulation of GaAs two-terminal devices seeks to replicate Gunn's original observations [10] and the subsequent variations. Briefly, Gunn observed that when a GaAs sample fitted with two contacts was subjected to a sufficiently high bias, spontaneous and coherent microwave frequency current oscillations appeared (see Fig. 1). The oscillations had specific characteristics in that the period was closely related to the time it took the majority carrier (electrons) to transit between the contracts. The explanation, as we know it today, is due to

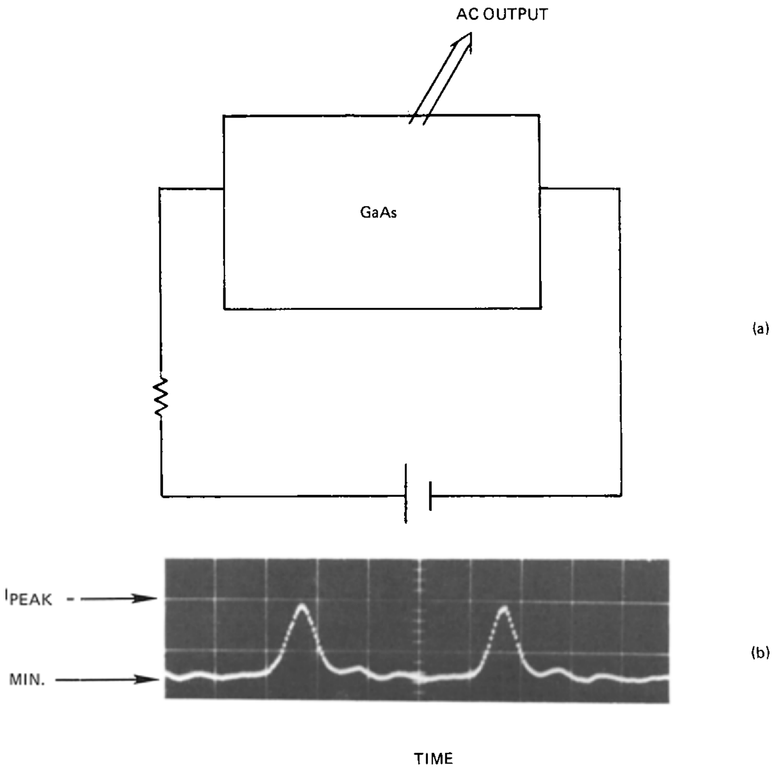


Fig. 1. (a) Schematic circuit for the appearance of Gunn oscillations. (b) One cycle of current versus time for a GaAs device sustaining dipole oscillations. Nucleation and extinction occurred during the "spiked" portion of the oscillation. Dipole transit is associated with the flat portion of the oscillation.

Kroemer [11], who proposed that when the oscillations occured the electric field within the gallium arsenide sample became highly nonuniform. This nonuniformity would manifest itself in the appearance of regions of high electric field surrounded by regions of low electric field, as in Fig. 2, and these high-field regions, often called *domains,* would move toward one end of the specimen. The oscillation was then a consequence of the following sequence of events. First, a domain would nucleate at one of the contacts (the cathode). At this point the current would drop, as in Fig. 1. Then the domain would leave the cathode region and travel down the sample toward the second contact (the anode). Here, the current would be approximately constant. Finally, the domain would be extinguished at the anode contact and the current would rise. The oscillation period T was determined mainly by the time it took the domain to travel between the

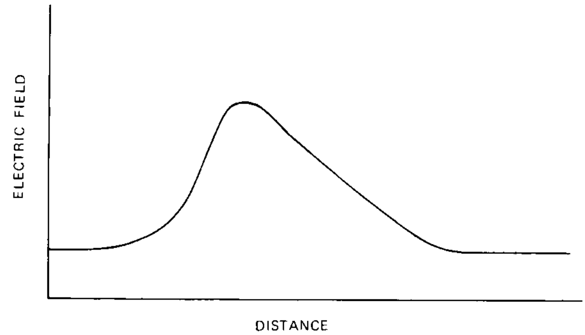


Fig. 2. Sketch of moving high electric field domain associated with the oscillation of Fig. 1.

cathode and anode contacts, and this was determined by the speed v of the moving electrons. Thus $T \simeq L/v$, where v is about 10^7 cm/sec and L the length of the specimen. Note that for $L \simeq 10^{-2}$ cm, $T \simeq 10^{-9}$ sec and $1/T \simeq 1.0$ GHz.

Kroemer's explanation of Gunn's experiments required that GaAs sustain a region of negative differential mobility, a fact demonstrated by Hilsum [12] in 1962. Here, by explicit calculation, it was shown that gallium arsenide, through electron transfer, would exhibit negative differential mobility, where for a range of increasing electric field, the electron velocity decreases rather than increases (Fig. 3).

Typically, simulations (see Fig. 4) of the behavior of negative differen-

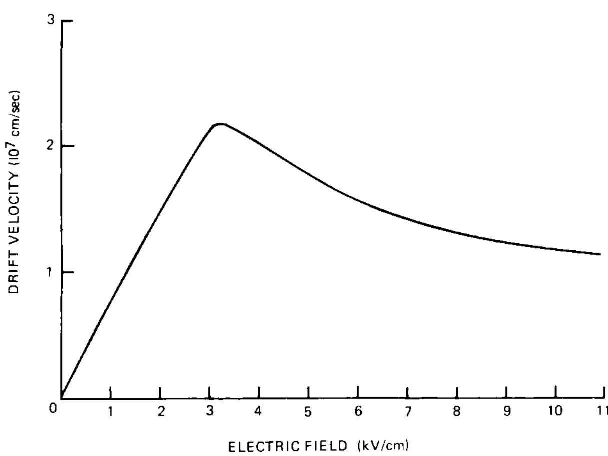


Fig. 3. Drift velocity versus electric field for GaAs. From Butcher [2], with permission of The Institute of Physics.

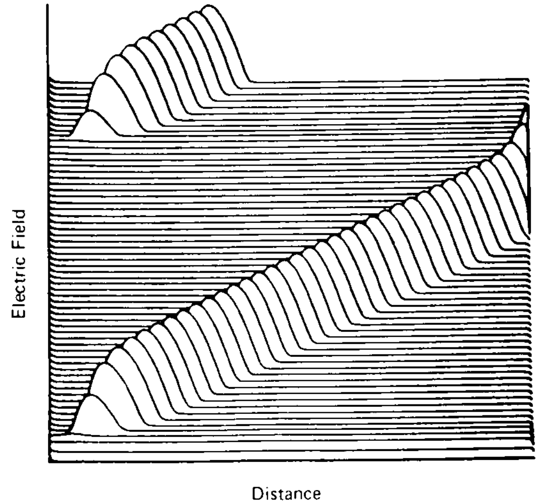


Fig. 4. Simulation of cathode-nucleated domain and subsequent transit-time oscillations. The NDM element is part of the resistive circuit, and the computations display $F(x,t)$ at successive instants of time. From Shaw *et al.* [8c], with permission.

tial mobility devices involve breaking down Eqs. (5), (6), and Poisson's equation and solving the following two sets of equations:

$$J_0(T) = v(F)\left(eN_0(x) + \epsilon\frac{\partial F(x,t)}{\partial x}\right) - D(F)\left(\epsilon\frac{\partial^2 F(x,t)}{\partial x^2} + e\frac{\partial N_0(x)}{\partial x}\right) + \epsilon\frac{\partial F(x,t)}{\partial t} \tag{11}$$

$$\Phi_B = \left(\mathscr{L}C_p\frac{d^2\Phi}{dt^2} + RC\frac{d\Phi}{dt} + \Phi\right) + \left(\mathscr{L}\frac{dI_0}{dt} + RI_0\right), \tag{12}$$

where $N_0(x)$ represents a spatially dependent donor density. Equation (12) is a representative circuit equation (see e.g., ref. 8c) with the device in parallel with a capacitor C_p, both connected serially to an inductor $\mathscr{L}$, resistor R, and bias Φ_B. Also,

$$\Phi = \int_0^L F(x,t)\,dx \tag{13}$$

and

$$I_0 = J_0(t)A, \tag{14}$$

where A is the cross-sectional area.

Equation (11) is the equation for total current density through the nonlinear semiconductor. The first collection of terms is the conduction-current contribution, the second and third contributions are the diffusion and displacement contributions, respectively. In the preceding equations, $v(F)$ represents the velocity–field relation for the carrier and $D(F)$ the diffusion–field relation. Within the framework of the mobility model $v(F)$ is represented by Fig. 3 for GaAs and $D(F)$ is given in Fig. 5.

Equation (11) is a partial differential equation rich in possibilities. Before discussing these, there are several points to be made with regard to the placement and use of the diffusion coefficient. First, note that D is to the left of the derivative. Within the framework of the Boltzmann transport theory, this is not entirely correct. For those situations where the distribution function departs slightly from equilibrium, Stratton [13] has demonstrated that the diffusion contribution should more generally appear as

$$\partial(Dn)/\partial x. \tag{15}$$

For highly nonequilibrium situations where such methods as the displaced Maxwellian are used, it is generally difficult to unequivocably identify a diffusivity. However, when it is defined, it usually emerges as a generalized Einstein relation. For multivalley semiconductors

$$D = \frac{1}{e}\sum \frac{\eta_i v_i T_i}{F}, \tag{16}$$

as in Fig. 5. Here v_i and T_i are, respectively, the carrier drift velocity of the ith valley, T_i its electron temperature, and η_i its fractional occupation. There are difficulties with this definition and we refer to Cheung *et al.* [14] for a more complete description.

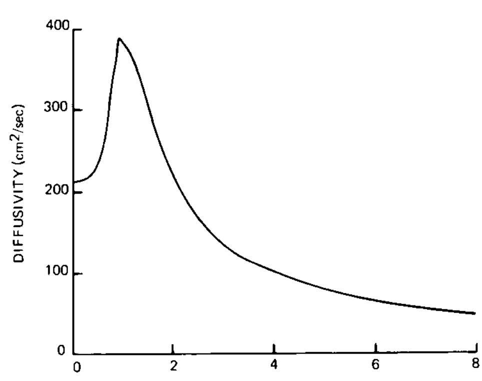


Fig. 5. Diffusivity versus electric field used in GaAs simulations. From Butcher [2], with permission of The Institute of Physics.

We also point out that Eq. (11) is written in "standard form" and retains the feature of treating electrons as positively changed particles. As long as we are treating one species of carrier, this does not present any real difficulties, but when electron and hole conduction occurs, care must be exercised. Among the difficulties to be aware of is the fact that the velocity–field curve in Fig. 3 represents the magnitude of the drift velocity since electrons move in a direction opposite to that of the electric field.

2. *Properties of Eq. (11)*

a. Length Dependence. Equation (11) is a two-point boundary-value problem and allows length-dependent effects to be studied. These effects are pronounced in GaAs and have their basis in the following: If an ostensibly homogeneous GaAs specimen is biased into the NDM (negative differential mobility) region, then ever-present space-charge fluctuations will grow. These growing fluctuations will simultaneously propagate and the amplitude of the disturbance in the simplest of cases [15] will be given by

$$G(x) = \exp(-x/v\tau), \tag{17}$$

where

$$\tau = \frac{\epsilon}{N_0 e \; dv/dF} < 0. \tag{18}$$

Significant gain will result in an instability when $G \gtrsim 1$ or

$$N_0 L > \left| \frac{\epsilon v}{e \; dv/dE} \right| . \tag{19}$$

For GaAs, $N_0 L$ is typically $\sim 10^{12}/\text{cm}^2$.

From another viewpoint, high field-domain propagation in devices greater than 50 μm is generally independent of the cathode and anode region and can be analyzed as though the device had infinitely separate boundaries. For $N_0 \simeq 10^{15}/\text{cm}^3$ and devices approximately 10 μm in length high field domains may fill a large percentage of the devices and proximity effects occur [8c].

b. Doping Variations. In the early stages of GaAs device fabrication, it was rare to find a uniformly doped device. The effects of these nonuniformities were generally accounted for through assumed variations in $N_0(x)$. For example, a significant drop in N_0 over a small distance was shown to lead to local high field-domain nucleation with subsequent propagation [16]. In another situation, Kroemer established, by numerical simulation [17], that the presence of small statistical variations in N_0 would generate transient nonuniformities and prevent accumulation-layer propagation in long samples. In short devices, ~10 cm long, these fluctuations are not as effective and accumulation-layer propagation is possible [8c].

c. Contact Effects. It has been established that the interface at the two-terminal active region dominates the subsequent behavior of the device. These contact effects have been treated by doping variations, mobility variations, and phenomenological interfacial fields. In many cases, through the combined use of simulations and experiments, one can determine whether doping variations or other contributions are responsible for device behavior [8c]. The effects of contact fields may, for example, be treated as boundary conditions to Eq. (11), where $F(x = 0, t)$ and $F(x = L, t)$ are specified [8c] or as a solution to a cathode differential equation, e.g.,

$$J_0(t) = \mathcal{J}(F) + \epsilon\, dF/dt \qquad \text{at} \quad x = 0, \tag{20}$$

where $\mathcal{J}$ is a phenomenological cathode current density. Our experience

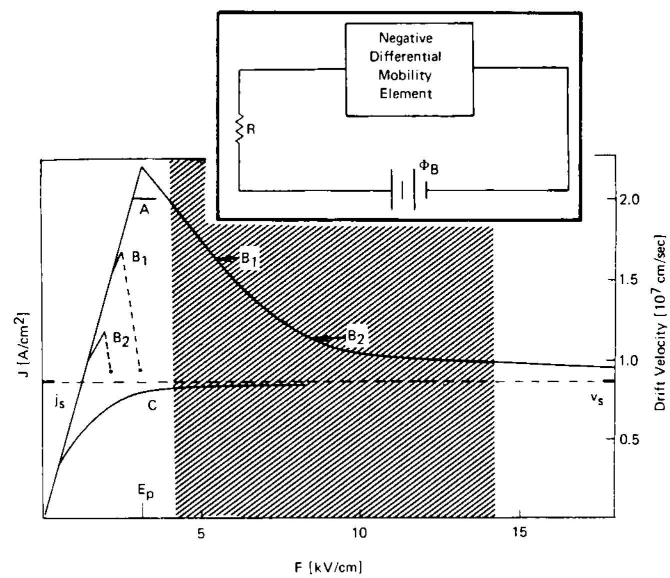


Fig. 6. The steady-state $v(F)$ curve and the simulated current density versus average field relation for various fixed values of carbode F_c. For the simulation, the NDM element is in the circuit shown in the inset. For curve A, $F_c = 0$, and the NDM element is quiet until the bulk field exceeds the NDM threshold. An instability then forms, and for relatively moderate doping, nonuniformities may damp. The result is a quiet element with a current level somewhat below the peak current density. The cathode fields for cuves B_1 and B_2 are indicated, and here cathode-nucleated domains result in distinct transit-time oscillations. For curve C, $F_c = 24$ kV/cm, and the NDM element is electrically stable. From Shaw *et al.* [8c] and related references therein.

indicates that the important contact to model is the cathode, and our results [8c] for a fixed cathode model are summarized in Fig. 6. More ambitious models are generated as the need arises, as for example with Eq. (20), which was generated to explain the unusual high-efficiency oscillations associated with InP [18,19].

In spite of the care one might exercise in solving Eq. (11), transient electrical instabilities require solutions to the circuit equations. For the device in a circuit containing reactive elements [e.g. Eq. (12)], the subsequent oscillation (see Fig. 7) may bear no resemblance to a propagating domain [18,19]. Another reason for insisting on the external circuit constraint is provided by the methods of characteristics and the field direction technique [20,21]. These show that there are an infinite number of time-independent solutions associated with a given value of J_0. It is the circuit constraint that picks out the correct distribution.

The preceding discussion has ignored the effects of generation–recombination on transient behavior of propagating high field regions. This is based on the assumption that instabilities develop on a time scale

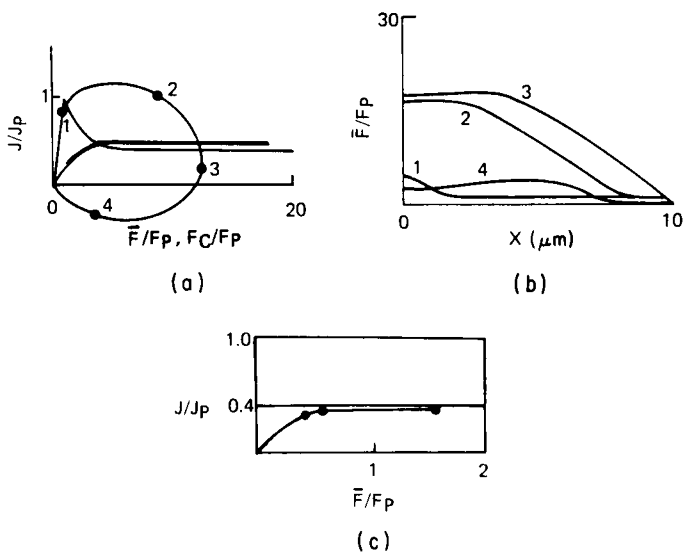


Fig. 7. Simulations using a time-dependent cathode condition [Eq. (20)]. The NDM elements is in a circuit [see Eq. (12)]. It is in parallel with a package capacitor and in series with an inductor, resistor, and dc source. (a) Simulation of current versus average field for a conduction current curve represented by the bold line. Superimposed on the bold line is the $v(F)$ curve for GaAs, scaled to current and voltage. (b) Electric field versus distance curves at four instants of time, keyed to part (a). (c) Preinstability current versus voltage. Filled circles denote computed points. From Shaw *et al.* [8c] and references therein.

that is short compared to the generation–recombination process. A recent modification of Eq. (11) was effected to include space- and time-dependent impurity ionization contributions expected if carrier generation–recombination becomes important [22].

C. Two-Dimensional Analysis of GaAs FETs

1. Introduction

Two-dimensional analyses of GaAs devices are almost exclusively devoted to simulations of three-terminal FETs in a variety of configurations. The second dimension in these simulations is transverse to the principal direction of current flow and its magnitude is often the determinant of whether a propagating instability will occur [23]. The need for simulations of two-dimensional problems is more generally accepted than that of one-dimensional ones because of the unavailability of widespread intuitive equations.

The FET in its simplest form is a semiconductor slab with three terminals. Two of these are usually low-resistance contacts, while the third is either a Schottky contact or a p–n junction with an accompanying region of charge depletion. For a unipolar conduction device, operation is based on modulation of the depletion region, which is usually accomplished by changes in the gate bias. Small and large signal gain are possible. Figure 8 is a sketch of the device and the connecting lumped elements.

The equations describing this device are Eq. (4), with

$$\rho = -e[N(x,y,t) - N_0(x,y,t)] \tag{21}$$

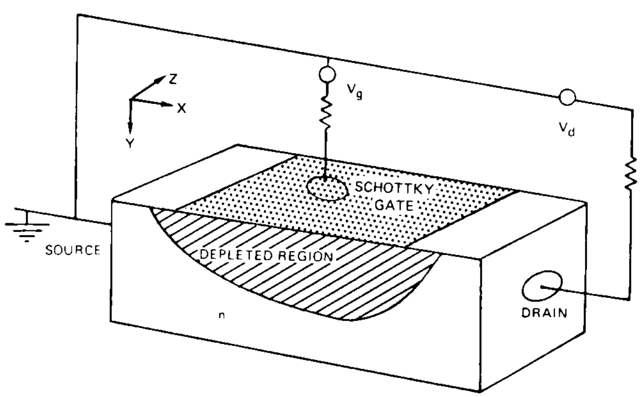


Fig. 8. Schematic representation of a Schottky gate field-effect transistor.

and

$$\mathbf{F}(x,y,t) = -\mathbf{grad}\ \phi(x,y,t), \tag{22}$$

and Eq. (5) with $G - R = 0$ and

$$\mathbf{J}_c(x,y,t) = -eN(x,y,t)\mathbf{v}(|F|) + eD\ \mathbf{grad}\ N(x,y,t). \tag{23}$$

Note that Eq. (5) requires point-by-point divergence-free total current density

$$\mathrm{div}\ \mathbf{i}(x,y,t) = 0, \tag{24}$$

where

$$-i(x,y,t) = \mathbf{J}_c(x,y,t) + \epsilon\ \partial\mathbf{F}(x,y,t)/\partial t. \tag{25}$$

The boundary conditions to these equations necessarily approximate the physical and electrical characteristics of the outer periphery of the semiconductor structure. In most GaAs FET simulations, the exposed surfaces are assumed to be ideal insulators and no current is permitted normal to these boundaries. Along the free surfaces and in the limit of zero permittivity for the space surrounding the device [24,8g],

$$\hat{n} \cdot \mathbf{grad}\ \phi = \hat{n} \cdot \mathbf{grad}\ N = 0, \tag{26}$$

where $\hat{n}$ is a unit vector normal to the free surface of the semiconductor.

The low-resistance source and drain contacts and the Schottky gate contact are approximated by equipotential surfaces with a prespecified charge density [24,8g]. For the calculations illustrated, the low-resistance source and drain contacts are neutral and located sufficiently far from the active region of the device that for the current levels involved they have no influence on the electrical properties of the device. For the sample calculations,

$$N_S = N_D = N_0 \tag{27}$$

and

$$N_G = N_0 \exp - [e\phi_{Bi}/k_B T_0] \tag{28}$$

where ϕ_{Bi} is the "built-in" potential [25] and N_S, N_D, and N_G indicate the source, drain, and gate, respectively.

To include circuit effects, the potentials on the contacts are needed. Generally we set the source potential to zero [8h] while the gate potential ϕ_G and drain potential ϕ_D are determined by simultaneous solution of the semiconductor equations and the circuit equations.

For the three-contact device of Fig. 8, the circuit equations are

$$\phi_S = 0, \tag{29}$$

$$\phi_{BD}(t) = I_D(t)Z_D + \phi_D(t), \tag{30}$$

$$\phi_{BG}(t) = I_G(t)Z_G + \phi_G(t), \tag{31}$$

$$I_S(t) = I_G(t) + I_D(t), \tag{32}$$

where ϕ_S is the source potential, ϕ_G the gate potential, ϕ_D the drain potential, ϕ_{BG} and ϕ_{BD} the drain and gate bias potentials, respectively, I_S the source current, I_G gate current, I_D drain current, Z_G the gate impedance, and Z_D the drain impedance. The external circuit impedances are modeled as lumped linear elements. The current passing across a contact is the integral over the contact area of the component of current density normal to the contact. For a device of dimension l_z in the z direction and a contact extending over a region l,

$$I(t) = l_z \int_l \hat{n} \cdot \mathbf{i}(x,y,t)\, dl'. \tag{33}$$

When simulating a GaAs FET, the important features are shape, dimension, and doping profile. Typically, effects of semi-insulating substrates may be examined by variations in $N_0(x,y,t)$ [24,26], alloyed contacts may be examined by increasing the doping level under the source and drain regions, and semi-insulating gate regions [27] may also be examined. We shall ignore these in the succeeding illustrations, concentrating instead on homogeneous doping profiles. Our intent is to illustrate the role of numerical simulations in classifying GaAs FET behavior.

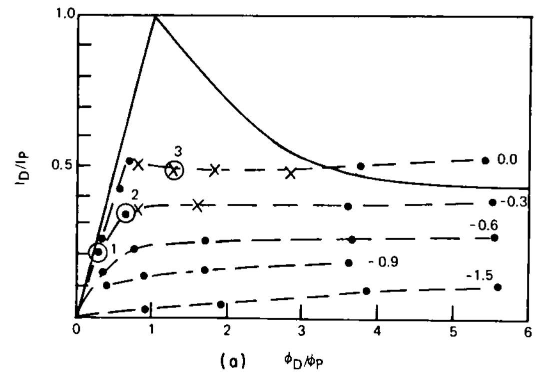
2. GaAs FET Classification

Numerical simulations and experiment [28] demonstrate that the GaAs FET can be placed in one of two groups as determined by the ratio

$$K = \frac{\text{gate voltage at cutoff}}{\begin{pmatrix}\text{drain voltage at the onset of current}\\ \text{saturation for zero gate voltage}\end{pmatrix}} \tag{34}$$

Devices with $K > 1$ sustain current oscillations, the origin of which lies in the presence of negative differential mobility in the semiconductor. Those with $K \lesssim 1$ are electrically stable. Devices with $K > 1$ are generally wider in the transverse dimension than those with $K \simeq 1$.

The electrical behavior associated with this classification is summarized in Fig. 9, where we sketch the simulated current–voltage relation for two GaAs FETs with a 10-μm source–drain separation. For reference, we have drawn the velocity–field relation for GaAs scaled to the current and voltage parameters. The first point to note about these results is that the current levels do not approach the peak current associated with



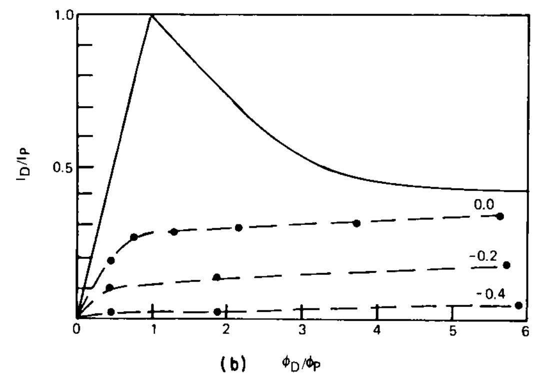


Fig. 9. Drain current–drain voltage relation: (a) $K > 1$ and channel height $l_y = 2.2\ \mu m$; (b) $K \simeq 1$ and $l_y = 1.2\ \mu m$. Filled circles denote computed points; ×s denote averages of current during an oscillation. The solid curve is the GaAs $v(F)$ relation scaled to current and voltage. Reprinted with permission from *Solid State Electron.*, **23**, H. L. Grubin, D. K. Ferry, and K. R. Gleason, "Spontaneous Oscillations in Gallium Arsenide Field Effect Transistors." Copyright 1980, Pergamon Press, Ltd.

GaAs. This is a consequence of the additional resistance supplied by the gate region as well as the velocity limitation. The second point is that for the wider-channel device an instability occurs. The instability is represented by the dashed I–V curves of Fig. 9a. The ×s in the diagram represent average current and voltage values for the instability, and the presence of negative conductance is due to the dynamic propagating domain. The filled circles in Fig. 9a represent stationary, time-dependent points, and we note that where there is an instability it is surrounded by regions of nonzero gate and drain bias for which there is no time-dependent behavior. The third point is that for the wider-channel device, the normalized current density at zero gate bias levels exceeds the sustaining current [8c], which is the minimum current necessary for stable domain propagation in two-terminal devices. The sustaining current plays a similar role in three-terminal devices, and in these wider-channel devices, domain propagation occurs. For the narrower-channel device, the current is below the sustaining current and no instability occurs. For this narrow device, and at sufficiently high drain bias levels, nonuniform domains form during an initial transient where they absorb most of the voltage and force a reduction in current level below the sustaining current. These results are consistent with the thickness dependence previously discussed [23].

We illustrate the internal distributions of charge and current at three points, 1, 2, and 3, of Fig. 9. Additional information is available in Grubin and McHugh [8h,26] and Grubin *et al.* [28]. The internal distribution of charge and current associated with the current and potential levels 1 and 2 of Fig. 9 are shown in Fig. 10 [8h]. Parts (a) and (b) show current-density streamlines through the device, with the length of each proportional to the magnitude of the vector's current density at that point. The maximum length of the individual x and y components before overlap is $J_p = N_0 e v_p$, where v_p is the peak carrier velocity. We note that in both cases the current density is greatest under the gate contact as required by current continuity. For the higher bias, the current density under the gate region is at least as great as J_p and velocity limitation introduces carrier accumulation. The density of charge particles in the FET is generally nonuniform and parts (c) and (d) of Fig. 10 are a projection of this distribution as it relates to the current-density profile of parts (a) and (b). We note that the particle density increases in the downward direction. Part (a) shows a

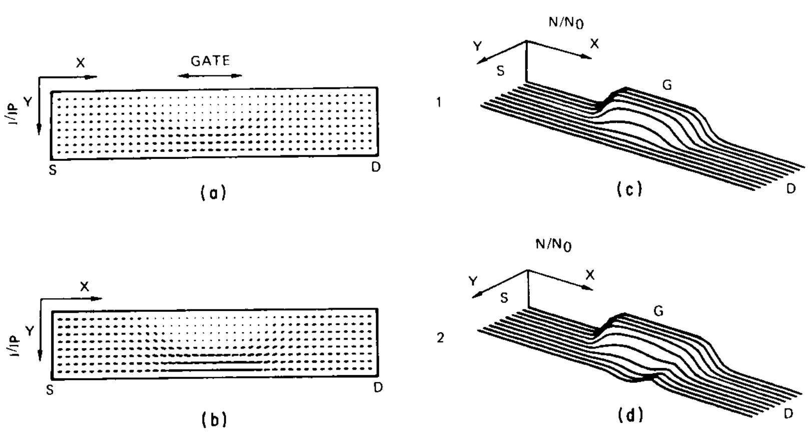


Fig. 10. The internal distribution of current and charge corresponding to the current and voltage levels represented by numbers 1 and 2 of Fig. 9. Note that the particle density surrounding the nonuniform distribution is uniformly distributed within the source–gate region and the gate–drain region. Reprinted with permission from *Solid State Electron.*, **21,** H. L. Grubin and T. M. McHugh, "Hot Electron Transport Effects in Field Effect Transistors." Copyright 1978, Pergamon Press, Ltd.

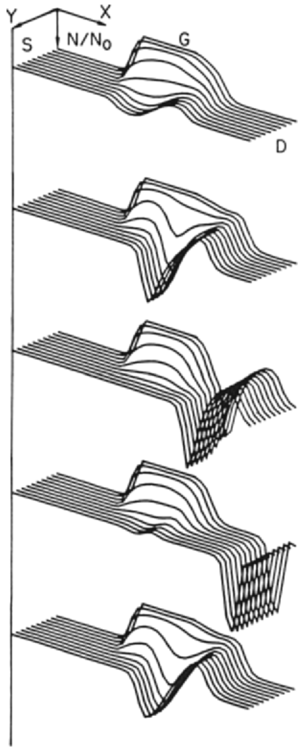


Fig. 11. Projection of the time-dependent particle density when an instability occurs. The parameters at which this occurs are represented by the number 3 in Fig. 10. Reprinted with permission from *Solid State Electron.*, **21,** H. L. Grubin and T. M. McHugh, "Hot Electron Transport Effects in Field Effect Transistors." Copyright 1978, Pergamon Press, Ltd.

region of charge depletion directly under the gate contact. Part (b) shows the formation of a weak stationary dipole layer under the gate contact. Here the x component of electric field has reached the NDM threshold field value at the drain edge of the gate contact.

We now consider the presence of an instability. We recall that for two-terminal devices the instability is determined by the value of the electric field at the cathode boundary [8c] and that the threshold current density for the instability is anywhere between J_s and J_p, where J_s is the current density associated with the high-field saturated drift velocity [8c]. In contrast, for three-terminal devices, the initiation of a domain instability generally occurs under the gate contact and at a local value of current density approximately equal to J_p. Figure 11 illustrates the sequence of events associated with an instability. Domain growth under the gate is accompanied by an increase in potential across the device. A corresponding decrease in current occurs throughout the device and circuit, as constrained by the dc load line. As the current decreases carriers with velocities below that of the peak velocity enter the accumulation layer, which subsequently begins to detach. The domain speeds as it leaves the gate region and settles into a value of current density somewhat in excess of that associated with the saturated drift velocity of the electrons. Prior to reaching the drain contact, the domain dynamics appear to be one-dimensional.

D. Two-Dimensional Analysis of a Silicon MOSFET

Although a large body of gallium arsenide work is still regarded as research and, consequently, only a relatively moderate amount of two-dimensional simulations have appeared, this is not the case with silicon devices. Here VLSI requirements introduce an urgency into the studies, and MOS simulations, involving two types of carriers, are often used as an alternative to experimental investigations.

For this case, the basic semiconductor equations are generalized so that the charge density in Eq. (1) becomes

$$\rho = e[P(x,y,t) - N(x,y,t) + N_{\mathrm{D}}(x,y,t) - N_{\mathrm{A}}(x,y,t)]. \tag{35}$$

The continuity equation is generalized so that

$$\operatorname{div} \mathbf{J}_n - e\, \partial[N(x,y,t) - N_{\mathrm{D}}(x,y,t)]/\partial t = G - R, \tag{36a}$$

$$\operatorname{div} \mathbf{J}_{\mathrm{p}} + e\, \partial[P(x,y,t) - N_{\mathrm{A}}(x,y,t)]/dt = -(G - R), \tag{36b}$$

and the current-density equation is

$$\mathbf{J}_n = -eN(x,y,t)\mathbf{v}_n + eD_n\, \mathbf{grad}\, N, \tag{37a}$$

$$\mathbf{J}_p = +eP(x,y,t)\mathbf{v}_p - eD_p\, \mathbf{grad}\, P. \tag{37b}$$

The boundary conditions, device shape, and material parameters specify the problems to be studied. Figures 12–14 summarize the results of a recent calculation [29] designed to show the dependence of the internal charge distribution on the impurity concentration of an MOS transistor. The device configuration is shown in Fig. 12, where the line segment BE represents the semiconductor–oxide interface. Within the oxide region,

$$\nabla^2\phi = 0, \tag{38}$$

and at the interface,

$$\epsilon_{\mathrm{ox}} \left.\frac{\partial\phi}{\partial y}\right|_{\mathrm{ox}} = \epsilon_{\mathrm{sem}} \left.\frac{\partial\phi}{\partial y}\right|_{\mathrm{sem}}. \tag{39}$$

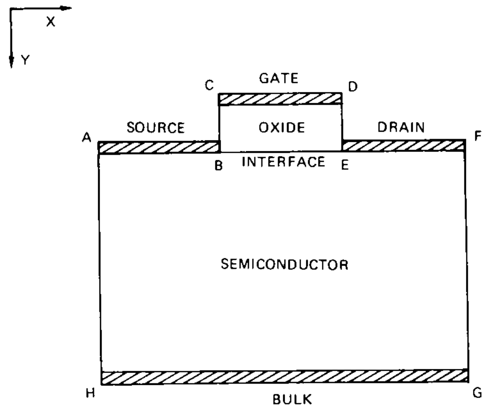


Fig. 12. Geometry for the MOSFET simulation. The regions under AB and EF are generally heavily doped n^{++} regions. From Selberherr *et al.* [29], with permission.© 1980 IEEE.

As in the MESFET calculation, the carrier densities at the source and drain contacts are set equal to the doping concentration. No current is permitted to flow across the interface BE, the bulk control HG, or the exposed surfaces AH and FG.

The doping profiles to which the mobile electrons respond are shown in

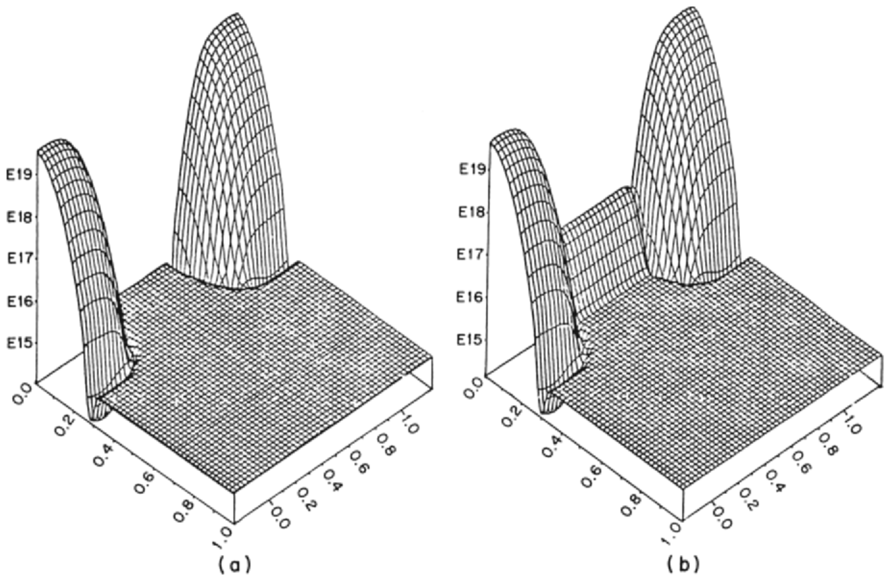


Fig. 13. Doping concentration of silicon MOSFET. Axes are lengths in microns. From Selberherr *et al.* [29], with permission.© 1980 IEEE.

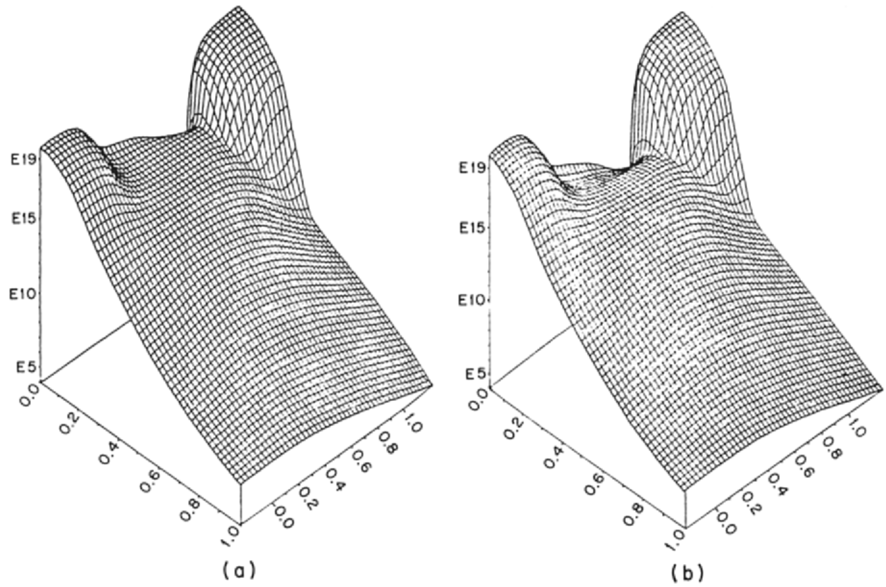


Fig. 14. Electron concentration of silicon MOSFET under strong inversion. Axes are lengths in microns. From Selberherr *et al.* [29], with permission. © 1980 IEEE.

Fig. 13. In Fig. 13a, the p-type dopant is uniformly $10^{15}/\text{cm}^3$, and in Fig. 13b, it increases at the oxide interface to approximately $5 \times 10^{16}/\text{cm}^3$. The surface concentration of the source and drain regions is $5 \times 10^{19}/\text{cm}^3$ and the depth of the p–n junction under the source and drain is approximately 3000 Å. Figure 14a shows the electron distribution for the first device under strong inversion. The surface concentration in the channel is high. Figure 14b shows the ion-implanted device (Fig. 13b) again under strong inversion. As expected, the surface concentration has decreased.

III. THE BOLTZMANN TRANSPORT EQUATION

A. Introduction

In general, the free carriers in a semiconductor gain energy from the electric field. This energy must be relaxed to the lattice through electron–phonon interactions. For vanishingly small values of the electric field, the energy gained from the field is negligible in comparison with the mean energy of the carriers, whether this latter quantity is represented by a thermal energy or by the Fermi energy in degenerate semiconductors.

For larger values of the electric field, the average energy of the carriers is increased by the field, and the carriers are said to be "hot." Because the average energy of the carriers increases in the field, the net rate of phonon emission must also increase to yield the energy loss.

The last factor of increased emission of phonons to balance the energy gain from the field is a consequence of detailed balance. Consider a non-degenerate distribution characterized by an electron temperature T_e. Phonon emission and absorption processes connect states at energy E with those at $E + \hbar\omega_0$. The rate of absorption of phonons out of the states at E is given by

$$AN_q \exp(-E/k_B T_e), \tag{40}$$

where N_q is the Bose–Einstein occupation factor of the phonons of wavelength q and k_B Boltzmann's constant; the rate of emission from the states at $E + \hbar\omega_0$ is given by

$$A(1 + N_q) \exp[-(E + \hbar\omega_0)/k_B T_e]. \tag{41}$$

Now $(1 + N_q) = N_q \exp(\hbar\omega_0/k_B T_0)$, where T_0 is the lattice temperature, so that emission and absorption processes establish a detailed balance when $T_e = T_0$. If additional energy is supplied to the electron gas by an external electric field, then the gain in average energy is characterized by a rise in T_e and the emission processes increase over the absorption mainly by the factor $\exp\{[\hbar\omega_0(1 - T_0)/T_e]/k_B T_0\}$, provided that the distribution function remains Maxwellian. The amount of rise in electron temperature T_e is governed by the energy gains and losses of the electron gas, and a detailed balance is established when the average rate of energy loss to the lattice $-\langle dE/dt \rangle$ equals the rate of energy gain from the electric field, or

$$e\mu F^2 = -\langle dE/dt \rangle. \tag{42}$$

The problem of evaluating Eq. (42) often reduces to one of determining the carrier-distribution function. At very low electric fields, the distribution function can be described as merely a small drift term superimposed on the thermal distribution, as

$$f(E,T_0) = f_0(E,T_0) + f_1, \qquad f_1 \ll f_0, \tag{43}$$

where $f_0(E,T_0)$ is the Maxwell–Boltzmann (or Fermi–Dirac) distribution at the lattice temperature T_0. When the electric field becomes sufficiently large, however, that the procedure of (43) can no longer be used, we are said to be in a high-field regime and hot-electron (or hole) effects must be considered [9]. The overriding theoretical concern in high-field transport is one of discerning the form that the distribution function takes in the presence of the electric field. In semiclassical transport, this distribution function is given as a solution to the Boltzmann transport equation (BTE). Because of the very complicated nature of the BTE—the fact that it is a nonlinear, integrodifferential equation—it is usually not possible to solve it analytically and several possible assumptions can be made. This aspect was recognized early, and much work was done in the mid-1930s by several Russian authors [30,31]. Since then a considerable amount of work has followed, both theoretical and experimental. One reason for this lies in the manner in which investigations of the high-field transport yield information on the details of the electron–phonon interaction and the interaction of the carriers with themselves and with impurities. Another major reason is the role played by hot-electron behavior in the operation of electron devices.

In spite of the many studies of hot-electron (and hot hole) behavior, the central problem underlying the entire area remains that of trying to understand the manner in which the distribution function of the electrons is modified by the presence of the electric field. This is true whether we are dealing with a bulk material or the current response in a device. It is also a formidable experimental problem [32]. In general, the Boltzmann transport equation can be expressed in its most general form as

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \frac{\partial f}{\partial \mathbf{r}} - e\mathbf{F} \cdot \frac{\partial f}{\partial \mathbf{p}} = \int d\mathbf{p}'[f(\mathbf{p}')W(\mathbf{p}',\mathbf{p}) - f(\mathbf{p})W(\mathbf{p},\mathbf{p}')], \quad (44)$$

where $f(\mathbf{r},\mathbf{p},\mathbf{t})$ is the carrier-distribution function. Generally for the steady-state response, the first two terms on the left are ignored and the distribution is a function of the carrier pseudomomentum $\mathbf{p} = \hbar\mathbf{k}$ and the energy. It is important to note that the Boltzmann transport equation assumes that the collisions are instantaneous in both space and time and that the field and scattering are different perturbations. In spatially varying problems, the addition of the position vector $\mathbf{r}$ brings about an effective lowering of the symmetry of the problem and complicates the solution of the Boltzmann equation (or an equivalent formulation). One can usefully classify the various phenomena or solutions upon the level of this symmetry. Where hot-carrier behavior is not considered, a single dimension, the electron energy E, is sufficient. If an axis of rotational symmetry exists for the hot-electron problem, perhaps along the electric field direction, then two variables, $\mathbf{p}$ (or E) and θ, are all that is required. In many cases the problem is more complicated, however. But, in some circumstances, an analytical form can be assumed for the distribution function and if this form depends only on a small number of parameters, then the zero-dimensional case results. In this latter case, simple equations for the evaluation of these parameters can be found, usually from moments of the

BTE [33]. In other cases, no approximations can be made, and detailed numerical techniques must be used. In modern transport theory, approximation techniques based on Legendre expansions for $f(\mathbf{p},t)$ no longer find much usage. Modern computers allow relatively rapid solution of detailed equations, and so the two approaches just discussed receive extensive usage. Which of the two methods is to be preferred depends on the rate of intercarrier energy exchange, discussed by Hearn [34].

B. Displaced Maxwellian Equation

Fröhlich [35] first pointed out that the isotropic part of the carrier-distribution function is Maxwellian provided that the carrier concentration exceeds a certain critical concentration; i.e., provided that the rate of intercarrier energy exchange is sufficiently large. Under conditions for which the anisotropic terms can be taken as small, Fröhlich and Paranjape [36] pointed out that a displaced Maxwellian distribution

$$f(E) = A \exp[-(E - \mathbf{v}_{\mathrm{d}} \cdot \mathbf{p})/k_{\mathrm{B}}T_{\mathrm{e}}], \tag{45}$$

containing the electron temperature T_{e} and drift velocity $\mathbf{v}_{\mathrm{d}}$ as parameters, could be utilized. This is then the hot-electron distribution in the approximation for which the critical carrier concentration is exceeded, and the parameters are then determined from balance equations, which, in turn, are obtained from the Boltzmann equation. Following the approach of Price [37], Eq. (44) is multiplied by any function $\phi(\mathbf{p})$ and integrated over $\mathbf{p}$. For electrons within a single valley, this gives

$$\frac{\partial}{\partial t}\langle\phi\rangle + \frac{\partial}{\partial \mathbf{r}} \cdot \langle\phi\mathbf{v}\rangle = -\left\langle e\mathbf{F} \cdot \frac{\partial}{\partial \mathbf{p}}\phi\right\rangle + \left\langle\int d\mathbf{p}'\,[\phi(\mathbf{p}') - \phi(\mathbf{p})]W(\mathbf{p},\mathbf{p}')\right\rangle. \tag{46}$$

The last term on the right arises from an interchange of variables involved in the double integration. The terms on the left-hand side vanish for the homogenous steady state. For ϕ equal to $\mathbf{p}$, the first term on the right is just the force $\mathbf{F}$ and (46) is the momentum-balance equation. For ϕ equal to the energy E, the first term on the right is $\mathbf{v}_{\mathrm{d}} \cdot \mathbf{F}$ and (46) is the energy-balance equation. In particular, the factor within the angle brackets in the last term on the right serves to define the average rate of energy loss to the lattice by collisions

$$\left\langle\frac{dE}{dt}\right\rangle_{\mathrm{coll}} = \int d\mathbf{p}'\,[E(\mathbf{p}') - E(\mathbf{p})]W(\mathbf{p},\mathbf{p}') \tag{47}$$

and the average rate of momentum loss due to collisions

$$\left.\frac{dp}{dt}\right|_{\text{coll}} = \int d\mathbf{p}' \,[\mathbf{p}' - \mathbf{p}]W(\mathbf{p},\mathbf{p}'). \tag{48}$$

Although Eq. (46) has been developed for a single valley, the procedure is readily extended to sets of nonequivalent valleys. This is done in Appendix A.

As an example, consider a material such as indium arsenide, where the scattering in the central valley is dominated by the polar optical phonon. In this case, the energy-balance equation becomes

$$ev_{\text{d}}F = A_1[\exp(y - x) - 1]x^{1/2}\exp(x/2)K_0(x/2), \tag{49}$$

where $A_1 = eF_0(2\hbar\omega_0/\pi m^*)^{1/2}/[\exp(y) - 1]$, $x = \hbar\omega_0/k_{\text{B}}T_{\text{e}}$, $y = \hbar\omega_0/k_{\text{B}}T_0$, and F_0 is an effective electric field describing the coupling between the electrons and the phonons. Similarly, the momentum-balance equation becomes

$$\begin{aligned} -eF = {} & (A_1 m^* v_{\text{d}}/3\hbar\omega_0)\{[\exp(y - x) + 1]K_1(x/2) \\ & + [\exp(y - x) - 1]K_0(x/2)\}x^{3/2}\exp(x/2) \\ & + 8v_{\text{d}}(2\pi m^* k_{\text{B}}T_{\text{e}})^{1/2}(3\pi l_{\text{ac}})^{-1} \end{aligned} \tag{50}$$

where the last term on the right takes into account the momentum loss to the elastic scattering by acoustic modes. Now, Eqs. (49) and (50) can be solved simultaneously to yield T_{e} for a given electric field F, and then this result can be used in Eq. (46) to find v_{d} and, hence, μ. It should be pointed out that this is an exceedingly simplified model, and factors such as intervalley transfer and band nonparabolicity should be considered. However, the results given here are useful as an illustrative example of the application of the displaced Maxwellian technique.

The accuracy of the balance equations obtained from (46) ranges, in its applications to solving transport problems, from very good to exceedingly poor, the latter in cases in which the preceding assumptions are just not valid. Perhaps the most easily violated condition is the critical carrier concentration required. In the cases in which the balance equations are good, one can use them to *infer* energy and momentum relaxation times as

$$-e\tau_{\text{E}} = d\langle E\rangle/d(\mathbf{v}_{\text{d}}\cdot\mathbf{F}), \tag{51}$$

$$-e\tau_{\text{p}} = d\langle\mathbf{p}\rangle/d\mathbf{F}. \tag{52}$$

It should be pointed out that although these definitions appear in the balance equations, their validity goes beyond the displaced Maxwellian. Equations (47) and (48) can be averaged over any distribution $f(E)$ to define effective energy and momentum relaxation times, but the connection of this to the electric field, as in Eqs. (51) and (52), must be used carefully. Since these times are based on the balance equations, the validity of their definitions is inherently tied into the results, and although any distribution function can be used, it must be done judiciously. One finds from these considerations of the relaxation times that the energy relaxation time τ_E is considerably longer than the momentum relaxation time τ_p, so that the drift velocity responds to a change in **F** faster than the electron temperature responds. This can lead to overshoot effects in the velocity, in which $\mathbf{v}_d$ rises to a value corresponding to an electron temperature at the starting time just before the change in **F**, then changes further as the temperature and distribution relax to the new temperature [3,38,39]. This time-dependent behavior and the steady-state ac cases can be treated by using the time-dependent form of (46), using, for example, a large, steady, dc $\mathbf{F}_0$ and a small, sinusoidal, ac field $\mathbf{F}_1$ superimposed upon it. In this case, one finds that the ac mobility includes terms like $(1 + i\omega\tau_p)$ and $(1 + i\omega\tau_E)$, reflecting the two time scales in the problem [40]. More complicated time variations have been examined by Butcher and Hearn [4], and spatial variations of the displaced Maxwellian have been considered by Bosch and Thim [41]. We shall expand considerably on these discussions, with particular attention paid to device simulation.

C. Semiempirical Transport Equation Method

In many cases, the BTE cannot be solved utilizing a displaced Maxwellian, but a set of balance equations is desirable for simplicity of application, especially in device modeling. An approach has built up that achieves this simplified goal and is called (by us) the *semiempirical transport equation method* (SETEM). Equation (46) can be rewritten as

$$\frac{\partial\langle\phi\rangle}{\partial t} + \nabla\cdot\langle\phi\mathbf{v}\rangle = -\langle e\mathbf{F}\cdot\nabla_p\phi\rangle - \frac{\langle\phi\rangle}{\tau_\phi}, \tag{53a}$$

where $1/\tau_\phi$ is an effective relaxation rate for $\langle\phi\rangle$. Of course, this quantity can be calculated directly from the collision integrals (47) and (48) if the distribution function is known. The approach of the SETEM is similar to that of the semiempirical tight-binding method in band-structure calculations, where the simple approach is used as an interpolation method. Relatively exact calculations of $\mathbf{v}(\mathbf{F},t)$ and $\langle E(\mathbf{F},t)\rangle$ can be made using the numerical techniques to be discussed. Then the set of equations given by (53) for different functions $\phi(\mathbf{p})$ can be fit to the exact calculations by using τ_p, τ_E, . . . as adjustable parameters that are functions of the field **F**. Once the relaxation times are known, the balance equations (53) can be used for device modeling (see e.g. Shur [42]), and we shall illustrate this next.

Carnez *et al* [43] applied the SETEM to the modeling of submicron FETs. They obtained simultaneous solution of Poisson's equation, the equation of continuity, and the time-dependent, but spatially independent, momentum and energy equations. For the latter, the SETEM equations are

$$d[m^*(E)v]/dt = -eF - [m^*(E)v/\tau_p(E)], \tag{53b}$$

$$dE/dt = -eFv - [(E - E_0)/\tau_E(E)], \tag{53c}$$

where $\tau_p(E)$ and $\tau_E(E)$ were obtained from the steady-state values of velocity and energy

$$-\tau_p(E) = m^*(E)\langle v(E)\rangle / e\langle F\rangle, \tag{53d}$$

$$-\tau_e(E) = E - E_0/e\langle F\rangle\langle v(E)\rangle. \tag{53e}$$

Here $\langle F\rangle$ relates F to energy, as in Eq. (53g). For a two-valley semiconductor,

$$m^*(E) = [1 - N_s(F)/N_0]m_c^* + [N_s(F)/N_0 m_s^*] \tag{53f}$$

and

$$E(F) = E_c(F)[1 - N_s(F)/N_0] + [E_s(F)N_s(F)/N_0]. \tag{53g}$$

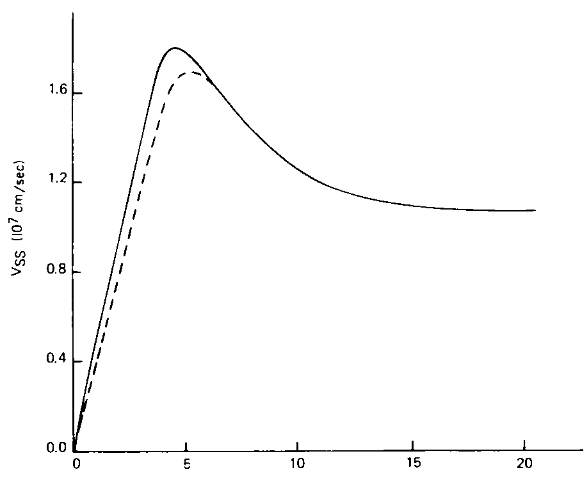


Fig. 15. Steady-state drift velocity versus electric field; (—)N_d = 10^{17} cm^{-3}, (---)N_d = 3×10^{17} cm^{-3}. From Carnez *et al.* [43], with permission.

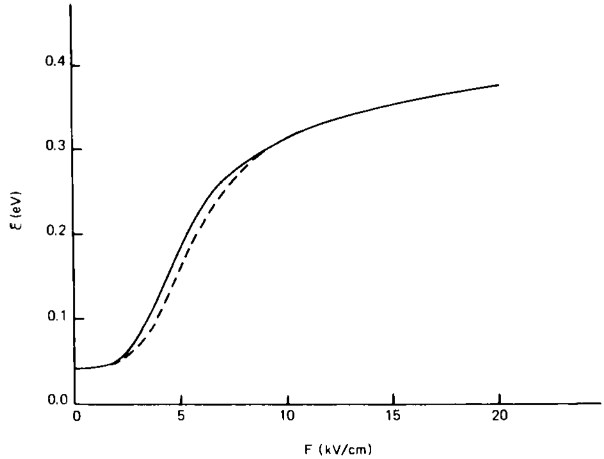


Fig. 16. Steady-state mean energy versus electric field; (—)N_d = 10^{17} cm^{-3}, (---)N_d = 3 × 10^{17} cm^{-3}. From Carnez *et al.* [43], with permission.

Here $N_s(F)$, $E_c(F)$, and $E_s(F)$ and the brackets ⟨ ⟩ denote steady-state relations.

Equations (53b)–(53g) contain the essential philosophy of SETEM as currently used; namely, nonequilibrium conditions can be obtained entirely from the steady-state parameters. The calculations require steady-state values for the field dependence of energy, velocity, and effective mass, which may be obtained from Monte Carlo method (see, e.g., Littlejohn *et al.* [44]). An illustration of this is given in Figs. 15–17.

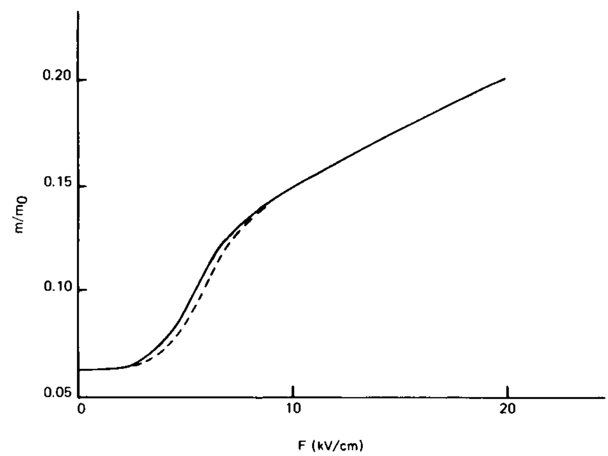


Fig. 17. Steady-state effective mass ratio versus electric field; (—)N_d = 10^{17} cm^{-3}, (---)N_d = 3 × 10^{17} cm^{-3}.

An important question concerning the use of Eqs. (53b)–(53g) is: How good is the approximation? Clearly, as used by Carnez *et al.* [43] and others, only two relaxation mechanisms, energy and momentum, are considered. An extremely important relaxation mechanism is intervalley carrier relaxation. While this, in principle, can be accounted for in a general SETEM treatment, this has not been addressed by Eqs. (53b)–(53g). In addition, all spatial relaxation has been ignored. In submicron devices where the fields are highly nonuniform, spatial relaxation may be the dominant transient contribution. Should this occur, Eqs. (53b)–(53g) would be inadequate. Nevertheless, as applied by Carnez *et al.*, surprisingly good results can be obtained (see Figs. 18 and 19).

D. Numerical Techniques

The truncation of the expansion for the distribution function to the first two terms, as is done in the displaced Maxwellian, is, in general, not a valid approach. It usually is justified only in the rare cases that eFl is small compared with the energy range over which f_0 varies appreciably, where l is a composite mean free path. The rapid variation of f_0 in the region about the optical phonon energy limits this truncation to small values of the field or to cases of very rapid energy exchange via carrier–carrier scattering. The problem is to derive the distribution function for free electrons in a semiconductor from a knowledge of the various scattering processes and the applied fields. Although it is possible to justify various numerical solutions of the electron-transport problem without specific reference to the

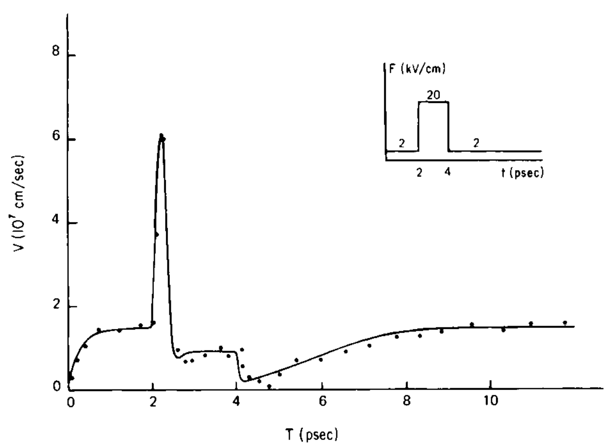


Fig. 18. Drift velocity versus time when applying an electric field pulse; (—) analytic formulation, (· · ·) Monte Carlo calculations. From Carnez *et al.* [43], with permission.

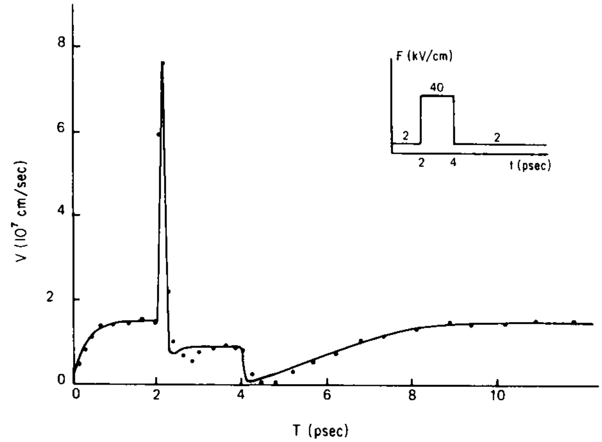


Fig. 19. Drift velocity versus time when applying an electric field pulse; (—) SETEM formulation, (· · ·) Monte Carlo calculations. From Carnez *et al.* [43], with permission.

integrodifferential form of the Boltzmann equation, it is more illustrative to take this as a starting point and, thus, to clarify the basic properties of the numerical methods. For a spatially uniform electron system in an external field **F**, the time-dependent Boltzmann equation is given from (44) as

$$\left[\frac{\partial}{\partial t} - e\mathbf{F} \cdot \frac{\partial}{\partial \mathbf{p}} + \lambda(\mathbf{k})\right] f(\mathbf{k},t) = \int W(\mathbf{k}',\mathbf{k}) f(\mathbf{k}',t)\, d\mathbf{k}', \qquad (54)$$

where $\mathbf{p} = \hbar\mathbf{k}$ and $\lambda(\mathbf{k})$ is the total out-scattering rate ($\lambda = 1/\tau$) and represents the second term on the right-hand side of (44). The term on the right-hand side of (54) represents just the in-scattering contributions. However, the definitions in $\lambda(\mathbf{k})$ are relatively incomplete since part, or all, of the $\lambda(\mathbf{k})f(\mathbf{k},t)$ term could be absorbed into the term on the right. The precise definition of $\lambda(\mathbf{k})$ and, consequently, of $W(\mathbf{k},\mathbf{k}')$ will usually depend on the particular calculation to be undertaken, but the formal theory is independent of these considerations. We shall therefore proceed as just defined.

The inverse of the differential operator (54) is just an integral operator. Budd [45] points out that the integration is a generalization of the Chambers [46] path integral, and the result can be written as

$$f(\mathbf{k},t) = \int_0^\infty ds \int d\mathbf{k}'\, f(\mathbf{k}',t-s)\, W\left(\mathbf{k}',\mathbf{k} + \frac{e\mathbf{F}s}{\hbar}\right) \times \exp\left\{-\int_0^s \lambda\left(\mathbf{k} + \frac{e\mathbf{F}y}{\hbar}\right) dy\right\}. \qquad (55)$$

The transformation of the BTE into this form, when supplemented by numerical techniques on a grid of points in **k** space, as may be peformed on modern digital computers, provides an extremely powerful technique for solving the Boltzmann equation. It is demonstrated, quite elegantly, by Rees [47,48] that an iterative approach can be utilized to great advantage. Moreover, he also points out that the iterative solution is a surrogate for the time evolution of $f(\mathbf{k},t)$, a point we shall discuss further.

An important technical innovation introduced by Rees [47] is the concept of self-scattering, a fictitious scattering process, which does not alter the physics, but allows a great simplification of the mathematical detail. To each side of (55) we add a term of the form

$$\Gamma(\mathbf{k})\,\delta(\mathbf{k}-\mathbf{k}'). \tag{56}$$

In particular, the simplification arises if we define this term as

$$\Gamma(\mathbf{k}) = \Gamma - \lambda(\mathbf{k}), \tag{57}$$

with Γ sufficiently large that $\Gamma(\mathbf{k}) > 0$ for all k, although adequate results can often still be obtained if this condition is relaxed. Then, (55) becomes

$$f(\mathbf{k},t) = \int_0^\infty ds \int d\mathbf{k}'\, f(\mathbf{k}',t-s)\,W^*\left(\mathbf{k}',\mathbf{k}+\frac{e\mathbf{F}s}{\hbar}\right)\exp(-\Gamma s), \tag{58}$$

where

$$W^*(\mathbf{k}',\mathbf{k}) = W(\mathbf{k}',\mathbf{k}) + \Gamma(\mathbf{k})\,\delta(\mathbf{k}-\mathbf{k}'). \tag{59}$$

The iteration presented by Rees [47,48] consists of two distinct parts, or steps. The first part of each iteration is represented in the time domain by the evaluation of an intermediate function $g_n(\mathbf{k},t)$ from the nth iterate $f_n(\mathbf{k},t)$ according to

$$g_n(\mathbf{k},t) = \int d\mathbf{k}'\, f_n(\mathbf{k}',t)\,W^*(\mathbf{k}',\mathbf{k}). \tag{60}$$

The second part of the iteration generates the $(n+1)$st iterate $f_{n+1}(\mathbf{k},t)$ as the causal solution of (44), recognizing that $g_n(\mathbf{k},t)$ can be the right-hand side and $f(\mathbf{k})$ is replaced by $f_{n+1}(\mathbf{k},t)$. This causal solution is then

$$f_{n+1}(\mathbf{k},t) = \int_0^\infty ds\, g_n\left(\mathbf{k}+\frac{e\mathbf{F}s}{\hbar},\,t-s\right)\exp(-\Gamma s). \tag{61}$$

Physically, this last integral represents integration along the trajectory (the path integral) and the exponential factor is just the probability that no scattering has occurred during the traverse of the path. The appeal to the stability of the steady state gives the result that the final distribution function arises from

$$f(\mathbf{k}) = \lim_{n\to\infty} f_{n+1}(\mathbf{k},t), \tag{62}$$

and since the scattering factors and path variables shape $f(\mathbf{k})$, the initial guess for $f_1(\mathbf{k})$ is not critical. There is the further useful result that with $\Gamma(\mathbf{k})$ defined as in (57),

$$\lim_{n\to\infty} f_n(\mathbf{k}) = f(\mathbf{k},n/\Gamma). \tag{63}$$

Rees [47], in arriving at this result, shows that each iteration is equivalent to a time step of $1/\Gamma$. However, whatever value of Γ is chosen, the resulting steady state $f(\mathbf{k})$ is the same. The time-development capability, however, allows a description of the approach to equilibrium from any given initial function to be ascertained, and by varying the field between iterations, the time-dependent response can be found.

It can be readily observed that the iterative approach represents a chain of integrations, representing alternate applications of a path integral and a scattering integral. The entire chain operates on an initial trial function. This chain suggests an alternative approach to the solution of (54), the Monte Carlo evaluation of the integrals. In this latter case, a two-step iteration is also followed. The first step is a path traversal, terminated at a time t selected on a random value of the function $\exp(-\Gamma t)$. The second step involves scattering from the state resulting at the end of this traverse to a new state. The new state is governed by the type of scattering process used and this latter quantity is randomly selected from those present. That is, a typical electron is considered at $t = 0$ to be accelerated to t_1, where $t_1 = \ln(R_1)/\Gamma$ and R_1 is a random number. At t_1, the electron has been accelerated to a state $(\mathbf{k}_1,E_1)$. At this point, the relative probability of the ith scattering event is $\lambda_i(E_1)/\Gamma$, where $\lambda_1 + \lambda_2 + \cdots + \lambda_n = \Gamma$, including self-scattering. The particular scattering event is selected by a second random number R_2 as the kth process when $\lambda_1 + \cdots + \lambda_{k-1} < \Gamma R_2 < \lambda_1 + \cdots + \lambda_{k-1} + \lambda_k$. The properties of this scattering event are then used to determine the final state $(\mathbf{k}_2,E_2)$, which is used as the new $t = 0$ state and the process repeated. If the states $(\mathbf{k}_1,E_1)$ are tabulated in a $\mathbf{k}$-space grid, then their distribution becomes a representation of $f(\mathbf{k})$. An estimator for the physical variable ϕ, such as the velocity, is generated as $\Sigma\phi(\mathbf{k}_1)/(\text{number of scatterings})$, where the sum runs over all of the scatterings. The validity of such an average lies in the ergodicity of the physical process, providing that a sufficiently large number of iterations has been used. The basic Monte Carlo technique was first put forward by Kurosawa [49], but its full capabilities were not evident until the introduction of self-scattering by Boardman *et al.* [50]. The application of Monte Carlo to time-dependent phenomena is restricted to cases where $\omega \ll \Gamma$, because of the need to establish equilibrium among the states.

Comparison of Techniques

In Fig. 20, a velocity-field curve is shown for electrons in the central valley of gallium nitride as found from two methods of calculation. The curve calculated from a Monte Carlo technique by Littlejohn *et al.* [53] is contrasted with that calculated by a displaced Maxwellian by Ferry [54]. The difference between the two lies in the anisotropy introduced by the polar optical phonon scattering. For low carrier concentrations, this anisotropy, due to the small angle of scattering in this situation, leads to streaming of the carriers. The streaming effect is generated by a spike in k space directed along the electric field. To model this adequately by a Legendre expansion would require an exceedingly large number of terms to be retained. At high densities, however, carrier-carrier scattering will cause the spike to be dissipated. Which of the two curves is correct depends on the degree to which the carrier-carrier interaction is significant in the development of the distribution function.

In Fig. 21, the opposite case is shown. Here the velocity-field curve of Si at $N_0 = 10^{17}$ cm^{-3} is shown. In this case, the nonpolar optical phonons are relatively isotropic scatterers and carrier-carrier scattering is large. Here, the two methods of calculation, Monte Carlo and drifted Maxwel-

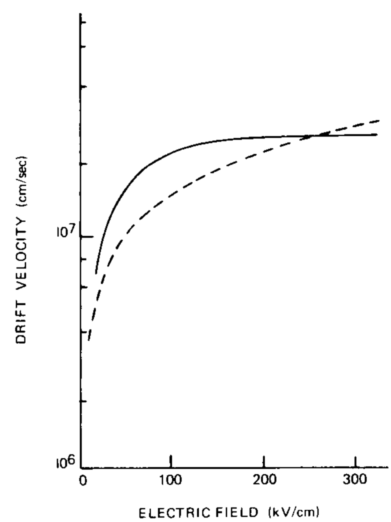


Fig. 20. The drift velocity in GaN at 300 K as calculated by a Monte Carlo technique [51] (—) and by a displaced Maxwellian [52] (---). The differences are discussed in the text.

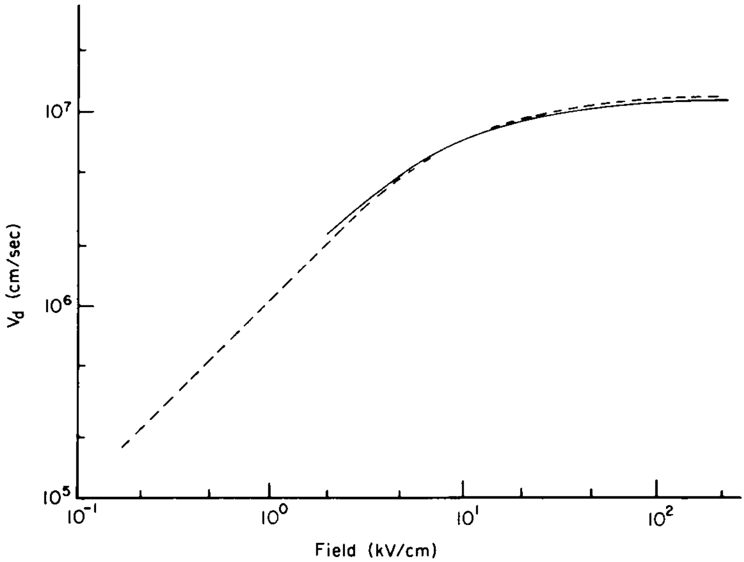


Fig. 21. The drift velocity in Si at 300 K as calculated by a Monte Carlo technique [55] (—) and by a displaced Maxwellian (---).

lian, agree quite well in this high-field region. We might also expect the two methods to agree well for GaAs, if the carrier concentration is high, such as 10^{17} cm^{-3}, and the scattering is dominated by the intervalley phonons. This is shown in Fig. 22 for the transient velocity in GaAs at 10 kV/cm. This figure also shows the importance of having $\Gamma \gg \omega_{max}$, as previously mentioned, where ω_{max} is the highest-frequency component in the transient response.

The explicit representation of the actual distribution function, such as occurs in the iterative solution and to a lesser extent in the Monte Carlo approach, has natural advantages. Effects that are nonlinear in $f(E)$, such as carrier–carrier scattering and degeneracy of $f(E)$, can readily be incorporated into the calculations. The details of the scattering are fully tied up in the terms $W^*(k,k')$ and $\Gamma(k)$, so no conceptual difficulty arises in incorporating nonphonon scattering events such as impact ionization [51], optical carrier generation [52], or even cyclotron-resonance-type transitions.

E. Velocity Transients

In traditional semiclassical approaches to solutions of the BTE, it is assumed that the response of the carrier to the applied force is simultaneous with the applied force, even though the system may undergo subsequent

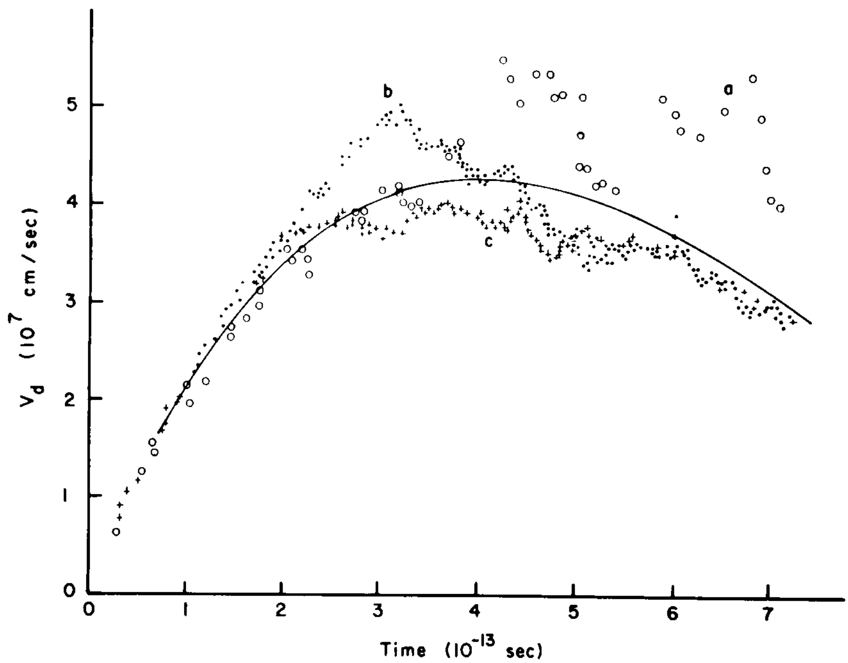


Fig. 22. Comparison of the transient dynamic response of electrons in GaAs for a field of 10 kV/cm at 300 K. The solid curve is the result of a displaced Maxwellian [59], while the data curves labeled a, b, and c are Monte Carlo results for $\Gamma = 1 \times 10^{14}$, 3×10^{14}, and 5×10^{14}, respectively [55].

relaxation. However, on the short time scale of the velocity transient, a truly causal theory introduces memory effects that lead to convolution integrals in the transport coefficients [55–62].

In this section, we shall provide the detailed derivations of this memory or retardation, showing in complete form both the momentum and energy-balance equations for a drifted Maxwellian approach. The energy-balance equation involves retardation of the collisional energy relaxation by the nonzero collision duration only. However, the collisional momentum relaxation is retarded, not only by this effect, but also through the normal memory effect. In order to allow completeness, we shall also include here the finite collision duration discussed in the next section.

The Boltzmann transport equation is valid in the weak coupling limit in which the collisions can be treated independently and perturbatively. It can readily be developed from more exact density matrix equations, in which the field contribution also appears as a differential superoperator in the momentum representation [55]. In the case where the field is large and/or the collision duration is significant, the collision integral in the BTE generalizes to a form involving the replacement of the energy-conserving δ function factors in the Golden Rule transition rates by a path integral over the time t into a collision [63–65], and the BTE becomes

$$\frac{\partial f}{\partial t} - e\mathbf{F}\cdot\nabla_{\mathbf{p}}f = \int_0^t dt'\left[\sum_{p'} S(\mathbf{p},\mathbf{p}',t')f(\mathbf{p}',t-t') - \sum_{p'} S(\mathbf{p}',\mathbf{p},t')f(\mathbf{p},t-t')\right]. \tag{64}$$

The right-hand side of (64) is defined as the collision integral, and the details of the transition terms take the form, for inelastic phonon scattering, of

$$S(\mathbf{p},\mathbf{p}';t,t') = \frac{1}{\pi\hbar}\,\mathrm{Re}\{W(\mathbf{p},\mathbf{p}')\exp\left(-\frac{t-t'}{\tau_\Gamma}\right) \times \delta_{\mathbf{p},\mathbf{p}'+\eta\mathbf{q}}\exp\left[-i\int_t^{t'}\frac{dt''}{\hbar}\beta(\mathbf{p},\mathbf{p}';t'')\right], \tag{65}$$

where

$$\beta(\mathbf{p},\mathbf{p}';t'') = E[\mathbf{p}(t'')] - E[\mathbf{p}'(t'')] + \eta\hbar\omega_{\mathbf{q}} \tag{66}$$

and the term in $\exp(-t/\tau_\Gamma)$ weights the time variation with the lifetime of the quasi-practical states and $W(\mathbf{p},\mathbf{p}')$ in $2\pi/\hbar$ times the square of the perturbing matrix element. The momenta $\mathbf{p},\mathbf{p}'$ are explicit functions of the retarded time t on the right-hand side through the relationships.

$$\mathbf{p}(t') = \mathbf{p} - \int_{t'}^{t} e\mathbf{F}(t'')\,dt'', \tag{67a}$$

$$\mathbf{p}'(t') = \mathbf{p}' - \int_{t'}^{t} e\mathbf{F}(t'')\,dt''. \tag{67b}$$

The two exponential factors in Eq. (65) are related to the joint spectral density function, which reduces to an energy-conserving δ function in the instantaneous collision, low-field limit. Here $E(p)$ is the quasi-particle renormalized electron energy, $\hbar/\tau$ the joint linewidth due to collisional broadening of the initial and final states, and η takes the values $+1$, -1 for phonon emission or absorption, respectively, in the in-scattering term. For the out-scattering term, the roles of p,p' are interchanged, although this does not upset detailed balance in the equilibrium sense.

In small semiconductor devices, where the dimensional scale is of the order of 1.0 μm or less, the carrier concentration will, in general, be relatively high.[4] Under these conditions, the anisotropic terms in the distribution function are small and a parameterized distribution function, the displaced Maxwellian, can be utilized. With this assumption, a hierarchy of moment equations can be generated from the BTE, from which the various parameters can be determined. A normal expansion of the displaced Maxwellian appears as Eq. (45), but for a form required for the memory functions, we specify $f(E)$ by the ansatz

$$f(E) \propto \int_0^t d\tau \left[\delta(t - \tau) - \mathbf{v}_d(t - \tau)\frac{d}{d\tau}\left(\mathbf{p}\frac{d}{dE}\right)\right] f_0(E,\tau), \tag{68}$$

where the two derivatives are required, respectively, to preserve the dimensionality and to bring β_e $(=1/k_B T_e)$ into the second expression in the bracket. In the limit of long times Eq. (68) reduces to (45). Here $f_0 = \exp(-\beta_e E)$.

We now take the Laplace transform of (64) as follows: assuming a single spherical isotropic conduction band,

$$\begin{aligned} &sf(\mathbf{p},s) - f(\mathbf{p},0) - e\mathbf{F}\cdot\nabla_p f(\mathbf{p},s) \\ &\quad = \frac{2}{\hbar^2}\sum_{p'}\{W(\mathbf{p},\mathbf{p}')f(\mathbf{p}',s) - f(\mathbf{p},s)W(\mathbf{p}',\mathbf{p})\}G(s) \end{aligned} \tag{69}$$

where

$$\begin{aligned} G(s) = \mathrm{Re}\int_0^\infty \exp(-st)\exp\bigg[&i(E - E' \pm \hbar\omega_0)\frac{t}{\hbar} \\ &+ i\frac{e\mathbf{F}\cdot(\mathbf{p}-\mathbf{p}')}{2m^*\hbar}t^2 - \frac{t}{\tau_\Gamma}\bigg]\,dt \end{aligned} \tag{70}$$

and the time dependence in (67) has been explicitly expanded. The second exponential in the integral is a very rapidly oscillating function and we can generate a first-order estimate by an asymptotic approximation to $G(s)$. We do this within the spirit of the method of stationary phase. The first exponential can be brought outside the integral and evaluated at τ_c, the time for which the phase in the second exponential is zero. We expect then that $s\tau_c$ will be small, the more so because we generally must take the limiting case of s small to assure that the collisions are complete. Under these assumptions, we can make the approximation

$$\exp(-s\tau_c) \simeq 1 - s\tau_c \simeq (1 + s\tau_c)^{-1}. \tag{71}$$

The remaining integral is just the field-shifted and -broadened joint spec-

[4] See, for example the arguments on size scaling in Hoeneisen and Mead [66].

tral density function [63–65], which, in the limit of small field and large τ_Γ, reduces to $\pi\,\delta(E - E' \pm \hbar\omega_0)$, the normal form. While the exact form of the integral is not easily analyzed in physical terms, a simple perturbation expansion allows us to model all but the irrelevant fast oscillatory behavior by approximations [67].

Using these forms for the function $G(s)$, we can rewrite (69) as

$$\begin{aligned} &sf(\mathbf{p},s) - f(\mathbf{p},0) - e\mathbf{F}\cdot\nabla_p f(\mathbf{p},s) \\ &\quad = \frac{2\pi}{\hbar}\sum_{p'}\left[W(\mathbf{p},\mathbf{p}')f(\mathbf{p}',s) - W(\mathbf{p}',\mathbf{p})f(\mathbf{p},s)\right]\frac{\delta_E(\mathbf{p}',\mathbf{p})}{1+s\tau_c}. \end{aligned} \tag{72}$$

We can now multiply by an arbitrary function $\phi(\mathbf{p})$ and eventually sum over all final states $\mathbf{p}$. Then a general approach, such as in Appendix A for developing moment equations, can be followed using the transformed version of Eq. (68). These balance equations are developed in the transformed domain, however, and must be inverted to obtain the time response. The results are: for the energy equation,

$$\frac{\partial}{\partial t}\left(\frac{3}{2}k_B T_e\right) = -e\mathbf{F}\cdot\mathbf{v}_d(t) - \frac{1}{\tau_c}\int_0^t \exp\left(\frac{-\tau}{\tau_c}\right)\langle\dot{\Gamma}_E(t-\tau)\rangle\,d\tau, \tag{73}$$

where we have recognized that $\langle\phi\rangle = 3k_BT_e/2$ when $\phi = E$, and $\dot{\Gamma}_E(t)$ is

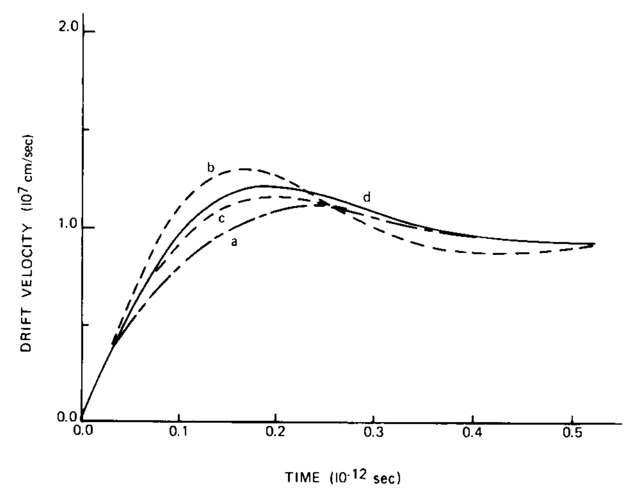


Fig. 23. Transient dynamic response of carriers in silicon at 300 K. It is assumed the carriers see a homogeneous field of 20 kV/cm that is applied at $t = 0$. Curve a shows the response neglecting the short-time effects but including the weakening of the collision in the high field. Curve b includes the effects of retardation due to the nonzero collision duration alone. Curve c includes the memory effects alone. Curve d includes all effects according to Eqs. (73) and (74).

now defined by (47); and for the momentum-balance equation,

$$m^* \frac{\partial \mathbf{v}_d}{\partial t} = -e\mathbf{F} - \frac{m^*}{\tau_c} \int_0^t \mathbf{v}_d(t - \tau) \int_0^\tau \langle \dot{\Gamma}_m(\tau - \tau') \rangle \exp\left(\frac{-\tau'}{\tau_c}\right) d\tau' \, d\tau. \quad (74)$$

If $\tau_c \to 0$ and v_d is slowly varying, then (74) reduces to the normal form (46) and Γ_m is recognized as the rate of momentum relaxation (48). The various convolutions introduce a significant temporal retardation in the rate of momentum relaxation. Although Eqs. (73) and (74) include significant corrections on a short time scale, there is reason to believe that further corrections still need to be included.

In Fig. 23, the transient response of the electrons in Si to a steady homogeneous field of 20 kV/cm is shown. Although the retardation due to the nonzero collision duration acts to speed up the response, the dominant factor is the memory effect of $\dot{\Gamma}_m$ and v_d. The collisional retardation speeds up the process, however, primarily because of the effect of slowing down changes in the effective energy via collisional relaxation.

F. Device Simulation from the Boltzmann Transport Equation

Based on the preceding discussion, it is clear that mobility models will have limited usefulness in submicron and high-frequency devices. The extent to which this is true is material-dependent and is most likely to be revealed by device simulations. Complete one- and two-dimensional simulations involving solutions of the BTE are only sparsely available. Rather, bits of the problem are treated. Here Monte Carlo, iteration, and moment methods are used and in the following we shall illustrate some results, drawing on GaAs, as the nonlocal spatial and temporal contributions are dramatically enhanced by the transferred electron effect. Our approach will emphasize the displaced Maxwellian moment equations, primarily because of intuitive advantages and for the relative ease with which space-charge contributors can be handled.

The device-moment equations [68] for a displaced Maxwellian of the form given by Eq. (A-1), are

$$(\partial N_i/\partial t) + \nabla \cdot (\mathbf{v}_i N) = (\partial N_i/\partial t)_c, \quad (75)$$

$$(\partial \mathbf{P}_i/\partial t) + \nabla \cdot (\mathbf{v}_i \mathbf{P}_i) = -eN_i\mathbf{F} - \nabla(N_i k_B T_i) + (\partial P_i/\partial t)_c, \quad (76)$$

$$(\partial W_i/\partial t) + \nabla \cdot (\mathbf{v}_i W_i) = -eN_i\mathbf{v}_i \cdot \mathbf{F} - \nabla \cdot (\mathbf{v}_i N_i k_B T_i) - \nabla \cdot \mathbf{q}_i + (\partial W_i/\partial t)_c, \quad (77)$$

where the subscripts identify a particular valley. Here

$$\mathbf{P}_i = m_i N_i \mathbf{v}_i, \quad (78)$$

$$W_i = \tfrac{3}{2}N_i k_B T_i + \tfrac{1}{2}m_i N_i v_i^2, \tag{79}$$

m_i is the effective mass of electrons in the ith valley, and T_i is the electron temperature of the ith valley, and W_i is regarded as the average total kinetic energy density.

These equations have simple physical interpretations. For Eq. (75), the increase of electron density plus the outflow of electrons equals the increase of density due to collisions. For Eq. (76), the left-hand side is the rate of change plus the outflow of momentum density of the ith valley. The right-hand side represents the forces exerted by the electric field and by the electron pressure $n_i k_B T_i$ and the rate of momentum density gained in collisions. For Eq. (77), the left-hand side contains the rate of change plus outflow of total kinetic energy density. The right-hand side represents the energy supplied by the electric field, the work performed by the electron pressure, the divergence of the heat flow $\mathbf{q}_i$, and the rate of change of total kinetic energy density due to collisions.

When the transport equations are derived by integration of the Boltzmann equation, the heat-flow vector $\mathbf{q}_i$ appears as a third moment of the distribution function. With the assumption of displaced Maxwellian distributions (or any symmetric distributions), these moments vanish. In spite of this, following Blotekjar [68], we may allow for heat conduction by assuming

$$\mathbf{q}_i = -K_i \nabla T_i, \tag{80}$$

where K_i is the heat conductivity of the electron gas in the ith valley. This term is believed to represent the most important effect of a non-Maxwellian distribution function [68].

The $(\partial/\partial t)_c$ terms in Eqs. (75)–(77) represent scattering integrals. They may be given in approximate form by Eq. (49), for example, for polar optical scattering or in exact form for the displaced Maxwellian. In the following simulations we chose the latter and use the integrals summarized by Butcher [2] (see also Blotekjar and Lunde [69]). The scattering integrals are then represented as

$$(\partial N_i/\partial t)_c = (-N_i/\tau_{N_{ij}}) + (N_j/\tau_{N_{ji}}), \tag{81}$$

$$(\partial \mathbf{P}_i/\partial t)_c = -\mathbf{P}_i/\tau_{P_i}, \tag{82}$$

$$(\partial W_i/\partial t)_c = -\tfrac{3}{2}(N_i k_B T_i/\tau_{E_{ij}}) + \tfrac{3}{2}(N_j k_B T_j/\tau_{E_{ji}}). \tag{83}$$

We illustrate a set of scattering curves for a Γ–X orientation in Fig. 24. The parameters used in the calculations are given in Table II. These results should be compared to those of Bosch and Thim [41]. We retain the Γ–X orientation because most of the early moment equations were evalu-

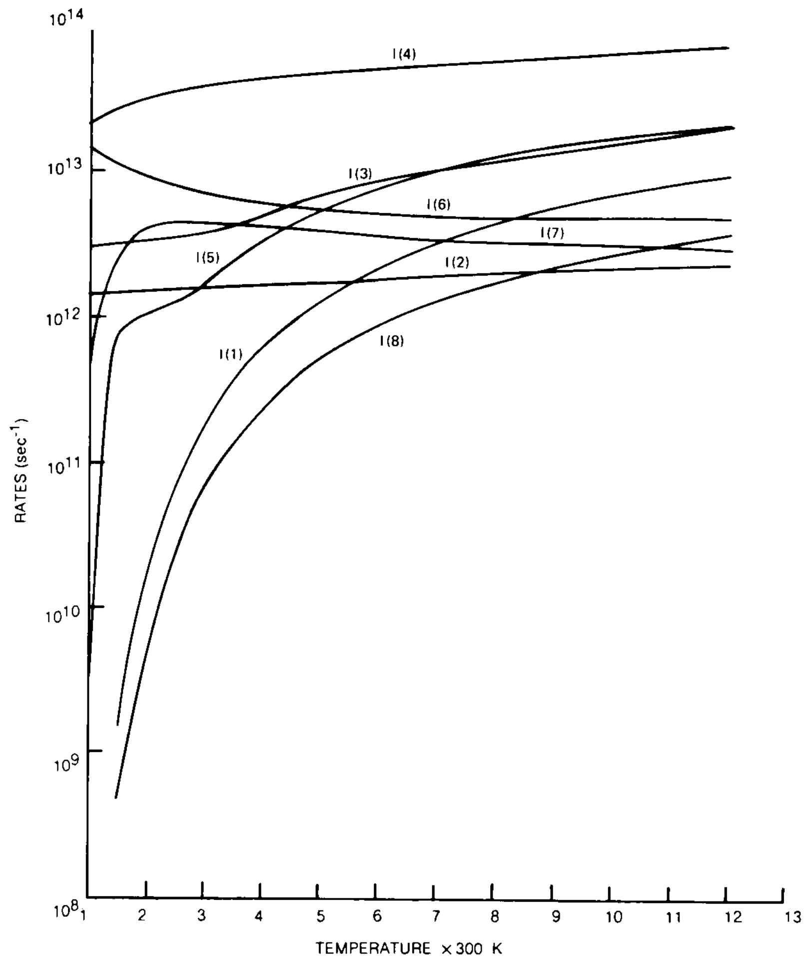


Fig. 24. Scattering rates for the parameters of Table II. Integrals are from Butcher [2]. Curves shown are I(1), CV particle; I(2), SV particle; I(3), CV momentum; I(4), SV momentum; I(5), CV energy; I(6) SV/CV energy; I(7), SV energy; I(8) CV/SV energy.

ated for this ordering. Changing to a Γ–L orientation (see, e.g., Littlejohn *et al.* [44], would offer only quantitative differences and might obscure some of the discussion.

1. Recovery of the Semiconductor Equations. Comparison to Nonlocal Equations

In analyzing transport from the nonlocal balance equation, we are sometimes interested in recovering the ordinary semiconductor equations

TABLE II
Parameters Used in Calculation

Parameters	Γ(000)	X(100)	Common
Number of equivalent valleys	1	3	
Effective mass (m_e)	0.067	0.40	
Γ–X Separation (eV)			0.36
Lattice constant (Å)			5.64
Density (gm/cm³)			5.37
Polar optical scattering			
Static dielectric constant			12.53
High-frequency dielectric C			10.82
LO phonon (eV)			0.0354
Γ–X Scattering			
Coupling constant (eV/cm)			0.621×10^9
Phonon energy (eV)			0.0300
X–X Scattering			
Coupling constant (eV/cm)		$1.0.64 \times 10^9$	
Phonon energy (eV)		0.0300	
Acoustic Scattering			
Deformation potential (eV)	7.0	7.0	
Acoustic velocity (cm/sec)			5.22×10^5

and, hence, mobility concepts. For a single parabolic conduction band, we can obtain the usual semiconductor equations through judicious neglect of select space and time derivatives. For example, in the case of continuity, neglect of avalanching allows us to set $(\partial N/\partial t)_c = 0$. We then obtain current continuity for one valley:

$$(\partial N/\partial t) + \nabla \cdot (\mathbf{v}N) = 0. \tag{84}$$

For the momentum term, we set $(\partial \mathbf{P}_i/\partial t)_c = -\mathbf{P}_i/\tau_p$, neglect the terms $\partial \mathbf{P}_i/\partial t$, and the contribution from $\nabla \cdot \mathbf{vP}$ and obtain

$$N\mathbf{v} = \mathbf{P}/m = -N\mu\mathbf{F} - (\mu/e)\,\nabla(Nk_B T), \tag{85}$$

where $\mu = e\tau_p/m$. Then using the Einstein relation $D(F) = \mu k_B T/e$, we obtain

$$N\mathbf{v} = -N\mu\mathbf{F} - \nabla ND. \tag{86}$$

We see that within the framework of the Boltzmann picture even the use of the "semiconductor" equation (86) is suspicious. Further complications arise in multivalley calculations where the mobility is taken as a weighted average

$$\langle \mu(F) \rangle = \frac{N_c\mu_c(F) + N_s\mu_s(F)}{N_c + N_s} \tag{87a}$$

and the diffusion coefficient is given, with limited validity, by

$$D(F) = \frac{k_B}{e}\left(\frac{N_c\mu_c(F)T_c + N_s\mu_s(F)T_s}{(N_c + N_s)}\right). \tag{87b}$$

Since we have a more general set of transport equations than the semiconductor equations, we are in a position to isolate differences between the two approaches, even on such a relatively direct topic such as domain propagation. Cheung and Hearn [70] examined this question by considering the mobility and scattering rates to be unique functions of electron temperature rather than of field. They are unique functions of field only in steady state. In their study, the particle current flux is given by a multivalley version of Eq. (85):

$$J_i = -N_i\mu_i(T_i)\left\{F + \frac{k_B}{eN_i}\frac{\partial}{\partial X}(N_iT_i)\right\}, \tag{88}$$

$$\frac{\partial N_i}{\partial t} = \frac{-\partial J_i}{\partial x} + \frac{N_j}{\tau_{N_{ji}}(T_j)} - \frac{N_i}{\tau_{N_{ij}}(T_i)}, \tag{89}$$

with

$$\frac{\partial}{\partial t}\left(\frac{3}{2}N_ik_BT_i\right) = -eJ_iF + \frac{3}{2}\left(\frac{-N_ik_BT_i}{\tau_{E_{ij}}} + \frac{N_jk_BT_j}{\tau_{E_{ji}}}\right) - \frac{\partial}{\partial x}\left(\frac{5}{2}J_ik_BT_i\right). \tag{90}$$

We note that Eq. (88) neglects inertial terms and hence overshoot effects. Figure 25 shows a comparison of domain size using the semiconductor equations and the BTE. We can see significant differences.

We consider some of these inertial effects more closely by solving the full set of equations, (75)–(77), for a one-dimensional 5000-Å GaAs element with a donor distribution as shown in Fig. 26. The element is part of a resistive circuit. The carrier dynamics are examined at two instants of time. (Note: Serious objections can be raised to the use of a "jellium" distribution, insofar as any combination of decreased donor density or size reduction will necessarily introduce effects due to the discrete nature of the donors. This will be ignored in the following discussion.

As the bias is turned on, there is an increase in potential across the device and a corresponding increase in current and field. The field is computed self-consistently and its slope reflects any incomplete screening of inhomogeneities, the mobile carriers. For the device in the schematic configuration of Fig. 26, as the field is rising, energy relaxation is incomplete

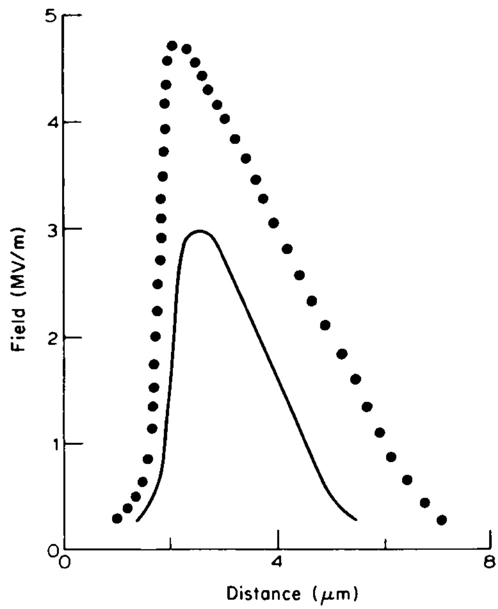


Fig. 25. Field profiles for a high-field-propagating domain. Dotted curve is obtained from the mobility equations. Solid curve is detained using Eqs. (88)–(90). From Cheung and Hearn [70], with permission.

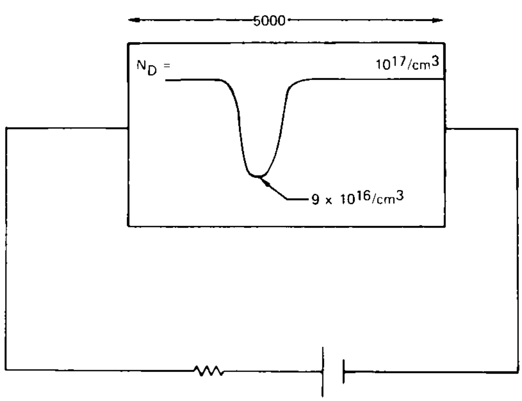


Fig. 26. Schematic representation of device and circuit configuration for submicron homogeneous field profiles. The inhomogeneous doping profile is treated as a "jellium" distribution. The beginning and end of this inhomogeneity are displayed in Fig. 28c. The circuit equation for this calculation is $\phi_B = \phi_D + IR$. We set R to the low-field resistance of the element and $\phi_B = \phi_B F_0 L$, where F_0 = 4.3 kV/cm and L = 5000 Å. Generally, ϕ_B is turned on at a finite rate. For these calculations, ϕ_B = 3.0.

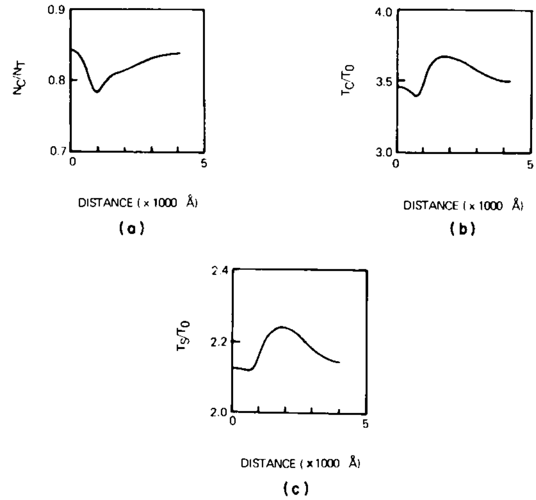


Fig. 27. (a) Fractional central valley population versus distance; (b) Central valley temperature versus distance; (c) satellite valley temperature versus distance. Computation occurs at time $t/\tau_0 = 4$, where τ_0 is the LO phonon intervalley scattering time and equals 0.32 psec [see Fig. 24, curve I(3)] and T_0 is room temperature. The gradient of N is zero at the boundaries. All nonuniformities are due to the notch.

and velocity overshoot contributions are dramatic (see Figs. 27 and 28). Here velocity is computed from the equation

$$\langle v(x,t)\rangle = \frac{P_c/m_c + P_s/m_s}{N_T(x,t)}. \tag{91}$$

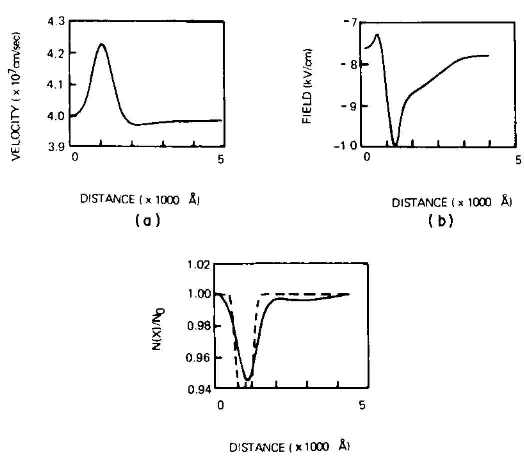


Fig. 28. (a) Mean velocity versus distance; (b) field versus distance; (c) free carrier density versus distance at time $t/\tau_0 = 4$. Dashed curve denotes background density.

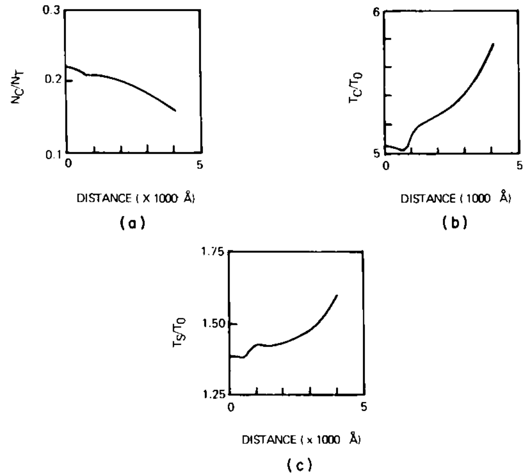


Fig. 29. As in Fig. 27, but at $t/\tau_0 = 16$.

An important point to make here is that when spatial gradients occur, the carrier velocity can overshoot its equilibrium value even though all time derivatives ($\partial/\partial t$) are zero. This was emphasized by Kroemer [71], who estimated substantial overshoot when $\partial F/\partial x \gtrsim F_{\text{th}}$/mean free path, where the subscript "th" designates the NDM threshold field. Thus spatial overshoot should be most pronounced in devices biased near the NDM threshold. The results in Figs. 29 and 30 are for bias fields substantially higher than the NDM threshold. In these figures, the fields are high enough to accommodate almost complete transfer. Here the carrier tem-

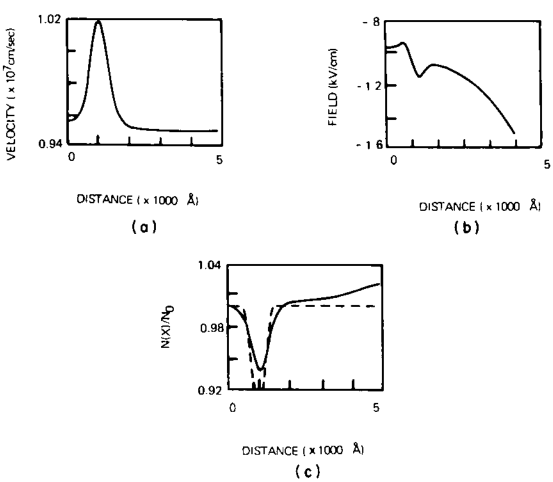


Fig. 30. As in Fig. 28, but at $t/\tau_0 = 16$.

perature is reduced, the energy relaxation rates are shorter, and there is virtually no overshoot. These results are virtually the same as we would obtain using the steady-state curves.

The nonuniform field calculations are the clear order of business in future numerical simulations but a catalog or results have yet to be produced. Until this is done, we are forced to rely heavily on uniform field analysis.

2. Uniform Field Transients

For uniform fields, the transport equations undergo considerable simplifications:

$$\frac{a_i\, dN_i}{dt} = \frac{-N_i}{\tau_{N_{ij}}} + \frac{N_j}{\tau_{N_{ji}}}, \tag{92}$$

where a_i denotes the number of equivalent ith valleys. For $\mathbf{P}_i = \mathbf{P}_i/N_i$,

$$\frac{\partial N_i \mathbf{p}_i}{\partial t} = -eN_i\mathbf{F} - \frac{N_i\mathbf{p}_i}{\tau_{\mathbf{p}_i}}, \tag{93}$$

$$\frac{\partial}{\partial t}\left(\frac{N_i p_i^2}{2m_i} + \frac{3}{2}N_i k_B T_i\right) = \frac{-eN_i\mathbf{F}\cdot\mathbf{p}_i}{m_i} \frac{-3N_i k_B T_i}{2\tau_{E_{ij}}} + \frac{3N_j k_B T_j}{2\tau_{E_{ji}}}. \tag{94}$$

It is instructive at this time to rewrite the momentum-balance equation as

$$\frac{dp_i}{dt} + p_i\left[\frac{1}{\tau_{p_i}} + \frac{d \log N_i}{dt}\right] = -eF. \tag{95}$$

In this form, the effect of intervalley transfer enters as an additional transient momentum scattering term [72].

As previously discussed, dynamic overshoot effects are consequences of differences in momentum and energy relaxation times. In multivalley semiconductors, the overshoot contributions therefore appear in the *momentum,* as well as the velocity, computations. We shall illustrate this for both the central and satellite valleys and for the situations where electrons starting with zero drift velocity are subjected to a sudden change in electric field.

For the central valley and at low values of bias, the electron temperature is approximately equal to room temperature and the ordinary time-dependent dynamic behavior occurs. At elevated bias levels, the electron temperature is substantially increased and the momentum-relaxation time, due to strong intervalley coupling, decreases with increasing temperature (Fig. 31). Thus, we see overshoot, in that the final momentum is below the peak momentum (Fig. 32a). (We point out that above moderate temperature increases, LO phonon intravalley and ionized impurity scat-

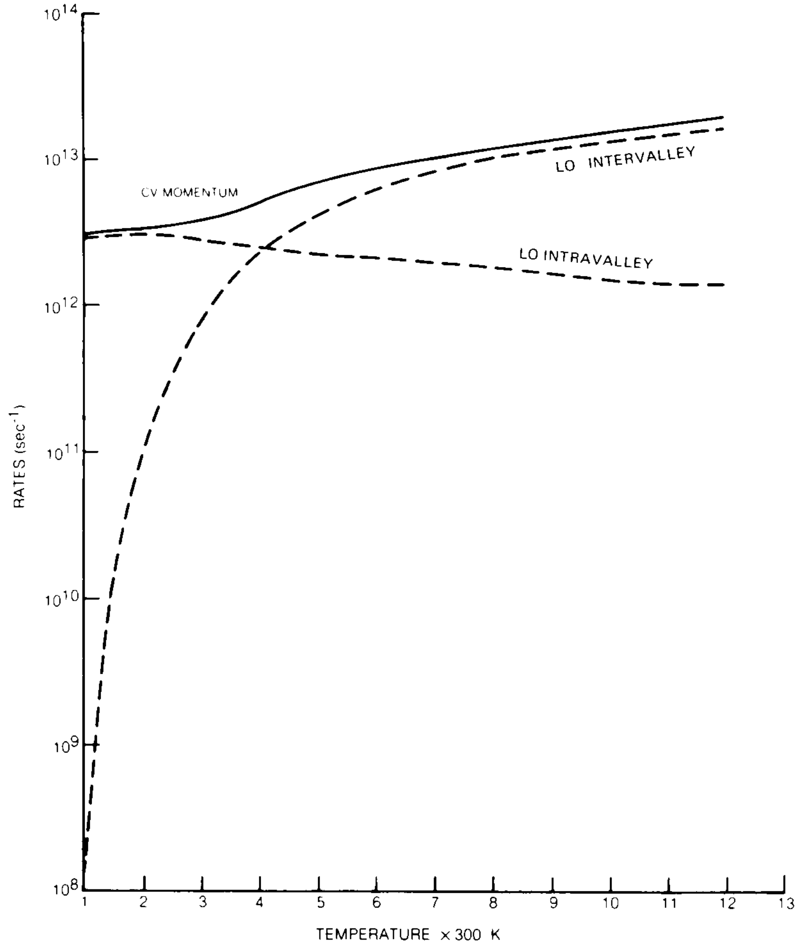


Fig. 31. Central valley momentum scattering rates versus electron temperature.

tering do not provide a momentum-relaxation time that decreases with energy. Intervalley phonons are required. Indeed, for ionized impurity scattering, the relaxation time increases with energy.[5] During this same time interval, the increasing central valley temperature (Fig. 32b) results in electron transfer, and the momentum density $N_c p_c$ shows an even greater overshoot (Fig. 32c). We now examine the contribution of the term $d(\log N)/dt$ appearing in Eq. (95). For the central valley, where at $t = 0$, $N_s \simeq 0$, and $p_c = p_s = 0$, this term is approximately zero. However,

[5] See, for example, Smith [1, Chapter 5].

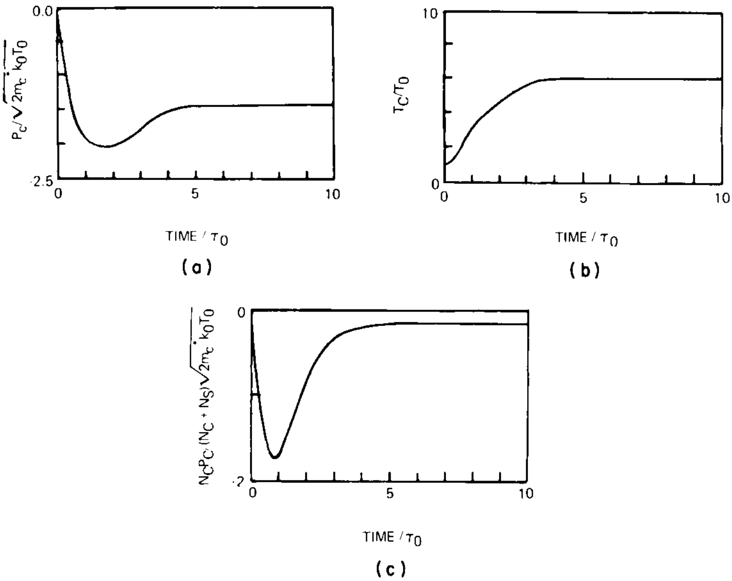


Fig. 32. Transient central valley behavior: (a) momentum versus time; (b) temperature versus time; (c) momentum density versus time. The circuit in this calculation is the same as that of Fig. 26. Here, because the field is uniform, we write the circuit equation as $F_B = F_D + R/R_0 i$, where $i = I/N_0 e v_p A$ with $v_p = \sqrt{2m_c^{-1}k_B T_0} = 3.7 \times 10^7$ cm/sec. For this calculation N_0 is $10^{17}/\mathrm{cm}^3$ and we have included geometrical capacitance. For this calculation, the normalized bias F_B is turned on suddenly to the value $F_B = 5$, which corresponds to an average field of 21.5 kV/cm ($\tau_0 = 0.32$ psec).

when we consider the transient behavior of the satellite valley, the time derivative of log N_S is important because the change in the satellite population, relative to the original number present, is quite large. In addition, we note that the satellite valley momentum-relaxation contribution is almost an order of magnitude larger than that of the central valley. Thus, in a time considerably shorter than that associated with the central valley, the satellite valley momentum reaches the value

$$p_S = -eF(t)\left[\frac{1}{\tau_{pS}} + \frac{d \log N_S}{dt}\right]^{-1},$$

where, because the satellite temperature remains close to room temperature for large changes in field [73], the scattering rates may often be taken as approximately constant. Combining both scattering contributions, we see some "overshoot" due to differential repopulation but none as dramatic as that associated with the central valley. Figure 33 illustrates this point. For part (a), 1 corresponds to the increasing momentum prior to

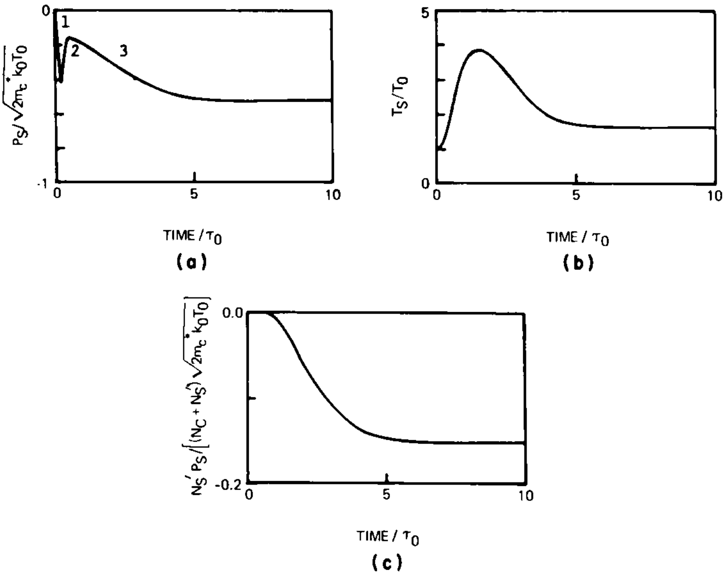


Fig. 33. As in Fig. 32, but for satellite valley behavior.

any significant electron transfer; 2 represents the scattering rate due to repopulation; 3 is the steady-state value. In Fig. 33c, we show the product of $N_s'P_s/(N_c + 3N_s')$. Here the prime indicates population of a single satellite valley ($N_s = 3N_s'$). We see that as far as the contribution to velocity overshoot is concerned, the electron transfer tends to wipe it out.

The average drift velocity versus time is shown in Fig. 34 and is computed from

$$v = \frac{(N_c p_c/m_c) + (N_s p_s/m_s)}{N_c + N_s}. \tag{96}$$

Perhaps the most remarkable aspect of this result is the very large peak velocity prior to steady state (see, e.g., Ruch [3]). This result has been one among many that has led some to suppose that narrow-channel devices will yield higher carrier velocities. To some extent, high overshoot velocities are illusory, as they are very sensitive to rise time. We shall illustrate this for a sequence of trapezoidal bias pulses, each with a varying rise time (see Fig. 35). (See also [74].)

The first set of results is for a relatively slow rise time and the dynamic curves come very close to the steady-state curves (see Fig. 36). A more significant departure from steady state occurs for the somewhat steeper rise time. In Fig. 37, we see some asymmetry in the time dependence of the central valley population and temperature and an increase in the peak

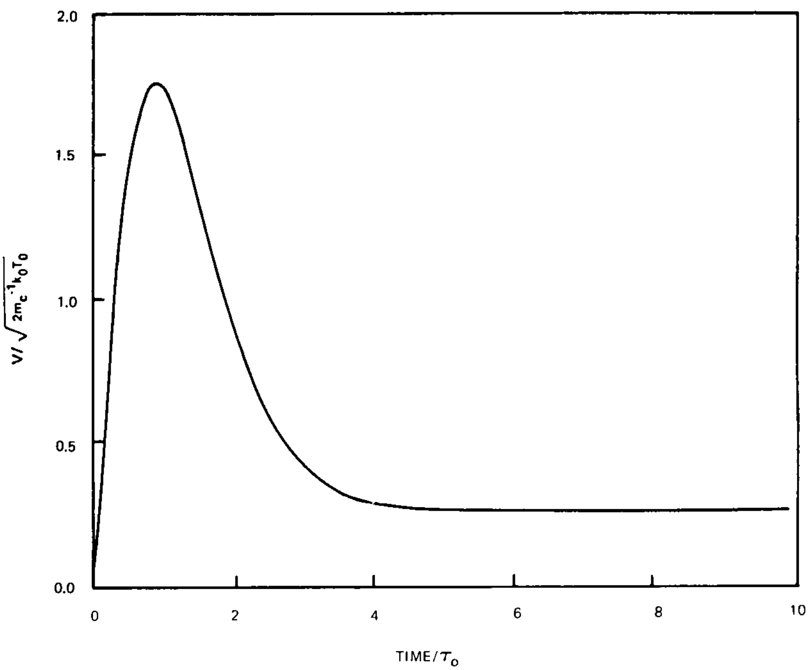


Fig. 34. Mean carrier velocity, from Figs. 32 and 33.

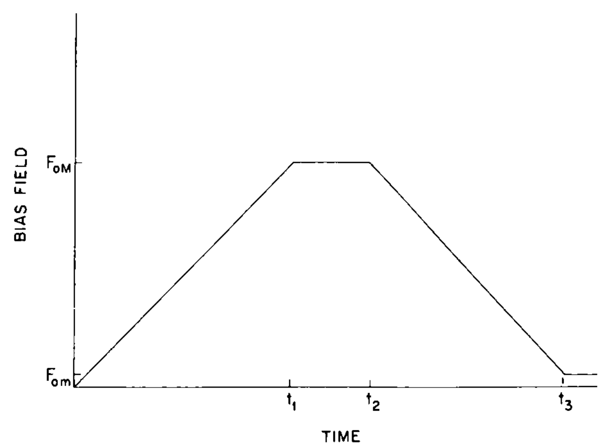


Fig. 35. Time-dependent bias field used to show rise-time dependence of transient velocity. For all of these calculations, $F_{OM} = 5$ (21.5 kV/cm) and $F_{Om} = 0.1$ (0.43 kV/cm).

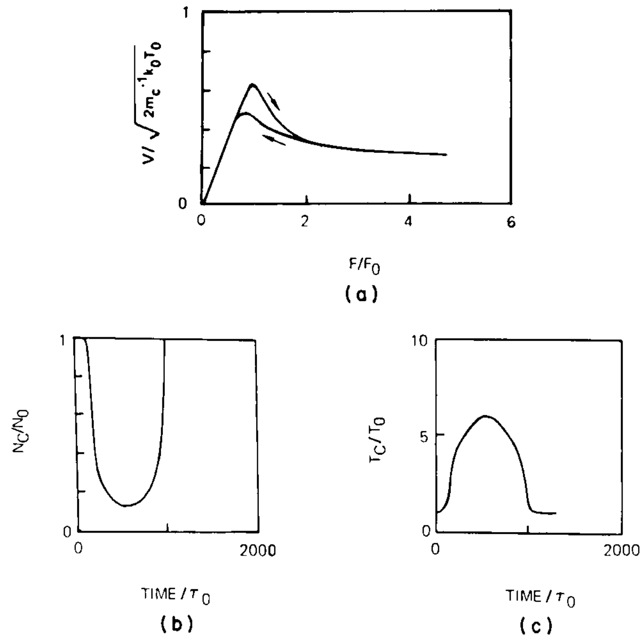


Fig. 36. For this pulse, $t_1 = 500/\tau_0$ (see Fig. 35), $t_2 = 600/\tau_0$, and $t_3 = 1200/\tau_0$. (a) Mean velocity versus field ($F_0 = 4.3$ kV/cm); (b) central valley populations versus time; (c) central valley temperature versus time.

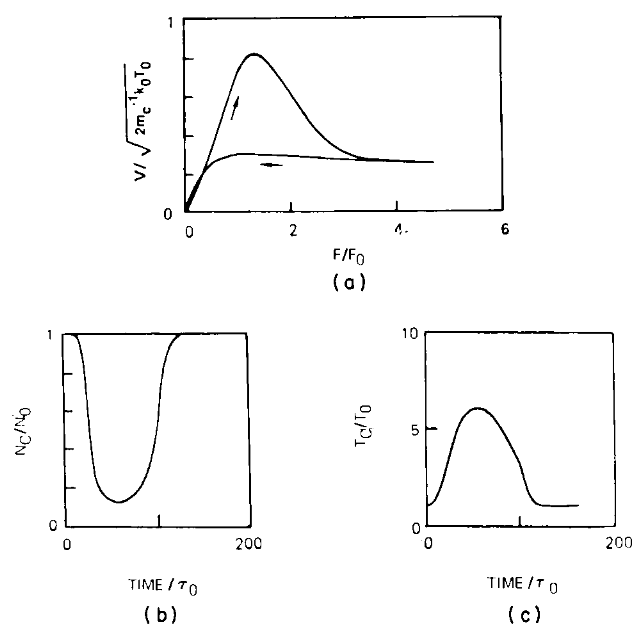


Fig. 37. As in Fig. 36, but here $t_1 = 50/\tau_0$, $t_2 = 60/\tau_0$, and $t_3 \simeq 110/\tau_0$.

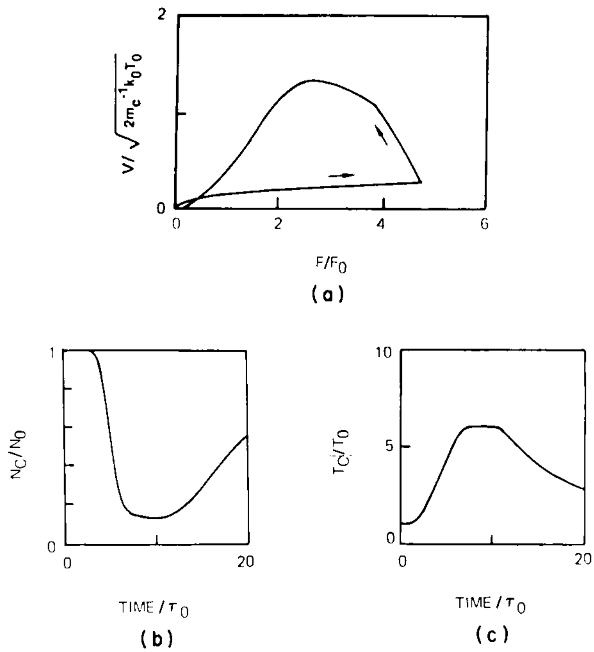


Fig. 38. As in Fig. 36, but here $t_1 = 5/\tau_0$, $t_2 = 10/\tau_0$, and $t_3 = 15/\tau_0$.

velocity. In the final sequence, we show results for a very short rise time. We see a dramatic increase in the peak velocity and clear asymmetry in the carrier dynamics (Fig. 38). Indeed, the final point of approximately zero field and velocity is not an equilibrium state. Rather, we have a dramatic example of velocity undershoot. A longer time is needed for the electron temperature to approach equilibrium. There are strong implications here for upper-frequency limits of device operation.

3. Determination of Maximum Frequency for Small-Signal, Large-Signal, and Self-Excited Oscillations

Perhaps the earliest attempt to examine the upper-frequency limit for large-signal oscillation was that of Butcher and Hearn [4]. Using a set of displaced Maxwellian electron distributions for each valley, the set of differential equations [see Eqs. (92)–(94) for the time-dependent electron temperatures, drift velocities, and valley populations was solved for a dc bias field plus rf field. The results of their study are shown in Fig. 39.

In Fig. 39, the mean drift velocity and satellite population are shown as functions of a total field consisting of a dc field of 15 kV/cm and a 60 GHz rf field with an amplitude of 13.1 kV/cm. The arrows indicate the

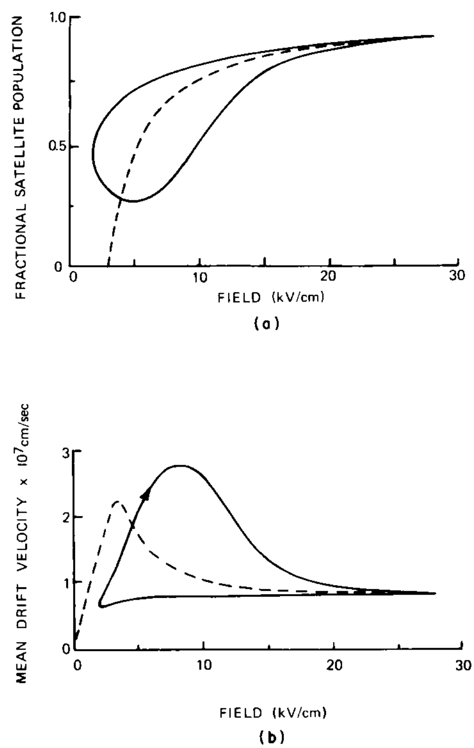


Fig. 39. Plots against field: (a) fractional satellite population; (b) mean drift velocity for a 60-GHz rf field of 13.1 kV/Vcm superimposed on a dc field of 15 kV/cm. The dashed curves give the static values. From Butcher and Hearn [4], with permission of IEE.

direction of increasing time and the dashed lines represent the static relationships. Near the maximum field of 28.1 kV/cm, the satellite population (Fig. 39a) approximates well the static value because the field is stationary and the variation of the population is nearly saturated. The curve is qualitatively similar those of Fig. 37. We see again the higher satellite population on the downswing and the absence of negative differential mobility. We point out that these curves very likely constitute the earliest attempt at including overshoot contributions. For dc bias levels of around 10 to 20 kV/cm and ac levels of 13.1 to 18.6 kV/cm, they obtained an upper-frequency limit of 100 GHz.

From the point of view of a device physicist, a driven oscillator probably lies somewhere between a small-signal and a large-signal self-excited oscillator. The self-excited oscillator is perhaps the most interesting of the three because it highlights the tenacious balance between electron

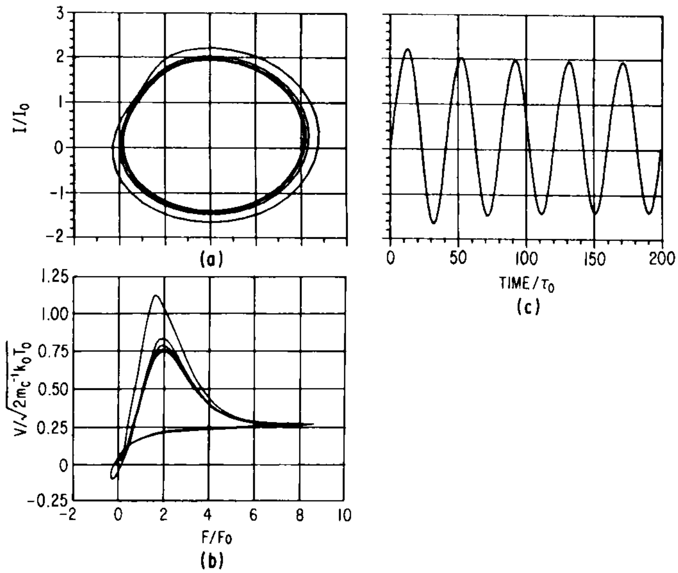


Fig. 40. Self-excited oscillation for the circuit displayed in Fig. 41, where $N_0 = 10^{15}/\text{cm}^3$. Oscillation frequency is 78 GHz.

transfer and sustained oscillations. It is extremely sensitive to contact, space-charge, and circuit conditions [8c]. We examine the upper-frequency limit for the device in a circuit with reactive elements. Figure 40 shows the oscillation.

The circuit differential for this oscillation is given by Eq. (11). For nonlinear devices, this equation offers difficulties in interpretation. For interpretative ends, we replace the nonlinear element by a nonlinear resistor with the current–voltage $I_0(V)$ relation. Then, Eq. (12) is conceptually replaced by [8c]

$$\begin{aligned} \phi_B &= \omega_0^2 \frac{d^2\phi}{dt^2} + \frac{Z_0}{\omega_0}\frac{dI_0}{d\phi}\frac{d\phi}{dt} + RC\frac{d\phi}{dt} + \phi + RI_0, \\ \omega_0 &= (LC)^{-1/2}, \qquad Z_0 = (L/C_T)^{1/2}, \end{aligned} \tag{97}$$

where $dI_0/d\phi$ represents resistance. When $dI_0/d\phi > 0$, Eq. (97) yields damped oscillations. When $dI_0/d\phi < 0$, the oscillations grow in amplitude. As $dI_0/d\phi$ changes sign during each cycle, the correct set of circuit, bias, and device parameters can yield sustained circuit-controlled oscillations [8c].

Several aspects of a self-excited oscillation are displayed in Fig. 40. The current through the load resistor is displayed in part (b); the dynamic voltage and I vs ϕ, obtained by eliminating time between current and device is shown in part (b). We also display the mean velocity (dynamic conduction current). The details of the oscillation will be discussed next.

As the field across the device increases and exceeds a threshold, value transfer begins to occur. However, because the field changes more rapidly than the electron temperature, more carriers are retained in the central valley, and with higher momenta than steady state would dictate. This effect is responsible for the higher peak conduction current. But if the increasing electric field sustains high fields for a sufficient duration, enough carriers will transfer to result in negative differential mobility (NDM), which must be of sufficient magnitude for sustained self-excited oscillations.

On the downswing, the field again changes more rapidly than the electron temperature and more carriers are retained in the satellite valley; i.e., we achieve transient undershoot. Now, although NDM is not necessary on the downswing [8c], enough carriers must be returned to the high-mobility valley for transfer on the upswing and NDM to occur. This means that the field must change slowly enough to allow a dump of carriers from the satellite to the central valley. If the field changes too rapidly, too many carriers are retained in the satellite valley and NDM is too weak to sustain steady-state oscillations. In Fig. 41, we plot the maximum frequency of self-excited oscillations as a function of dc bias.

The large variations in field and carrier temperature for both self-excited and large-signal-driven oscillations should result in quantitatively different upper-frequency limits than that obtained for small-signal oscil-

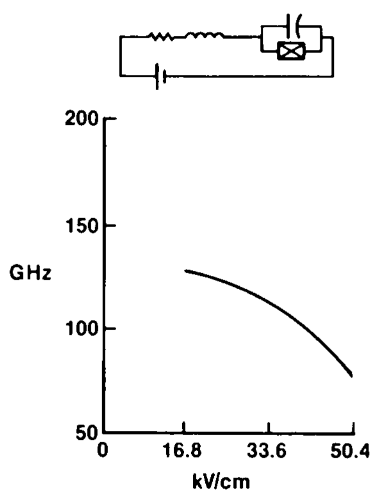


Fig. 41. Maximum frequency for self-excited oscillations.

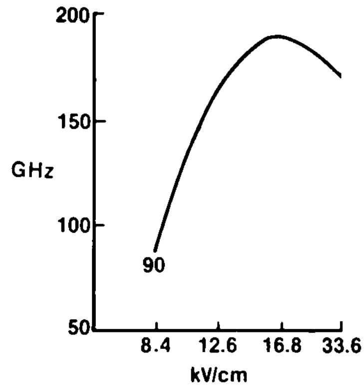


Fig. 42. Maximum frequency for small-signal oscillations

lations. We have examined the latter by perturbing a nonlinear element in steady state. The perturbation is a square wave voltage pulse and the resulting response is then Fourier-analyzed. The results are shown in Fig. 42, where we plot the maximum frequency of small-signal negative conductance as a function of dc bias. The most significant feature here is that f_{max} for small-signal operation is significantly greater than that for large-signal operation. We note that when similar calculations with the nonlinear element driven by an ac source of controlled amplitude and frequency are performed the results bridge the small- and large-signal oscillation calculations.

The explanation for differences in the large- and small-signal results lies in the energy scattering rates (see Fig. 24). For large-signal oscillations, the transient temperature in both the central and satellite valleys oscillates over a larger range than that for the small-signal oscillations and sample lower scattering rates. This gives rise to the reduced maximum frequency for the large-signal oscillation.

Detailed analysis of the frequency limitation of the transferred electron effect in GaAs using the SETEM approach was recently discussed by Rolland *et al.* [75]. They were able to show that reasonable efficiencies could be obtained with frequencies up to 150 GHz and offered the uniform field mode as an alternative means for circumventing the drastic size restrictions usually associated with multimeter wave devices.

4. Length Dependence of Negative Differential Mobility

We have been discussing the upper-frequency limit of transferred electron devices from the circuit viewpoint and the transfer and return of electrons between the central and subsidiary valleys. In this analysis we have

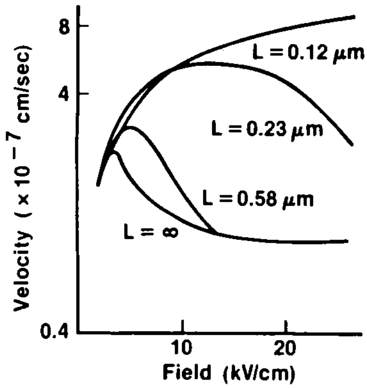


Fig. 43. Length dependence of negative differential mobility.

not explicitly considered the following problem: When an electron enters the active region of a device, it accelerates in the presence of an applied electric field. If the initial drift velocity of the carrier is low, is the transit length sufficient to cause electron transfer and negative differential mobility? The answer lies in earlier calculations. If the carrier experiences a sudden change in field, the mean initial transient ($t \ll$ LO phonon scattering time) will increase approximately linearly with time, followed by a region where v will approach $-e\tau F/m^*$ for a single valley. If the transit time is short enough to prevent significant transfer, NDM will be weak, if at all. Figure 43 summarizes where, for uniform fields, device length is a derived quantity [74,76]. Here

$$L = \int_0 v(t)\, dt. \tag{98}$$

Now the velocity versus field curves and the velocity versus time curves of the type shown here provide an indication of why there is interest in submicron devices. The possibility exists for achieving very high velocities over very short distances. But, again, a word of caution. The calculations of Fig. 42 are for carriers subjected to sudden changes in field. As we have seen, finite rise time dramatically reduces this peak, so the results will be somewhat less important.

G. Ballistic Transport

On the basis of the discussion associated with Fig. 42, there exists an interesting conceptual possibility of a "mean" ballistic transport, where a "typical" electron may "apparently" travel without scattering. (In the context of semiconductor carrier dynamics, this concept of an electron lucky enough to escape collisions was first suggested by Shockley [77] in studying impact ionization phenomena.)

Although the physical concepts associated with ballistic transport have not been completely discussed, one point should be emphasized. The equations used tend to treat the mean particle as if it were an isolated electron uncorrelated with any other electrons. In actual fact, the correlation among electrons is very important and governs the transient response. As a consequence, ballistic transport based on mean electrons may not be appropriate to semiconductors. With this caveat, we nevertheless discuss some aspect of this transport. We neglect statistics and assume that the number of electrons involved has a monoenergetic distribution and that electron–electron scattering has not contributed to the broadening of the distribution (see Hess [78]). Under these conditions, the following three equations are necessary: the conduction current equation,

$$J = Nev, \tag{99}$$

the Poisson equation, and the equation for electron velocity (conservation of energy),

$$\tfrac{1}{2}m^* v^2(x) - e\phi(x) = \tfrac{1}{2}m^* v^2(0) - e\phi(0), \tag{100}$$

where $v(0)$ corresponds to some initial velocity.

It is clear that these equations yield a set of current–voltage characteristics for any spatially dependent doping level. The simplest case to consider is that for which the charge density injected at the cathode is considerably greater than the background doping density, which is assumed to be uniform. In this case N_0 is ignored, and solutions are borrowed from the analyses of electrical phenomena in gases [79]. The point we shall emphasize here is the role of the cathode on the resulting current–voltage characteristics.

First we take the special case in which the cathode is an inexhaustible source of electrons, i.e., $N(x = 0) \to \infty$. The $v_0 = 0$, and we obtain Child's law (see, e.g., Shur *et al.* [80]),

$$J = (4\epsilon/9)(2e/m)^{1/2}[\phi(x)^{3/2}/x^2]. \tag{101a}$$

(Note: Assuming energy loss by collisions, Eq. (101a) no longer applies. In the simplest case where $v = \mu F$, where μ is constant, the current–voltage relation is the same as that for unipolar flow in dense gases[79][6]:

$$J = (9\epsilon/8)\mu(\phi(x)^2/x^3), \tag{101b}$$

[6] Ivey [79] reviews space-charge-limited corrents.

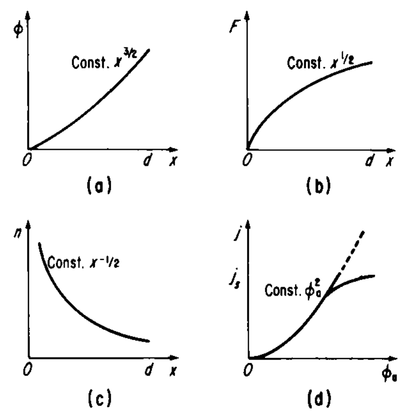


Fig. 44. Characteristics of space-charge-limited current with $v = \mu F$ and zero cathode velocity: (a) voltage versus distance; (b) field versus distance; (c) carrier density versus distance; (d) current versus voltage. Dashed line signifies possible thermionic-limited cathode. See also Papandor [81].

where x is the distance from the source. Figure 44 [81] illustrates the potential and carrier distribution for this case.) In general, we may expect the electrons to be emitted with a finite velocity. In this case, $n(x = 0)$ is finite. Among the consequences of this are that the field at the cathode is no longer zero. Then, if a potential minima ϕ_m occurs at a small distance from the origin, as shown in Fig. 45, for example, beyond this minima the system behaves as if there were a potential difference $\phi_{(L)} + \phi_{\mathrm{m}}$ (see Fig. 45) between the anode and a virtual cathode at X_{m}. Thus [79],

$$J \simeq \frac{\epsilon}{9}\left(\frac{2e}{m^*}\right)^{1/2}\left(\frac{\phi(L) + \phi_{\mathrm{m}}}{d - X_{\mathrm{m}}}\right)^{3/2}. \tag{102}$$

Since we may expect the cathode velocity and, hence, X_{m} to depend on the current, the $J(\phi)$ characteristics may be expected to depart significantly from a "$\frac{3}{2}$" relation.

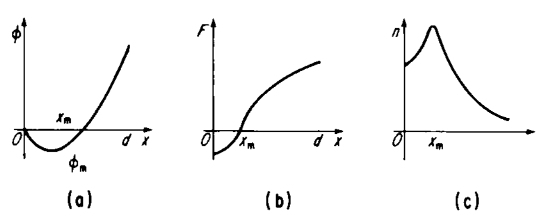


Fig. 45. Space-charge-limited current flow, showing the influence of finite cathode velocity: (a) voltage versus distance; (b) field versus distance; (c) carrier density versus distance. See also Papondar [81].

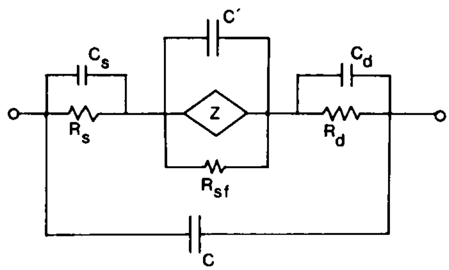


Fig. 46. Suggested simple model for a semiconductor device showing the role played by the contact, space-charge, and transport properties. The parameters are discussed in the text. From Ferry *et al.* [82], with permission.

The situation is even more complicated than just stated. The cathode may be thermionically (see Fig. 44) limited, there may be two-dimensional variation in the cathode structure, the distribution of carriers at the cathode may vary, etc. What this means is that for collisionless transport the $I-V$ characteristics are dominated by boundary conditions and not by ballistics. From a systems viewpoint, the ballistic device has an equivalent circuit like Fig. 46 [82]. Here R_d and C_d define the drain characteristics, R_s and C_s define the source characteristics, C is the total device and package capacitance, C' is a space-charge capacitance, R_{sf} is the resistance due to surface scattering, and Z is the carrier dynamic impedance, which is mostly resistive in long channels but will have an inductive contribution in the ballistic regime due to carrier retardation (see also [83]). Here, however, ac contact impedance may camouflage this contribution. The impedance Z is further complicated in ultrashort channels where transport is likely to be dominated by size-quantization effects [7,84]. It is unlikely that Z can be observed in dc measurements, but the inductive nature should be observable in microwave experiments [83], or time-of-flight [85] techniques can be used to observe the change of $v(F)$ characteristics to the characteristic $\phi^{1/2}$.

Since the possibility of ballistic transport offers interesting device applications, care must be exercised to avoid oversimplifying the criteria for its existence. As indicated, the simplest condition for ballistic transport is that the channel length d be smaller than the bulk mean free path l_b. However, a number of effects conspire to make the problem more complicated. First, if the overall device dimensions are such that $d = l_b$, carriers injected at a nonzero angle to the channel may traverse trajectories whose path length exceeds l_b and will, therefore, scatter. This will be particularly important for wide devices. Second, if the overall device dimensions are less than l_b, some carriers injected at nonzero angles to the channel direction may intersect the device boundaries and undergo surface elastic or inelastic scattering. Depending on the critical dimension l_b, surface scattering may extend through a significant portion of the device volume. Consequently, l_b should be replaced by l_s, which depends on the finite geometry, the electric field strength, and the surface/environ scattering. Third, if intradevice scattering is negligible, the current flow may be space-charge-limited because of Coulomb scattering. As we have seen, if space-charge exists, it is almost impossible to unfold the ballistic efforts from the space-charge-limited current. A fourth problem relates to devices configured to dimensions comparable to the de Broglie wavelength; carrier reflection effects may then become important and carry an enhanced probability of scattering. In this case we need to examine quantum transport theory (QTT) and apply it to submicron devices. Generally, QTT involves solving the equation of motion of the density matrix. We shall introduce this briefly in Appendix B (p. 290).

IV. QUANTUM TRANSPORT THEORY

A. Introduction

Boltzmann transport theory (BTT) is an ideal theory. It has the twin virtues of conceptual and mathematical simplicity. Quantum transport theory (QTT) [60,86–91] enjoys no such status; it is neither conceptually nor mathematically simple and often reduces to the Boltzmann picture [92,93] only after considerable labor. Even if there were no outward experimental manifestation of quantum transport phenomena, QTT would still be necessary to explain how the phenomenological BTT picture and its related concepts actually arise from the underlying framework of reversible quantum statistical mechanics. Thus QTT is necessary as an explanatory and supportive theory for the Boltzmann picture (where such a picture is applicable), for setting confidence limits for the application of BTT, and for developing the novel concepts and transport kinetics necessary for describing manifest quantum transport phenomena (those effects that depend explicitly on the quantum-mechanical nature of the electron as well as those processes for which the local Boltzmann description fails).

Boltzmann transport theory models the conduction electrons as an approximately independent-particle dilute gas in which the electronic states are nearly stationary and free-electron-like with a well-defined momentum $\mathbf{k}$ [92]. Nonstationarity arises from the assumption that perfect crystal periodicity is violated by imperfections, impurities, and phonons. These sources of crystal imperfection are assumed to cause weak, infrequent scattering of the electrons among the states $\{\mathbf{k}\}$. The applied electric field serves to accelerate carriers through the momentum states without distorting the states or interfering with the scattering process. It then seems meaningful to describe the carriers by a classical distribution function $f(\mathbf{r},\mathbf{k},t)$ over a vaguely defined phase space $\{\mathbf{r},\mathbf{k}]$ in which $\mathbf{r}$, the position coordinate for the carrier, is not too well defined in lieu of the uncertainty principle [94].

Usually, one deals with macroscopic systems so that the intuitive concepts of τ_d, the transit time through the channel length; τ, the mean free time between collisions; and τ_c, the atomic duration of a collision, are assumed to satisfy the inequality

$$\tau_c \ll \tau \ll \tau_d. \tag{103}$$

Transport processes are thus conventionally viewed on a coarse-grained time scale $\tau \gg \tau_c$ so that many independent collisions are assumed to occur in the passage of a carrier through a channel length. Moreover, each collision event is treated as an irreversible process that occurs locally in space, locally in time (instantaneously), and independently of any driving fields and other scattering processes. Given these assumptions, the time evolution of the distribution function f is governed by

$$(\partial f/\partial t) + (\partial f/\partial t)_{\mathrm{diff}} + (\partial f/\partial t)_{\mathrm{fields}} = -(\partial f/\partial t)_{\mathrm{coll}}. \tag{104}$$

The left-hand side of Eq. (104) is time-reversible, but the equation overall is irreversible due to the gain–loss structure of the collision integral on the right-hand side.

From the preceding discussion, it is clear that the central concept of classical transport physics is the assumption that a single carrier-distribution function exists that may be used to compute statistical expectation values for macroscopic current flow. In the quantum formulation of transport physics, the concept of a distribution function that depends on position and momentum of the particle is not possible inasmuch as the Heisenberg uncertainty principle precludes simultaneous specification of position and momentum. From the quantum viewpoint, it is therefore necessary to conceptually view phase space as coarse-grained if a distribution is to be regarded as a simultaneous function of carrier momentum and position. The conceptual difficulties introduced by the noncommutativity of the carrier position and momentum operators are not too critical for large devices (channel length greater than 1 μm), for, in this case, it is generally assumed that spatial variations in the distribution function occur over distances that are large compared to the de Broglie wavelength of the carrier.

The temporal development of the distribution function is also an important facet of the transport picture. When considering carrier transport in large devices, the temporal scale of the distribution function is very long compared to the collision duration but very short compared to the total transit time of the carriers, in which case the response of the distribution function to fields and collisions is instantaneous. For medium-sized devices (channel length of about 2500 Å), the temporal scale of the distribution function is of the order of the collision time but short compared to the transit time of the carriers, so that retardation due to the finite duration of collisions becomes possible. For very small devices (channel lengths less than 250 Å), the collision time is of the same order of magnitude as the transit time, so that the applicability of a classical distribution function is questionable.

In order to continue the distribution-function approach, it must be possible to define such a distribution function over suitable momentum, spatial, and temporal variables. The concept of the carrier-distribution function is expected to retain a useful role in medium-sized device transport, but with major modification in the Boltzmann transport equation. However, for "very small device" transport, where the analog of a distribution function certainly exists, an alternative approach to BTT, which emphasizes the role of device environment, size quantization, and fluctuations seems to be necessary. There has been some success in establishing the existence of distribution functions by employing the concepts of the statistical density matrix and the Wigner- density matrix (see Appendix B). In the remainder of this section, the density matrix formalism and its relation to quantum transport will be briefly discussed; an illustrative example of the usefulness of quantum transport will be given for an array of very small devices.

B. Quantum Transport Formulation

Features that are neglected in BTT provide the warning signs for the failure of the semiclassical approach. These features will now be cited and discussed.

(1) *Nonlocality of scattering processes.* Each collision event is actually extended in space and time. If the spatial and temporal variations (described by wavevector $\mathbf{q}$ and frequency ω) of the applied driving forces approach the microscopic scale, then collisions will be only partially completed. Normally, BTT assumes $\omega\tau < 1$ and $qL < 1$, where L is the mean free path, i.e., many collisions are completed in one cycle of the applied forces. If, however, $\omega\tau_c \gtrsim 1$ and $qv\tau_c \gtrsim 1$ (where τ_c is estimated by $\hbar/E$, E is a characteristic carrier energy, and $v\tau_c$ is the de Broglie wave-

length), appreciable quantum effects can arise and the irreversible character of completed collisions will be lost. Interband effects may also occur if $\hbar\omega, \hbar^2 q^2/m^* \gtrsim E_b$, where E_b is the vertical energy separation to the band controlling m^*. Very high frequencies correspond to a quantal behavior more reminiscent of optical response [95–97]. The elementary treatment of collisions must also be reconsidered if the mean free time becomes comparable to τ_c; multiple scattering involving at least two scatterers is then possible.

(2) *Strong driving forces.* Once the extended nature of a collision is recognized, it becomes obvious that applied fields can transfer energy and momentum to the carrier during the collision; an interference, or intracollisional, field effect results [63,65,98]. The effect will be very large if $eFv\tau_c \simeq E$. The reverse effect may also occur; scattering can forestall the instantaneous accelerative effect of the driving field (this effect also occurs for low fields). In general, the driving and scattering terms in BTT cannot be independent.

(3) *Strong scattering.* Strong scattering magnifies the previous problems and weakens the assumption that the electronic states are of long lifetime and are free-electron-like. Polaron and cooperative effects are typical consequences.

(4) *Dense systems.* Many-body effects and a single-carrier description fail.

(5) *Small systems.* Size quantization or surface-limited transport effects become important [99]. Ultimately, the condition implicit in Eq. (103) breaks down.

(6) *Nonclassical influence of driving fields.* Sufficiently strong electric and magnetic fields lead to Stark or Landau quantization of the electronic states [99].

These features and the concepts discussed therein are intuitive but can be given precise meaning within QTT, to which we shall now turn.

Quantum transport theory is generally based on the Liouville–von Neumann equation for the statistical density matrix $\rho(t)$, which is given as

$$i\, \partial\rho(t)/\partial t = \hat{H}_F \rho(t), \tag{105}$$

$$\hat{H}_F \rho \equiv [H_F, \rho] \equiv H_F\rho - \rho H_F, \tag{106}$$

where $H_F = H + \mathcal{F}$, and the Hamiltonian H describes the full system in the absence of the coupling $\mathcal{F}$ to the externally applied driving forces. The usual boundary condition is that $\rho = \rho_0(H)$ for $t < 0$, where ρ_0 is a thermal equilibrium solution (e.g., the grand canonical density matrix). The driving perturbation is initiated at $t = 0$. This starting point needs modification for the description of small systems embedded in an interactive environment [59]; this aspect of QTT will be discussed later in this section. (Note: The units are chosen so that $\hbar = 1$ throughout.)

All observable properties, e.g., the current or charge densities, labeled generically by J_i may be evaluated as quantum-statistical expectation values determined by $\rho(t)$ as

$$\begin{aligned} \langle J_i(t)\rangle = \mathrm{Tr}[J_i\rho(t)] &\equiv \sum_{\lambda,\lambda'} \langle\lambda|J_i|\lambda'\rangle\langle\lambda'|\rho|\lambda\rangle \\ &= \sum_{\lambda,\lambda'} J_i^{\lambda\lambda'}\rho_{\lambda'\lambda}. \end{aligned} \tag{107}$$

Here $\{|\lambda\rangle\}$ is any complete set of states. Usually, the Hamiltonian H_F is partitioned into "free" carrier, "free" scatterer, carrier–scatterer interaction, and driving force components as

$$H_F = H_e + H_s + V + \mathcal{F} \equiv H_0 + V + \mathcal{F}. \tag{108}$$

This partitioning is not unique. The component H_e might describe small polaron states by incorporating part of the electron–phonon interaction, or H_e might include the coupling to a magnetic field and describe Landau states. The basis states $|\lambda\rangle$ are then chosen to diagonalize $H_0 = H_e + H_s$, usually via $|\lambda\rangle \equiv |e\rangle|S\rangle$, where $\{|e\rangle\}$, $\{|s\rangle\}$ diagonalize H_e and H_s, respectively. The choice of representation $\{|\lambda\rangle\}$ decides the character and interpretation of the subsequent transport theory.

If the current–density operator $\mathbf{J}$ depends only on electronic variables and commutes with H_e (which is true for extended-state, homogeneous transport in zero magnetic fields), the observable response is

$$\langle \mathbf{J}(t)\rangle = \sum \mathbf{J}^{ec} f(e), \tag{109}$$

where

$$f(e) = \langle e| \sum_s \langle s|\rho|s\rangle |e\rangle \equiv Tr_s\langle e|\rho|e\rangle \equiv \langle\langle e|\rho|e\rangle\rangle_s \tag{110}$$

defines a real, time-dependent, generalized, electron-distribution function over the free-carrier states $\{|e\rangle\}$. A transport equation for $f(e)$ may then be constructable from the Liouville equation. It may turn out to have Boltzmann-like form, although the quantum nature of the states $|e\rangle$ will be reflected in the detailed forms for the collision rates. However, for inhomogeneous transport and for transport in quantizing magnetic fields, for example, $\mathbf{J}$ is not necessarily diagonal and we must consider the off-diagonal matrix elements of the electron density matrix $f = \langle\rho\rangle_s$. Various methods exist [99] for expressing $f(e)$ in terms of $f(e,e')$, but the subsequent transport theory will not have a Boltzmann-like form. In general, a closed equation of motion for $f(t)$ can only be obtained for the special case of independent carrier transport in a stationary scattering system (i.e., where the scatterers remain in thermal equilibrium at all times). This situation has been extensively studied for homogeneous, nonlinear transport.

Wigner [100] has shown that QTT can get quite close to the classical concept of a phase-space distribution function (see Appendix B). The Wigner one-electron distribution function, for example, is defined for free-carrier states by

$$\begin{aligned} f_\sigma(\mathbf{k},\mathbf{r},t) &\equiv \int d^3\mathbf{y}\, \exp(-i\mathbf{k}\cdot\mathbf{y})\, \mathrm{Tr}\{\rho(t)\psi_\sigma^\dagger(\mathbf{r}-\tfrac{1}{2}\mathbf{y})\psi_\sigma(\mathbf{r}+\tfrac{1}{2}\mathbf{y})\} \\ &\equiv \int d^3\mathbf{y}\, \exp(-i\mathbf{k}\cdot\mathbf{y}) f_\sigma(\mathbf{r},\mathbf{y},t), \end{aligned} \tag{111}$$

where $\psi_\sigma^\dagger(\mathbf{r})$ and $\psi_\sigma(\mathbf{r})$ are the second quantized creation and annihilation operators, respectively, for a carrier of spin σ at location $\mathbf{r}$. Here ρ is second quantized and the trace is a many-body trace. Similarly, a phonon Wigner distribution may be defined by

$$N_\alpha(\mathbf{k},\mathbf{r},t) = \sum_{\mathbf{K}} \exp i\mathbf{K}\cdot\mathbf{r}\langle b_\alpha^\dagger(\mathbf{k}-\tfrac{1}{2}\mathbf{K})b_\alpha(\mathbf{k}+\tfrac{1}{2}\mathbf{K})\rangle \tag{112}$$

where $\langle\cdot\cdot\cdot\rangle \equiv \mathrm{Tr}(.\ .\ .\ .)$ and $b_\alpha^\dagger$ and b_α are creation or annihilation operators for a phonon of type α and momentum $\mathbf{k}$. The Wigner construction utilizes a partial, generalized, Fourier transform over the full (off-diagonal) density matrix and is easily generalized to other basis states (e.g., Bloch states or Landau states). One may easily prove that f_σ and N_α are real-valued, generalized distributions that give the correct statistical expectation values, e.g., the carrier density and current density in an inhomogeneous system are given by

$$\langle n(\mathbf{r},t)\rangle = \sum_\sigma \int \frac{d^3\mathbf{k}}{(2\pi)^3} f_\sigma(\mathbf{k},\mathbf{r},t), \tag{113}$$

$$\langle \mathbf{J}(\mathbf{r},t)\rangle = \sum_\sigma \int \frac{d^3\mathbf{k}}{(2\pi)^3} e\mathbf{v}(\mathbf{k}) f_\sigma(\mathbf{k},\mathbf{r},t), \tag{114}$$

where $\mathbf{v}(\mathbf{k}) \equiv \nabla_{\mathbf{k}} E(\mathbf{k})$ is the (c-number) group velocity of the electron momentum state $|\mathbf{k}\rangle$. Homogeneous systems are translationally invariant [$f_\sigma(\mathbf{k},\mathbf{r})$ is independent of $\mathbf{r}$], which implies that

$$f(\mathbf{k},\mathbf{K},t) \equiv \int \exp(-i\mathbf{K}\cdot\mathbf{R}) f_\sigma(\mathbf{k},\mathbf{R}) d^3\mathbf{R}$$

is independent of $\mathbf{K}$ and the equivalent electron-density matrix is diagonal in momentum space.

The are some difficulties with interpreting Wigner distributions as probability densities; they are not necessarily positive-definite (usually a sign of strong quantum interference effects). However, the Wigner method does allow one to construct a gauge-invariant transport theory [97]. For independent carriers, f_σ reduces to $\langle r - \frac{1}{2}y|\langle \rho\rangle_s|r + \frac{1}{2}y\rangle$ where ρ is now a functional of the one-electron Hamiltonian. Quantum transport theory has been extensively developed for this case [88].

We shall now sketch the general features of QTT. The electron-density matrix $f(t)$ is determined from the full-density matrix by $f(t) = \mathrm{Tr_s}[\rho(t)]$. For the case of stationary phonon and impurity distributions, we factorize approximately the initial thermal-equilibrium density matrix as

$$\rho(t = 0) \simeq f_0(H_e + V)\Omega_s(H_s),$$

where f_0 will be taken as the Maxwellian equilibrium form and Ω_s describes the equilibrium distribution of scatterers. Let us now Laplace-transform the Liouville equation (105), rearrange the terms using projection calculus, and then retransform back to the time domain to obtain a general master equation in the form

$$\frac{\partial f}{\partial t} + i[H_e, f] + i[\mathcal{F}, f] = i\int_0^t d\tau\ \hat{C}_F(\tau)f(t - \tau) + M_F(t). \qquad (115)$$

The left-hand side of (115) describes the collision-free diffusion and acceleration of the carriers. For homogeneous systems, f is a function of momentum only and if $H_e = p^2/2m^*$, the term $[H_e, f]$ vanishes (for inhomogeneous transport it gives rise to a term $\mathbf{v}(\mathbf{k})\ \partial f/\partial \mathbf{r}$; $[H_e, f]$ is also nonvanishing if H_e includes coupling to a quantizing magnetic field). With our previous model assumption, the coupling to the electric field is simply $\mathcal{F} \equiv -eF \cdot r$, and in the momentum representation, $i[\mathcal{F}, f]$ reduces to eF $\partial f/\partial \mathbf{k}$. The right-hand side is proportional to the scattering interaction and describes collision effects via C_F and memory effects via M_F.

Boltzmann transport theory may be exactly recovered under the following conditions [63]: (1) weak, infrequent scattering; (2) point collisions; (3) translational invariance of the scattering system; and (4) asymptotic time scale $t \gg \tau_c$ [actually related to (104)]. Following these assumptions, it can be shown that the right-hand side of Eq. (115) reduces to

$$\left(\frac{\partial f}{\partial t}\right)_{\mathrm{e-p}} = \sum_{\mathbf{k}'} [R(\mathbf{k},\mathbf{k}')f(\mathbf{k}') - R(\mathbf{k}',\mathbf{k})f(\mathbf{k})], \qquad (116)$$

where the $R(\mathbf{k},\mathbf{k}')$ are the usual second-order perturbation theory scattering rates. Thus, under these conditions, the master equation of Eq. (115) reduces to the Boltzmann equation.

The master equation of Eq. (115) contains a memory term $M_F(t)$, which is one consequence of the interference between the electric field and the scattering processes and is a function of the initial equilibrium state $f_0(H_e + V)$. It represents the correction to the otherwise instantaneous accelerative effect of the field due to the field having to break up the correlations in the electron states induced by scattering processes. Indeed, M_F may be interpreted as a renormalization of the driving-force term in the kinetic equation [99], because its nonvanishing part is proportional to **F**.

The influence of the electric field within a collision event has been called an intracollisional field effect (ICFE). The mathematical details of the ICFE have previously been given [88], so we shall not go into detail here. Two major modifications of the scattering integral occur as a result of this intracollisional process. First, the total energy-conserving δ function is broadened by the presence of the electric field. Second, the threshold energy required for the emission of an optical phonon is modified, which causes an energy shift of the δ function. This latter process is easily understood in physical terms. The argument of the energy-conserving δ function is just

$$E_f - E_i \pm h\omega_0 = E(\mathbf{p}_f) - E(\mathbf{p}_i) \pm h\omega_0, \tag{117}$$

but the initial and final momenta evolve during the collision as

$$\mathbf{p}(t') = \mathbf{p} - \int_{t'}^{t} eF(t'')\,dt'', \tag{118a}$$

$$\mathbf{p}'(t') = \mathbf{p}' - \int_{t'}^{t} eF(t'')\,dt''. \tag{118b}$$

In the emission of an optical phonon, where the electron is scattered against the electric field, the field will absorb a portion of the electron energy during the collision, and, hence, a reduction in energy loss to the lattice will be favored. The opposite effect, an enhancement in energy to the lattice, occurs for emission along the electric field.

C. Synergetic Effects from Device–Device Interactions

Preliminary theoretical studies [59] of the quantum-mechanical operation of an array of very small devices suggest that synergetic effects are possible when the individual feature size decreases below 1000 Å. In this case, novel device possibilities become available.

Let us summarize some previous findings [59]. Small semiconductor devices (<0.1-μm channel length) are controlled by: short spatial scales, very fast temporal response, very high fields ($\lesssim$600 kV/cm), high carrier densities ($\lesssim 10^{18}/\mathrm{cm}^3$), and strong size-related effects (coupling to the environment of contacts, interfaces/boundaries/surfaces, interconnects, and other devices). The ultrafast transit times ($\lesssim 1-2$ psec) within any one device plus the collision-quenching intracollisional field effect preclude any significant dissipation of energy within the device volume; traditionally, this is the so-called ballistic regime). However, studies of the device-density matrix equations [59], which take into account coupling to the device environment, show that dissipation will predominantly occur over the extended region surrounding the device. This mechanism admits a second-order device–device correlations interaction, provided the spatial extent of interconnect regions becomes comparable to the de Broglie wavelengths (indeed, for metallized interconnect regions on the order of 100 Å, de Broglie waveguide modes should dominate the current flow). By using projection calculus methods on the full system-density matrix equations, it has been shown that the interaction of a particular device with its host VLSI system may be classified into coherent (time-reversible) and incoherent (dissipative, irreversible) components. The coherent effects include state renormalization and size quantization arising from the short-range interaction with the regular part of the finite device boundaries; these effects are enhanced by very high, inhomogeneous, controlling electric fields within the device volume. There is also a coherent long-range device interaction with the environment, which stems from effects of device replication and gives rise, for example, to super-lattice phenomena and the consequent overriding of the bulk intradevice carrier dynamics. The latter have already been observed in intercalated structures prepared by molecular-beam epitaxy [102]. The incoherent processes include surface–interface roughness scattering, surface phonons and plasmons, long-range ($\lesssim 1$ μm) device electron, insulator phonon scattering, and phonon-mediated electron–electron interdevice scattering.

The joint action of many subsystems so as to produce structure and functioning on the full-system scale are well known in physical science and are characteristic of nonlinear systems [103]. In recent years, considerable attention has been devoted to so-called dissipative structures [104]: a class of spatially inhomogeneous, ordered structures in which order may be created spontaneously in open systems far from equilibrium and which obey specific nonlinear kinetic (transport) laws. For such nonequilibrium structures, stability is not self-sustaining but is maintained by a continuous exchange of energy and matter with the surroundings. Biological systems, chemically reacting mixtures under open-system conditions, and Benard cell phenomena provide well-known examples.

Comparably complex signal-processing VLSI systems may also support synergetic phenomena as the feature size and complexity approach that of natural systems. Analogies with many-body phase-transition and synergetic theory lead us to expect strong qualitative differences in stability between sequential and concurrent (array) processing signal systems. The argument is simple; we know from model studies that one-dimensional systems cannot support phase transitions; but, as soon as two- and three-dimensional cross-interactions are introduced, systems may condense or "lock into" ordered macrostructures and make transitions between them. Concurrency has a strong equivalency to multidimensionality, which is, in turn, a prerequisite for cooperative phenomena that are stable against local fluctuations. All the necessary features for synergetic phenomena in VLSI structures appear to exist when feature sizes approach a few hundred angströms, such as in multidevice slaving interactions, nonlinear slaving of intradevice variables by self-consistent control fields, input-current open-system operation.

Of course, device–environment interactions are not unknown even in present scale systems. For example, gated logic systems and programmable logic arrays have device–environment control exercised via the interconnect matrix. A true device–device interaction also appears in large MOS memory chips (near-cell interference in READ–WRITE situations), which is currently treated as a reliability problem rather than an effect on which to capitalize. Of course, capacitative coupling between devices is also very familiar. The true device–environment coupling envisaged here, however, is only possible on almost atomically small scales for which superlattice effects and dissipative coupling become possible.

The main theoretical tools to explore synergetic VLSI already exist: nonlinear quantum transport theory, synergetic theory, renormalization group theory, etc. However, applications will require (a) knowledge of the intended function and skeletal VLSI structure, (b) characterization of the relevant interdevice coupling, and (c) selection of the appropriate control fields and currents. These do not as yet exist and must be the target for future research. We have, however, sketched the outline for such a theory using many-body compaction techniques [105]. Such sophistication is not necessary to understand the qualitative role of these synergetic effects. We can illustrate the basic principles by a simple circuit-theoretic analogy utilizing component–connection-type approaches. First, we examine a special case of isolated devices. Then, we introduce a connection function to describe the system in terms of the devices and show how the properties of the connection function can alter the system dynamics. As the individual device dynamics and connections will be nonlinear, we expect that, although the equations used here are linear, the general nonlinear results will admit of synergetic responses for the system. Thus. Thus, for example, a regular, replicated device structure in the environment is expected to give a superlattice modulation of the individual device dynamics. Now, "superlattice" in the normal sense is usually used to refer to multiple thin layers of different materials in which the layers are thinner than the electron wavelength so that the atomic potential variation between layers introduces minigaps, which represent a renormalization of the energy spectrum of the electrons within a single material. Here, however, we emphasize the device–device or device–environment coupling, which can be an essentially time-reversible interaction [59], to renormalize the functional behavior of a single device within the array and to change the system dynamics of the array itself. In particular, if the spatial extent of interconnected regions becomes comparable to the range of possible device–device interactions, a second-order device–device correlation or interaction arises, from which the coherent long-range component admits to possible functional superlattice phenomena. In these cases, in very small devices, the bulk intradevice dynamics may be of secondary importance to the system dynamics taken as a whole.

In particular, though, it is apparent that the system equations are functionally similar to those utilized for nonlinear structures, [104], and for such nonequilibrium structures, stability is not self-sustaining but is maintained by a continuous exchange of energy with the surroundings. Comparably complex signal-processing VLSI systems may also support synergetic phenomena as the feature size and complexity approach that of natural systems. Analogies with solid-state and synergetic theory therefore lead us to expect qualitative differences in stability between sequential and concurrent (array) signal-processing systems.

We examine first the state equations for an isolated integrator with input/output conditioning and then examine the applicability of the example. The state equations are then, for the ith device,

$$\dot{u}_i = a_i u_i + b_i y_i, \tag{119a}$$

$$z_i = c_i u_i, \tag{119b}$$

where u_i is the state variable and y_i and z_i the input and output variables, respectively. For an ensemble of N devices, these become

$$\dot{\mathbf{U}} = \mathbf{AU} + \mathbf{BY}, \tag{120a}$$

$$\mathbf{Z} = \mathbf{CU}, \tag{120b}$$

where $\mathbf{A}, \mathbf{B}, \mathbf{C}$ are square diagonal matrices and $\mathbf{U}, \mathbf{Y}, \mathbf{Z}$ column matrices. Solving for the transfer function gives (in the Laplace transform domain with relaxed initial state)

$$\mathbf{Z} = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\,\mathbf{BY}. \tag{121}$$

So far, we have considered that each device was isolated from the others. If we describe the system input and output as $\mathbf{G}$ and $\mathbf{H}$, respectively, we can describe the connection matrix $\mathbf{F}$ through the following [106]:

$$\mathbf{Y} = \mathbf{FZ} + \mathbf{LG}, \tag{122a}$$

$$\mathbf{H} = \mathbf{MZ}. \tag{122b}$$

The connection matrix $\mathbf{F}$ describes generally how the input of a particular device is related to the outputs of other devices. We can combine Eqs. (121) and (122) to yield the system transfer function as

$$\mathbf{H} = \mathbf{MCB}(s\mathbf{I} - \mathbf{A} - \mathbf{FCB})^{-1}\,\mathbf{LG}, \tag{123}$$

where we have used the fact that $\mathbf{A}$ and, hence, $(s\mathbf{I} - \mathbf{A})^{-1}$ are diagonal.

Equation (123) for the system-transfer function is a special case of the connection-function theory of systems [106] applicable to the integrator. Although we have used this special case, the approach is far more general and is applicable to arbitrary circuits. Furthermore, even though we have assumed an analog signal approach by employing the Laplace transformation for the time variation, the technique is extendible to the class of digital circuits known as linear sequential circuits through the description of system dynamics in an abstract extension field [107]. However, this simple case is adequate to illustrate the major points just discussed.

The quantity $(s\mathbf{I} - \mathbf{A} - \mathbf{FCB})^{-1} = \mathbf{S}^{-1}$ plays the conceptual role of a resolvent for the system and the zeros of det($\mathbf{S}$) define the various modes of operation. Since $\mathbf{B}$ and $\mathbf{C}$ are diagonal, any deviation of the system response from that defined by $\mathbf{A}$ must arise through the structure of $\mathbf{F}$. For example, if we consider that the system is logically connected, i.e., y_i is connected only to $z_i, j < i$, then $\mathbf{F}$ has elements only in the lower triangle below the main diagonal. Since $\mathbf{A}$ is diagonal, $\mathbf{F}$ does not modify the modes determined by $\mathbf{A}$, i.e., $\det(\mathbf{S}) = \det(s\mathbf{I} - \mathbf{A})$. Only when $\mathbf{F}$ has entries across the main diagonal does this change. For example, if $y_i = z_{i-1}$, then $\mathbf{F}$ has entries along the diagonal just below the main diagonal. If, however, the last stage is fed back to the first, an entry appears in the upper right corner of $\mathbf{F}$ and one new mode is generated, the collective ring-oscillator mode.

In general, the connection function $\mathbf{F}$ can be divided into two parts, $\mathbf{F}_1$ and $\mathbf{F}_2$, where $\mathbf{F}_1$ is the portion of $\mathbf{F}$ that represents the desired metallizations, i.e., the designed architectural circuit yielding

$$\mathbf{S}_1 = s\mathbf{I} - \mathbf{A} - \mathbf{F}_1\mathbf{CB}. \tag{124}$$

Then, $\mathbf{F}_2$ represents the parasitic interactions (the parasitic device–device couplings) that arise from the line-to-line coupling capacitance, for example. Thus, a new resolvent $\mathbf{S}_2$,

$$\mathbf{S}_2 = \mathbf{S}_1 - \mathbf{F}_2\mathbf{CB}, \tag{125}$$

arises with a new set of eigenmodes given by det($\mathbf{S}_2$). Thus, the structure of the system is altered in the presence of $\mathbf{F}_2$. As $\mathbf{F}_2$ depends on the states of $\mathbf{U}$ (voltages, for example) as well as the inputs $\mathbf{G}$, it is entirely conceivable that the system is now strongly nonlinear. In large-scale systems, where sizes are more than 1 μm in scale, $\mathbf{F}_2$ may reasonably be assumed to be negligible. In future VLSI and ULSI systems of submicron dimensions, this is no longer the case, and the presence of $\mathbf{F}_2$ will have to be accounted for in design.

It is clear from the preceding discussion that the structure of the connection function is instrumental in determining the collective modes of system operation. Moreover, the connections need not be the deliberately wired interactions but must include the parametric device–device interactions that can occur in arrays of small devices. Such interactions could occur, for example, from capacitative coupling, charge spill-over, or potential barrier lowering, such as is present in subthreshold currents due to drain-induced barrier lowering [108]. Finally, if the connection matrix $\mathbf{F}$ is functionally dependent on the state variables or control signals, i.e., $\mathbf{F} = \mathbf{F}(\mathbf{U},\mathbf{G})$, then a nonlinear interaction is possible that can lead to synergetic restructuring of the system function. It can be stated that the general results presented in this simple example have been known for some time. However, the importance of device–device correlation–interaction in dense arrays lies in the role this essentially parasitic long-range interaction can play in restructuring $\mathbf{F}$ and hence the system dynamics. Essentially, $\mathbf{F}$ can be split into design and parasitic fractions. This is the principal point of the present discussion; changes in $\mathbf{F}$ due to device–device interactions can lead to a restructuring of $\mathbf{F}$ and, therefore, to a restructuring of the entire system dynamics. Since these interactions are principally expected to be nonlinear, synergetic effects can be expected in the system dynamics.

V. DIFFUSION

A. Introduction

One of the most fundamental parameters required for modeling semiconductor devices is the diffusion coefficient $D(F,\omega)$, where F is the electric field and ω the frequency. Not only is the diffusion coefficient necessary for evaluating operating and high-frequency characteristics, it provides a fundamental characterization of velocity fluctuations in the system and their contribution to noise in the device [111,112]. If diffusion is relatively well understood for low fields, this situation does not carry over to the case of high electric fields [112]. The general case for high-field transport in semiconductors differs in that relaxation of the velocity fluctuations is to a nonequilibrium steady state [113–115], and the process is nonlinear [116,117].

Diffusion is a general result of the Brownian motion of the carriers, and early work centered on the calculation of the mean square displacement of the carriers. This led to the Einstein relation for a free particle [118]:

$$D = \lim_{t\to\infty} \langle(\Delta x)^2\rangle/2t. \tag{126}$$

This form has been utilized considerably in studies of Brownian motion in many systems, as is apparent from the many review articles and original papers [56; 60; 111, p. 415; 113; 119–125]. On the other hand, it has also been suggested that the mean square displacement is related to D through the derivative

$$D = \tfrac{1}{2}\, d/dt\langle(\Delta x)^2\rangle, \tag{127}$$

and this form has also found widespread use [114,126,127]. In the steady state, however, in both equilibrium and nonequilibrium situations, $\langle(\Delta x)^2\rangle \simeq t$ for long times. Thus, both Eq. (126) and (127) give the same result for D [128].[7] In this regime, we also overlook the long-time tails that are observed in some hydrodynamic systems [127,128]. For short times, however, differences can arise between Eqs. (126) and (127). We shall return to this later.

B. Diffusion Formalism

We shall begin with a brief review of the formal theory of generalized diffusion and mobility. For the present purposes, we shall neglect the influence of magnetic fields and assume that the electric field $F(x,t)$ is a slowly varying function of position when compared to the spatial extant of the de Broglie wavelength or to the radius of the collision sphere [88]. Transport in a medium that is macroscopically homogeneous with respect to scattering centers is then described by a kinetic transport equation of the form

$$\left[\frac{\partial}{\partial t} + \mathbf{v}\cdot\frac{\partial}{\partial \mathbf{x}} + \mathscr{F}\cdot\frac{\partial}{\partial \mathbf{v}}\right] f(\mathbf{v},\mathbf{x},t) = -\hat{C}f(\mathbf{v},\mathbf{x},t), \tag{128}$$

[7] This is not a totally general result, since non-Gaussian, non-Markovian diffusion is known in some disordered systems.

where $\mathcal{F} = e\mathbf{F}/m^*$. For Boltzmann–Bloch transport, f is the usual distribution function defined over velocity–position phase space (we are assuming a simple, parabolic, isotropic band so that v is interchangeable with $\hbar k/m^*$, although the Monte Carlo calculations of the next section assume the proper nonparabolic band structure) and $\hat{C}$ is the collision operator, which is independent of space, time, and electric field but is velocity dependent. If the field $\mathbf{F}$ is a rapidly varying field of very short wavelength, the driving term $\mathcal{F} \cdot \partial f/\partial \mathbf{v}$ requires modification [88, p. 126; 96].

The first integral of Eq. (128) yields the particle-continuity equation

$$\frac{\partial n(x,t)}{\partial t} = -\nabla \cdot \left[\int d^3v \; \mathbf{v}f(\mathbf{v},\mathbf{x},t)\right], \tag{129}$$

where

$$n(x,t) = \int d^3v\, f(\mathbf{v},\mathbf{x},t). \tag{130}$$

Equation (129) may be given the *form* of the traditional phenomenological semiconductor equation by employing the formal causal solution to (128) using the collision operator to form an integrating factor. The result is

$$f(\mathbf{v},\mathbf{x},t) = f_{\mathrm{tr}} - \int_0^t d\tau \exp\left[-\int_\tau^t d\tau'\, \hat{C}(\tau')\right] \times \left[\mathbf{v} \cdot \nabla + \mathcal{F} \cdot \frac{\partial}{\partial \mathbf{v}}\right] f(\mathbf{v},\mathbf{x},\tau), \tag{131}$$

where the transient term is

$$f_{\mathrm{tr}} = \exp\left[-\int_0^t d\tau\, \hat{C}(\tau)\right] f(\mathbf{v},\mathbf{x},0). \tag{132}$$

The transient term will be ignored for now, but we should pointed out that it will have an important role in transport on short time scales, such as for ballistic transport or in very short channel devices. Inserting Eq. (131) into (129) gives, for the classical case,

$$\partial n(\mathbf{x},t)/\partial t = -\nabla \cdot \mathbf{J}(x,t), \tag{133}$$

$$J_\mu(\mathbf{x},t) = -\partial[n(\mathbf{x},t)D_{\mu\nu}(\mathbf{x},t)]/\partial x_\nu - n(\mathbf{x},t)\mu_{\mu\nu}(\mathbf{x},t)F_\nu(\mathbf{x},t), \tag{134}$$

where the generalized diffusion and mobility terms are defined by

$$D_{\mu\nu}(\mathbf{x},t) = \frac{1}{n}\int_0^t d\tau \int d^3v\, v_\mu \exp[-(t-\tau)\hat{C}]v_\nu f(\mathbf{v},\mathbf{x},\tau), \tag{135}$$

$$\mu_{\mu\nu}(\mathbf{x},t) = \frac{e}{m^* n}\int_0^t d\tau \int d^3v\, v_\mu \exp[-(t-\tau)\hat{C}]\left(-\frac{\partial f}{\partial v_\nu}\right). \tag{136}$$

Although these results superficially appear reminiscent of classical diffusion and Doob's theorem [130], Eqs. (131)–(136) provide a rigorous comprehensive formulation of the semiconductor equations for the description of diffusion and device modeling. It should be noted that, in general, the generalized mobility and diffusion do not admit of an Einstein relation [117], except for systems close to thermal equilibrium. Because of the complexity of the collision operator, or, more properly, the resolvent $(\hat{C} + s)^{-1}$ (where s is the Laplace transform variable), it transpires that expressions (135) and (136) are not trivial to evaluate, *even if* the exact solution of $f(\mathbf{v},\mathbf{x},t)$ is known. As with other nonlinear systems, expansions can be made in order to approximate low-order corrections to the Fokker–Planck equation [116]. The quantum case is similar but involves nonlocal diffusion and mobility densities. It will be discussed elsewhere.

Equations (135) and (136) are formally of the correlation-function-type and constitute a more proper definition of the diffusion coefficient itself [60,137]. For longitudinal diffusion, it may then be shown that

$$D(t) = \int_0^t d\tau\, \Phi'(\tau), \tag{137}$$

where $\Phi'(\tau)$ is the reduced velocity-autocorrelation function. On the other hand, we can multiply Eq. (133) by x^2 and integrate over all space to obtain an expression for $D(t)$. This relation is then conveniently compared with Eq. (137) in the Laplace domain to yield

$$\Phi'(s) = s \int_{-\infty}^{\infty} dv\, v[\hat{C} + s]^{-1} v f(v,s)/N, \tag{138}$$

where $f(v,t)$ is the velocity-distribution function (normalized to N) that satisfies

$$\left[\frac{\partial}{\partial t} + \mathcal{F}\frac{\partial}{\partial v}\right] f(v,t) = -\hat{C}f(v,t), \tag{139}$$

and v the reduced velocity. Equation (138) relates $\Phi'(s)$ to $sD(s)$, the longitudinal diffusion coefficient as defined by Eq. (135). The appearance of the resolvent $(\hat{C} + s)^{-1}$ is noteworthy.

C. Correlation Functions for Hot Electrons

The fluctuation response is, in general, complicated because of the many physical processes involved, but the velocity fluctuation can be considered as having two main contributions: $v' = v(t) - \langle v \rangle = v'' + u'$. The first of these, u', is the velocity fluctuation arising from a fluctuation in carrier energy: $u' = u(E + \Delta E) - u(E)$ and the second, v'', arises from velocity fluctuations about u' [117,132]. These various factors can be observed by studying, not the diffusion coefficient itself, but rather the velocity autocorrelation function $\phi'(t)$, which is the inverse Fourier-cosine transform of $D(F,\omega)$. If we define $\phi'(t)$ as

$$\phi'(t) = \langle [v(t + t_0) - \langle v \rangle][v(t_0) - \langle v \rangle] \rangle, \tag{140}$$

then it is found that, for high electric fields, $\phi'(t)$ decreases initially as an exponential, becomes negative, passes through a minimum, and relaxes finally to zero [117]. This process is basically related to the fact that, in general, energy relaxation is slower than momentum relaxation; and this behavior can be expected to occur via the same processes that lead, for example, to velocity overshoot [133]. Such general behavior was observed in the recent work of Fauquemberque *et al.* [112] but was not adequately explained. We should remark here that such behavior for $\phi'(t)$ is also found generally in hydrodynamic systems [129]. The detailed behavior of $\phi'(t)$ assumes more than academic interest as semiconductor devices begin to assume submicron dimensions. In Si, for example, the time duration of $\phi'(t)$ can be of the order of 1 psec, so as channel lengths drop below, say, 0.1 μm, correlated electron motion and enhanced noise in the devices can be expected to occur.

In general, the diffusion coefficient $D(\omega,F)$ depends on the velocity fluctuations in the electron system and is related to the noise spectral density $S_v(\omega)$ associated with these fluctuations. These are related as (for longitudinal diffusion)

$$\begin{aligned} D(\omega,F) &= \frac{S_v(\omega)}{4} = \int_0^\infty \phi'(\tau)\cos(\omega\tau)\,d\tau \\ &= \int_0^\infty \langle [v(\tau + t) - \langle v \rangle][v(t) - \langle v \rangle] \rangle \cos(\omega\tau)\,d\tau, \end{aligned} \tag{141}$$

where all velocities are understood to be longitudinal. The principal difficulty in calculating transport parameters, particularly $\phi'(t)$, in these systems lies in the complicated energy dependence of the many scattering processes. In the past few years, however, ensemble Monte Carlo techniques have been developed that can be used to calculate these transport parameters with high resolution. As developed by Lebowhol and Price [134] and subsequently used by Ferry and Barker [53,135], the ensemble Monte Carlo technique is a hybrid method in which an ensemble of electrons is adopted. This ensemble is composed of N electrons, with variables $\{R_i\}$, $i = 1,2, \ldots, N$, where the set $R_i = \{k_i,x_i, \ldots\}$ includes all necessary descriptors of each electron's state. At each time step, all R_i are calculated by a Monte Carlo process, and the set $[R_i\}$ is treated as an ensemble evolving in time. The ensemble Monte Carlo method has advantages over the normal Monte Carlo technique in that an ensemble-distribution function exists and evolves with $\{R_i\}$. Variables such as velocity or position are calculated from an ensemble average over $\{R_i\}$ at each time step and the variance is controlled by a sufficiently large value for N. One should be aware, however, of the vagaries of stochastic simulations on a computer, and we have used a variety of fields, number of electrons, and seeds for the random number generators without affecting these results. In the calculations reported here, a value of $N = 2500$ was used. This value is sufficiently large to give high-resolution results for the transient dynamic response of the electrons to a high electric field [55,135], for example. Thus the method is capable of yielding good results for the transport characteristics.

The ensemble Monte Carlo method was used to calculate the correlation function in Si for the total velocity, $\phi(t) = \langle v(t + t_0)v(t_0)\rangle = \phi'(t) + \langle v\rangle^2$ (all calculations shown in the figures have $\phi(t)$ normalized to $\langle v^2\rangle$).[8] The ensemble of electrons was initialized as a Maxwellian at the lattice temperature 300 K and was assumed to reside at $x = 0$ at $t = 0$. A homogeneous electric field was applied at $t = 0$, and the ensemble allowed to evolve in time. After a reasonable period of time, the ensemble was in pseudoequilibrium with the field and had a steady drift velocity.[9] After this pseudoequilibrium was achieved, the longitudinal velocity autocorrelation function $\phi(t) = \langle v(t_0 + t)v(t_0)\rangle$ was calculated for several initial times t_0. The stationarity of the system was, therefore, verified as well, and the averaging process was carried out over the ensemble as well as over various initial times. In Fig. 47 is shown the variation of $\phi(t)$ as a function of time t, for several values of the electric field. The initial fall of $\phi(t)$ is primarily due to momentum relaxation, with the local minimum and subsequent rise due to energy relaxation as suggested by Price [117]. The error bars indicate the spread of data points from the calculations and averaging procedures.

The lowest field in Fig. 47, 10 kV/cm, lies below the knee of the velocity–field curve and does not really correspond to hot electrons. In this case, there is no hint of a negative-going portion [where $\phi(t) << \langle v\rangle^2$, the steady-state result]. There is evidence, however, of a tailing behavior

[8] The use of $\phi(t)$, rather than the more normal $\phi'(t)$, was adopted as this allows log–log or semilog plots to be used without zero-crossing complications. The two are, of course, identical for equilibrium cases, but the second equality follows if the process is at least wide-sense stationary. The latter is not a foregone conclusion in nonlinear processes [3] and must be checked in each case. As discussed in the text, this was done and calculations using either $\phi'(t)$ or $\phi(t)$ were found to agree.

[9] It is found, for example, that at 25 kV/cm, the transient and overshoot velocity effects have decayed in less than 0.3 psec.

away from the initial exponential. This behavior is well developed in Fig. 47b, for 25 kV/cm. It is clear from Fig. 47b that the initial decay, the momentum-relaxation portion, deviates substantially from an exponential for times greater than about 0.1 psec, and it is evident that although $\phi(t)$ decays initially as an exponential, it deviates noticeably from this behavior at long times and begins to decay as $t^{-3/2}$. This behavior differs from that reported by Fauquemberque *et al.* [112] but appears to be intrinsic to the momentum-relaxation process [136]. At still higher electric fields, this tailing behavior is washed out because of the much faster energy-relaxation process.

The initial exponential decay portion is significant. The time constant of this portion of the decay of $\phi(t)$ is closely related to and slightly larger than the momentum-relaxation time τ_m associated with the chordal mobil-

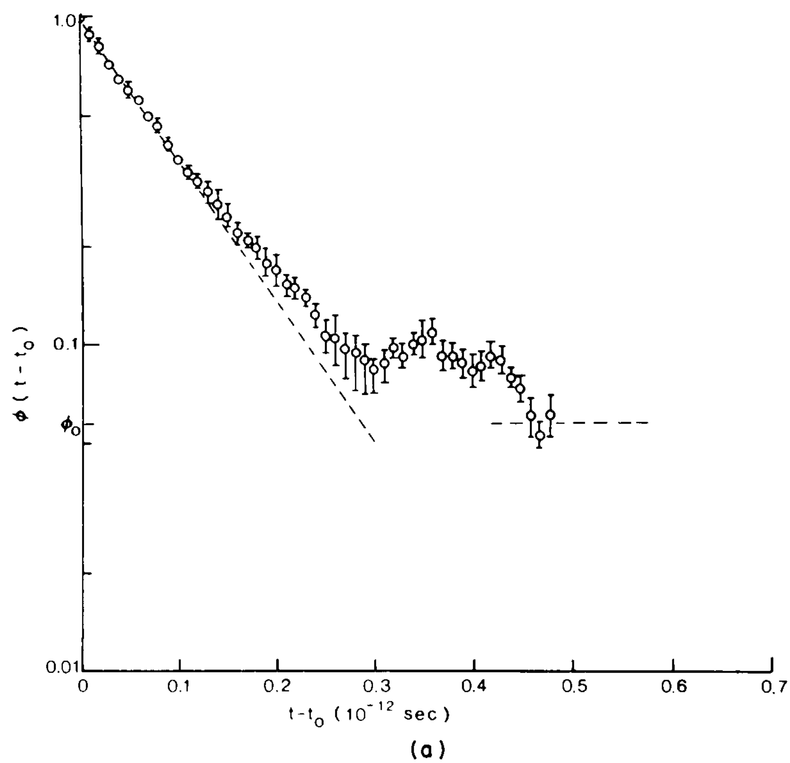


Fig. 47a. The total velocity-correlation function, $\phi(t - t_0) = \langle v(t)v(t_0)\rangle$, as a function of $t - t_0$ for Si at 300 K: (a) field at 10 kV/cm. The curves are normalized to $\langle v^2\rangle$ and ϕ_0 is $\langle v\rangle^2/\langle v^2\rangle$, the final value. The individual curves are discussed in the text.

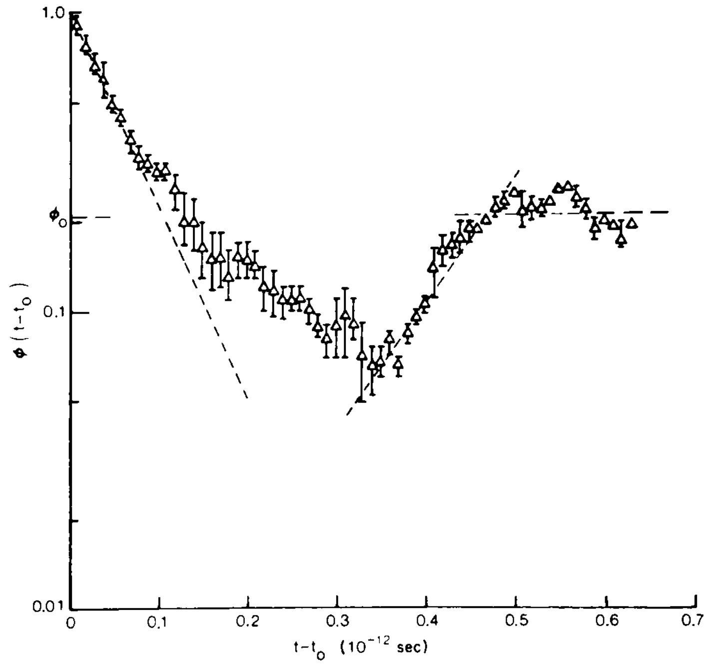


Fig. 47b. Field at 25 kV/cm.

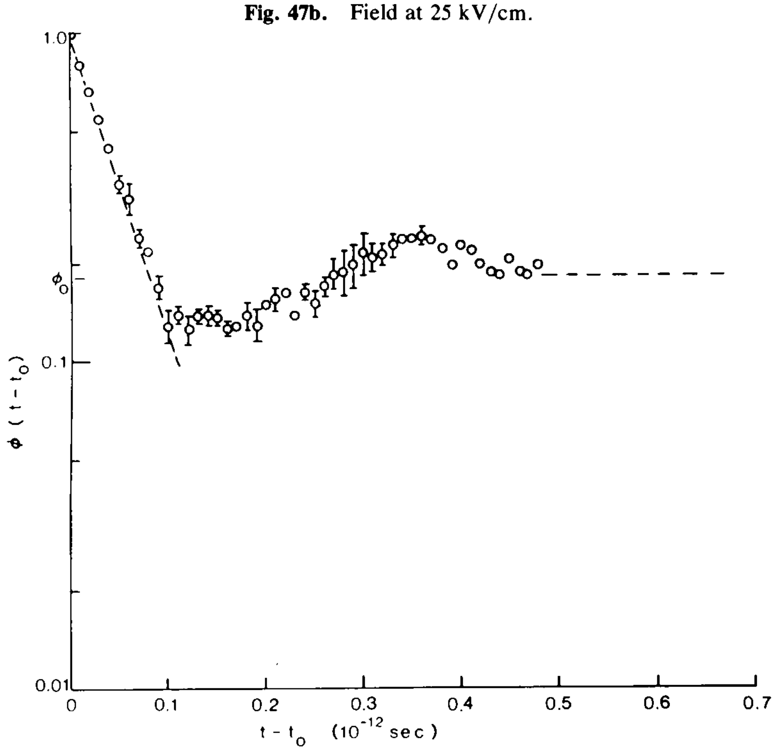


Fig. 47c. Field at 50 kV/cm.

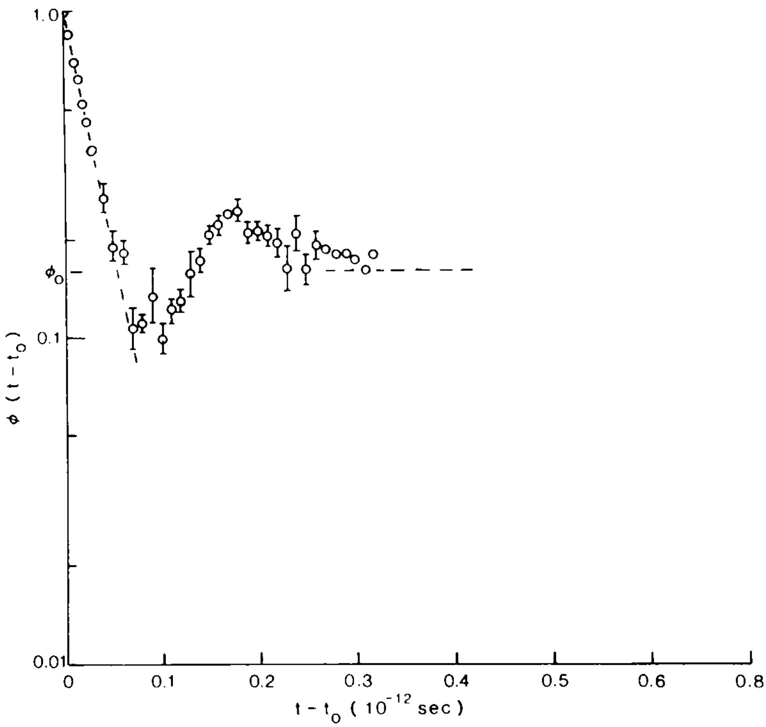


Fig. 47d. Field at 100 kV/cm.

ity $\mu = v_d/F$, rather than the differential mobility dv_d/dE (here we define the effective or average $\tau_m = m^*\mu/e$). The latter quantity has been suggested as the appropriate quantity for longitudinal diffusion [111]. At 25 kV/cm, the velocity is becoming very nearly saturated, so that the differential mobility is more than an order of magnitude smaller than the chordal mobility. This difference is readily distinguished from the data in Fig. 47. The decay of $\phi(t) \simeq \exp(-t/\tau_0)$ and at 25 kV/cm, for example, is best fit with a τ_0 of 7×10^{-14} sec, while $\tau_m \simeq 5.4 \times 10^{-14}$ sec. The results of the decay constant of the exponential portion of $\phi(t)$ being slightly larger than τ_m appears to be a general result, as it was checked at several other values of electric field. Van Kampen [116] has suggested such a difference would occur as a general result of nonlinear relaxation. If $v(t)$ decays as $\exp(-t/\tau_m)$, he suggests that a fully nonlinear treatment of noise would have the correlation function decay with a characteristic time $\tau_0 = \tau_m/(1 - \epsilon)$, where $\epsilon \simeq 1.5\langle v\rangle^2/\langle v^2\rangle$. This gives $\tau_0 \simeq 6.8 \times 10^{-14}$ sec at 25 kV/cm, for the τ_m just given, which is within the accuracy of the

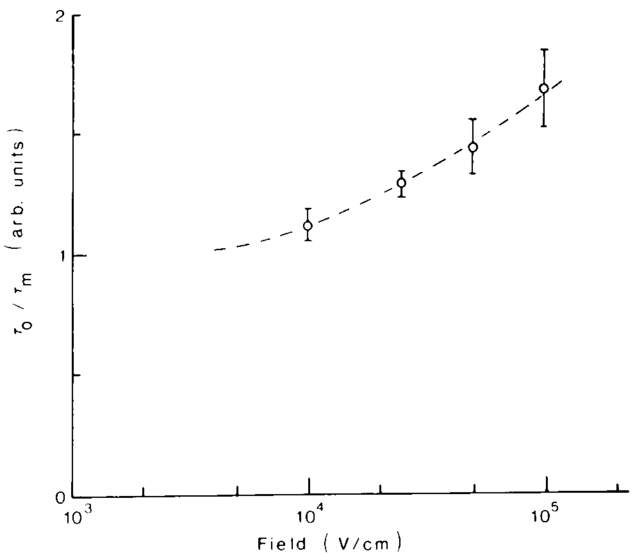


Fig. 48. The deviation of τ_0 from τ_m for Si at 300 K. The velocity-autocorrelation function $\phi(t)$ decreases initially as $\exp(-t/\tau_0)$. Although this initial fall corresponds to momentum relaxation, $\tau_0 > \tau_m$.

present calculations. However, this correction does not show the explicit field dependence of τ_0/τ_m. Both $\langle v \rangle^2$ and $\langle v^2 \rangle$ can be expected to increase (almost quadratically) with F until saturation. Then $\langle v \rangle^2$ should become independent of F. This would imply a reduction of E. It is observed, however, that E increases with F. In Fig. 48, we plot the variation of τ_0/τ_m with electric field. It is observed that τ_0/τ_m increases with the field, although not quite linearly.

In Fig. 49, the Laplace transform and Fourier cosine transform of $\phi'(t)$ are shown for a field of 25 kV/cm. Contrary to linear transport, these functions are not simple, monotonically decreasing functions for large ω and s. Rather, they exhibit peaks at high frequency. The origin of these peaks lies in the enhanced high-frequency conductivity [88, p. 126] in regions where the energy-relaxation process can no longer follow the ac field. Thus, these peaks have their origin in the same processes that lead to velocity overshoot. The oscillations in $S_v(\omega)$ at high frequency appear to be related to the oscillations at long time on $\phi'(t)$. While these oscillations may not be real, their presence and the shape of $S_v(\omega)$ has also been observed by Grondin in GaAs [137]. From this figure, it is apparent that enhanced noise will appear in Si devices at frequencies above $\sim 10^{11}$ Hz and that correlated carrier motion can be expected for times on the order of 1 psec.

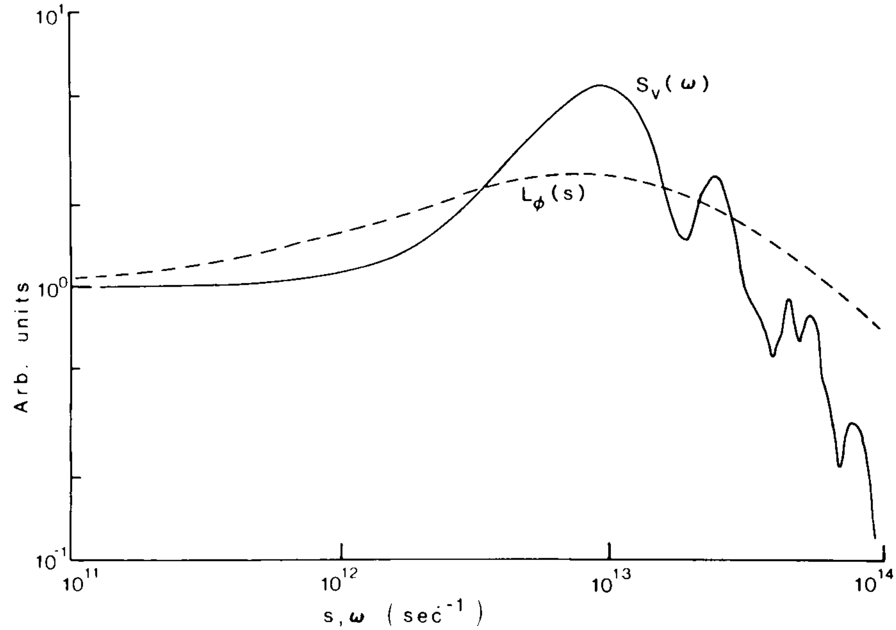


Fig. 49. The Laplace transform $L_\phi(s)$ and Fourier cosine transform $S_V(\omega)$ (noise spectral density), of $\phi'(t) = \langle[v(t + t_0) - \langle v\rangle][v(t_0) - \langle v\rangle]\rangle$ for Si at 300 K, 25 kV/cm.

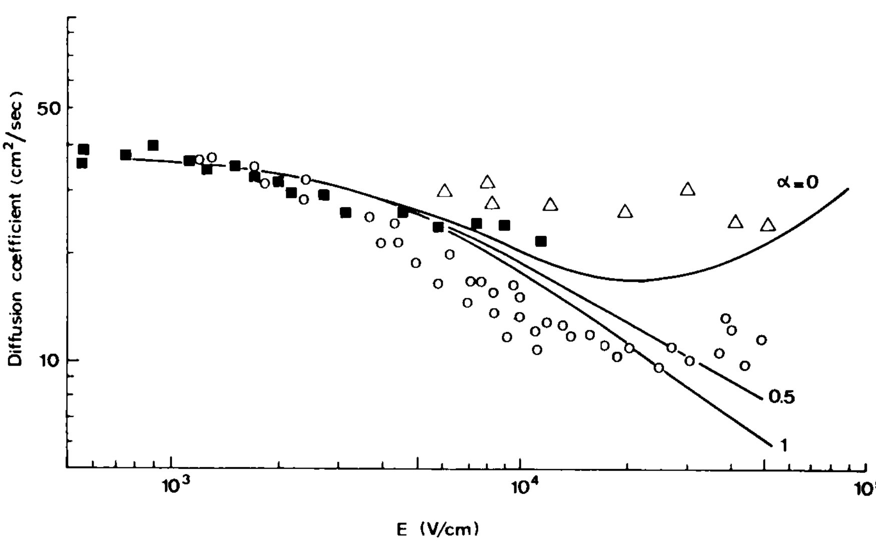
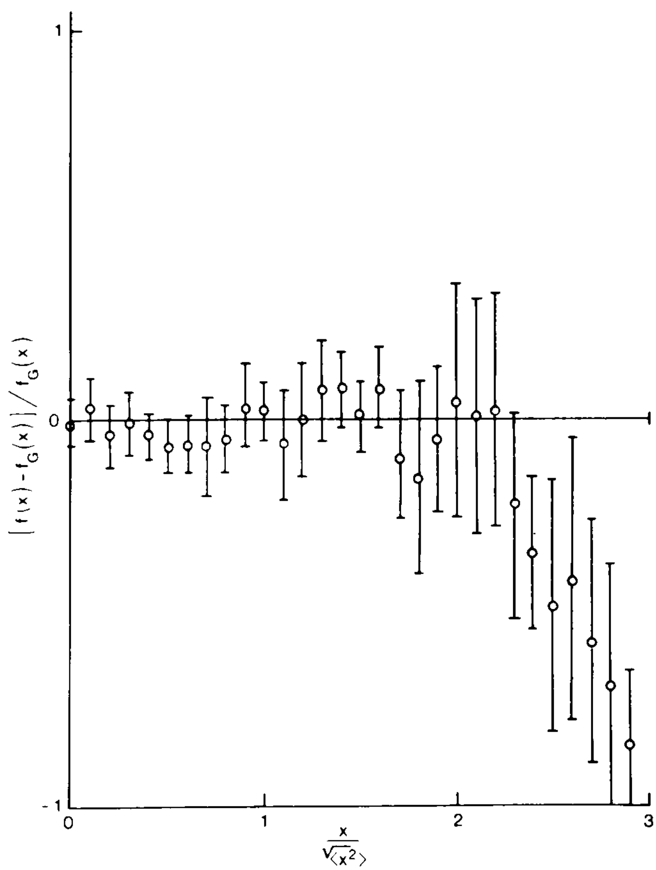


Fig. 50. Comparison of theoretical and experimental values of D for Si at 300 K. (After Ferry [135]).

Approaches such as this can be used to calculate the diffusion coefficient as well, through Eq. (141) for $\omega \to 0$. This has been done theoretically and compared with experiments by the group in Modena [138]. The results are shown in Fig. 50. The curve with $\alpha = 0.5$ is the preferred nonparabolicity-corrected curve for Si. The agreement with experiment is good, but not great. Part of the difference is in the assumption that the packet of electrons diffuses as a Gaussian, which is not a valid assumption.

The spatial distribution shows strikingly non-Gaussian behavior. In Fig. 51a, we illustrate $n(x)$ normalized to a Gaussian with the same value of $[\langle(\Delta x)^2\rangle]^{1/2}$ (i.e., same σ values) for 10 kV/cm. When compared to the



(a)

Fig. 51. The spatial distribution function $n(x)$ normalized to a Gaussian for (a) 10 kV/cm and (b) 100 kV/cm for Si at 300 K. The non-Gaussian nature is discussed in the text.

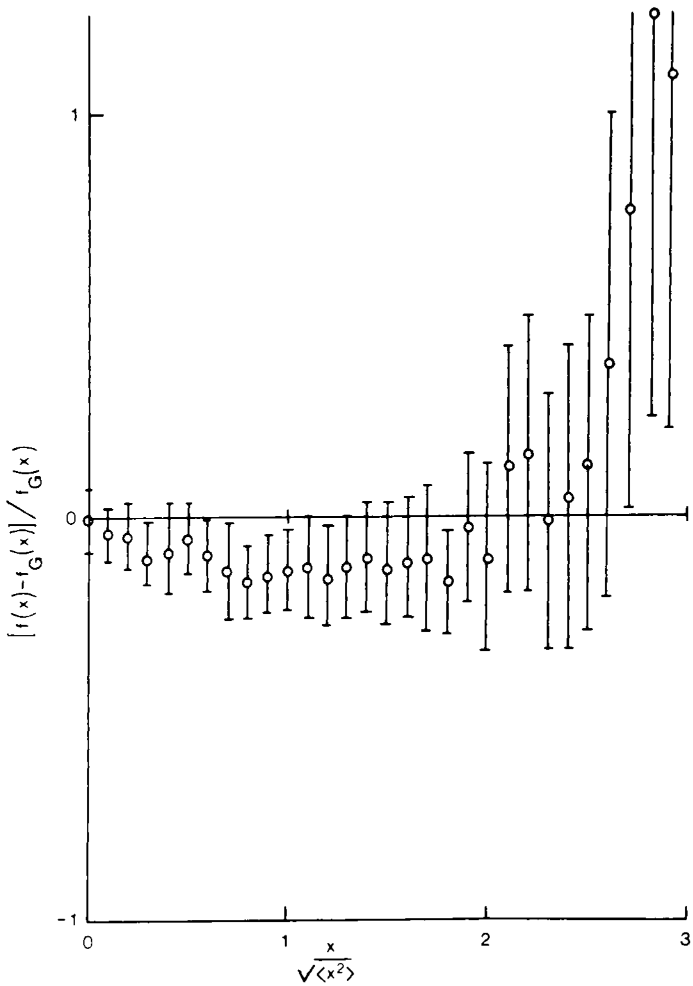


Fig. 51b.

Gaussian, we find $n(x)$ truncated in the tail region. This probably arises from the fact that the electrons in the tail have much higher energy and get hot faster. Once into the velocity-saturation region ($F \simeq 50$ kV/cm), a Gaussian again appears; but at higher fields, there are indications that the tailing is reversed. In Fig. 51b, we illustrate this by showing $n(x)$ for 100 kV/cm. At this field, we are approaching breakdown, and there are a significant number of "lucky" electrons [77] in the tail of $n(x)$. These variations agree with the trends seen in the data of Fig. 50 and may be the cause for the lack of agreement.

D. Transient Diffusion

In recent years, much interest has centered on the transient dynamic response of electrons, especially as it impacts carrier transport through small spatial regions in which the electric field is high. In particular, in the pinch-off region of the channel in a field-effect transistor, the carriers move by drift and diffusion in a very high electric field. Considerable interest has centered on the velocity response, especially that of the overshoot velocity [3,38,39]. The transient response in these high-field conditions is significant in that carriers may completely transit the region prior to obtaining a steady-state high-field distribution. Thus, the transient velocity can be more significant than the steady-state velocity. Considerably less attention has been focused on the transient diffusion that also occurs under these conditions. The lack of attention paid to transient diffusion is easily understood when it is recognized that diffusion is actually a process depending on velocity correlation [139], and the relationship between diffusion and drift, as expressed by the Einstein relation, is a steady-state (stationary distribution) relation [122]. The problem is complicated by the fact that the random-walk equations governing diffusion do not reduce to normal Fick's law behavior on time scales comparable to relaxation processes [122], a result of the general non-Markovian nature of transport on these time scales [62]. The purpose of this section will be to highlight some of these problems and to illustrate them with results calculated by the ensemble Monte Carlo technique.

Diffusion is related to the spatial spreading of an ensemble of carriers with time, as the ensemble responds to both applied drift forces and random forces, such as are generated by collisions. In general, the diffusion coefficient is related to the ensemble position distribution through Eq. (126). In the case of a transient dynamic response, however, the problem is more complicated. A Fokker–Plank equation can be generated whose solution is the transition probability for a particle at x_0,t_0 to transition to x,t. On the short-time scale over which relaxation processes occur, this equation does not reduce to the normal diffusion equation [130]. The problem lies in the fact that the Langevin equation, from which the former equation is obtained, is second order in position. While it remains Markovian in phase space, its projection onto real space does not, except for long times when the ensemble is stationary. Thus Eq. (126) must be corrected for the nonlocal (in time) behavior. Then [122,130],

$$\langle (\Delta x)^2 \rangle = (2\langle v^2 \rangle/\gamma)\{t - (1/\gamma)[1 - e^{-\gamma t}]\}, \tag{142}$$

where $1/\gamma$ is the relaxation time and $\langle v^2 \rangle$ the mean squared velocity about the mean.[10] From a fluctuation–dissipation theorem, we generally define $D = \langle v^2 \rangle / \gamma$. If we use Eq. (126), we find

$$D = \frac{\langle (\Delta x)^2 \rangle}{2t\,\{1 - [1 - e^{-\gamma t}]/\gamma t\}}. \tag{143}$$

If, however, we use Eq. (127), then

$$D = \frac{d\langle (\Delta x)^2 \rangle / dt}{2(1 - e^{-\gamma t})}. \tag{144}$$

The difference in these two is, of course, in the denominator. The denominator of (143) goes as t^2 for small t and that of (144) goes as t plus whatever variation comes from the derivative term. We cannot say which is correct, but if $\langle (\Delta x)^2 \rangle$ varied as t^2 for small t, both would give a constant D. In fact, this does not occur for the general nonlinear relaxation to a nonequilibrium steady state.

In a stationary distribution, the diffusion coefficient can be related to a velocity-correlation function from which the mobility is also derived. This leads to the classical Einstein relation, which, for a nondegenerate semiconductor, is

$$D = \mu k_B T_e / e, \tag{145}$$

where μ and T_e are, respectively, the chordal mobility and electron temperature at a particular electric field. In general, the Einstein relation does not hold for hot electrons, and it is of interest to know how far this result differs from Eqs. (126) or (143) in the case of the transient dynamic response.

We have carried out calculations for the longitudinal diffusion by the ensemble Monte Carlo process. In this case, an ensemble of 10^4 electrons is subjected to transport via a Monte Carlo technique. At each time step, the transport parameters are obtained by an ensemble average. The time evolution of these ensemble averages yields the time evolution of the transport parameters. This technique has previously been shown to yield excellent argeement for the transport coefficients and to agree well with parameterized distribution approaches, which explicitly define selected coefficients. In the present calculations, the carrier ensemble was assumed to have been injected in GaAs at $x = 0$ at $t = 0$, i.e., as a δ-function ensemble. The spread of the distribution and its drift under an applied, homogeneous electric field of 25 kV/cm was calculated. The kinetic temperature ($T_L = 300$ K) and drift velocity were also calculated

[10] Strictly speaking, Eq. (142) is valid only for linear relaxation to equilibrium. Under certain conditions, however, it is not a bad approximation for quasi-linear relaxation to a nonequilibrium steady state. Unfortunately, there exists no good theory for nonlinear, transient relaxation to a nonequilibrium steady state.

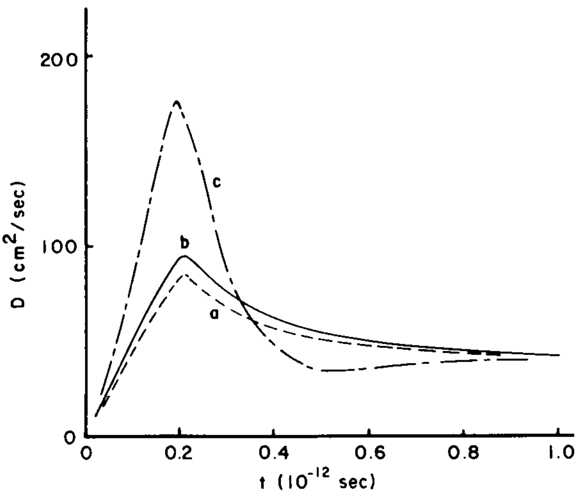


Fig. 52. Diffusion coefficients calculated from Eqs. (126), (143), and (145) (curves a–c, respectively) for an ensemble of carriers injected into GaAs at $x = 0$, $t = 0$, $T = 300$ K, at 25 kV/cm.

during the calculation. Gallium arsenide was chosen because of the complications arising from intervalley transfer and negative differential conductivity. In such material, the transient dynamic response is dominated by the differential repopulation between nonequivalent valleys [72,140]. The diffusion coefficient was calculated by each of Eqs. (126), (143), and (145), and the results are shown in Fig. 52. In using (145), the mobility μ is taken as v_d/F and is not a differential mobility, a definition in keeping with the correlation-function approach. If a derivative approach [Eq. (127)] were used, the results would be closer to those of (145) on the rising portion of the response ($t < 0.2$ psec). However, the derivative approach would give a negative diffusion during the falling portion of the curve, a result not at all in keeping with (145), although understandable on physical terms. In this region, the carrier ensemble appears to actually be contracting as the faster, more energetic carriers transfer to the heavy mass satellite valleys. Within the accuracy of the Monte Carlo method, all three approaches converge as the ensemble approaches a stationary distribution in phase space.

APPENDIX A. DERIVATION OF THE BALANCE EQUATIONS

The balance equations given in Section III are obtained by taking the moments of the Boltzmann transport equation, assuming that the distribution function in each valley is a drifted Maxwellian. Thus

$$f_j(\mathbf{k}) = C_j \exp[-\hbar^2(\mathbf{k} - \mathbf{k}_{dj})^2/2m_jk_BT_{ej}], \tag{A-1}$$

where $\hbar k_{dj} = m_jv_{dj}$ is the average drift momentum of the electron gas, C a constant for normalization purposes, and T_{ej} the electron temperature, with the subscript j referring to the jth valley. From Eq. (A-1), it follows that $f_j(E) \approx f_{j0}(E) + f_{j1}(E)$, with

$$f_{j0}(E) = C_j \exp(-E/k_BT_{ej}) \tag{A-2}$$

and

$$f_{j1}(E) = v(m_cv_{dj}/k_BT_{ej})f_{j0}(E). \tag{A-3}$$

The actual moments themselves are complicated by the fact that the semiconductor conduction band is describable by a multivalley structure, which often comprises nonequivalent valley sets, and, hence, is a coupled system. We must also account for the repopulation effects that can occur.

The Boltzmann transport equation is given as

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla f + e\mathbf{F} \cdot \nabla_p f = \sum_{\mathbf{p}'} \{W(\mathbf{p},\mathbf{p}')f(\mathbf{p}') - W(\mathbf{p}',\mathbf{p})f(\mathbf{p})\}. \tag{A-4}$$

In Eq. (A-4), $W(\mathbf{p},\mathbf{p}')$ is the scattering rate from state $\mathbf{p}'$ to state $\mathbf{p}$. In the following, we shall ignore the inhomogeneity term $\mathbf{v} \cdot \nabla f$, although it can readily be incorporated. We define the average of $\phi(\mathbf{p})$ in the ith valley as

$$\langle \phi(\mathbf{p}_i) \rangle_i = \frac{1}{n_i} \int \phi(\mathbf{p})f_i(\mathbf{p})\, d\mathbf{p}. \tag{A-5}$$

To begin, we multiply Eq. (A-4) by an arbitrary function of $\phi(\mathbf{p}_i)$ assuming $\phi(\mathbf{p}_i)$ and $f_i(\mathbf{p}_i)$ represent the ith valley (or set of equivalent valleys in this case). Then, integrating over $\mathbf{p}_i$ yields

$$\begin{aligned} &\frac{\partial}{\partial t}(n_i\langle\phi\rangle_i) - n_ie\mathbf{F} \cdot \langle \nabla_{pi}\phi\rangle_i \\ &\quad = \sum_{\mathbf{p}_i} \Big\{ \sum_{\mathbf{p}_i'} [W(\mathbf{p}_i,\mathbf{p}_i')f(\mathbf{p}_i') - W(\mathbf{p}_i',\mathbf{p}_i)f(\mathbf{p}_i)] \\ &\qquad + \sum_{p_j} [W(\mathbf{p}_i,\mathbf{p}_j')f(\mathbf{p}_j') - W(\mathbf{p}_j',\mathbf{p}_i)f(\mathbf{p}_i)] \Big\} \phi(\mathbf{p}_i). \end{aligned} \tag{A-6}$$

Here, we have separated the intravalley (with equivalent intervalley) and nonequivalent intervalley contributions to the scattering processes. When the initial and final states lie in the same valley (or an equivalent valley), they are describable by the same local distribution function. With this in mind, the intravalley terms on the right-hand side of Eq. (A-6) (summation over $\mathbf{p}_i'$) can be treated by a simple change of variables and this term becomes

$$n_i\langle\Gamma_\phi(\mathbf{p}_i)\rangle_i, \tag{A-7}$$

where

$$\Gamma_\phi(\mathbf{p}_i) = \sum_{\mathbf{p}'} W(\mathbf{p}_i',\mathbf{p}_i)[\phi(\mathbf{p}_i') - \phi(\mathbf{p}_i)]. \tag{A-8}$$

If $\phi = C$, this term vanishes and, as expected, makes no contribution to a density-balance equation.

The nonequivalent intervalley terms can be written as

$$\begin{aligned} n_i\left\langle\frac{d\phi}{dt}\bigg|_{\text{coll}}\right\rangle &= \sum_{\mathbf{p}_i}\phi(\mathbf{p}_i)\left\{\sum_{\mathbf{p}_j'} W(\mathbf{p}_i,\mathbf{p}_j')f(\mathbf{p}_j') - W(\mathbf{p}_j',\mathbf{p}_i)f(\mathbf{p}_i)\right\} \\ &= \sum_{\mathbf{p}_j'} f(\mathbf{p}_j')\phi(\xi_j)\sum_{\mathbf{p}_i} W(\mathbf{p}_i,\mathbf{p}_j') \\ &\quad - \sum_{\mathbf{p}_i} f(\mathbf{p}_i)\phi(\mathbf{p}_i)\sum_{\mathbf{p}_j'} W(\mathbf{p}_j',\mathbf{p}_i), \end{aligned} \tag{A-9}$$

where we have used the energy-conserving δ function inherent in $W(\mathbf{p}_i,\mathbf{p}_j')$ to define the renormalized momentum ξ_j through the relation

$$\epsilon_j' = \epsilon_i \pm h\omega_0 = \xi_j^2/2m_j^* \pm h\omega_0, \tag{A-10}$$

so that ξ_j is a function of $\mathbf{p}_j'$. Introducing the scattering rate as

$$\Gamma(\mathbf{p}) = \sum_{\mathbf{p}'} W(\mathbf{p}',\mathbf{p}), \tag{A-11}$$

Eq. (A-8) becomes

$$n_i\left\langle\frac{d\phi}{dt}\bigg|_{\text{coll}}\right\rangle = n_j\langle\phi(\xi_j)\Gamma(\mathbf{p}_j')\rangle_j - n_i\langle\phi(\mathbf{p}_i)\Gamma(\mathbf{p}_i)\rangle_i. \tag{A-12}$$

By inserting the details of the various scattering processes, the individual moment equations can be readily set up. However, because of the multiplicity of scatterers, the individual equations are quite complicated. We shall not delve deeper into their structure here.

APPENDIX B. THE WIGNER DISTRIBUTION FUNCTION

Classical transport physics is based on the concept of a probability distribution function, which is defined over the phase space of the position and momenta of all the particles concerned. The time rate of change of this distribution function is governed by the Liouville equation; the classical expectation value of any physical observable, which is a function of position and momentum, is obtained by integrating the product of the observable and the distribution function over all of phase space.

In the quantum formulation of transport physics, the concept of a phase-space distribution function is not possible inasmuch as the noncommutation of the position and momentum operators (the Heisenberg uncertainty principle) precludes the precise specification of a point in phase space. However, within the matrix formulation of quantum mechanics, it is possible to construct a "probability" density matrix, which is often interpreted as the analog of the classical distribution function in phase space. The time rate of change of this probability density matrix is governed by the quantum analog of the Liouville equation. Moreover, the expectation value of a physical observable is obtained by taking the trace of the matrix, which is the product of the probability density matrix and the matrix that corresponds to the physical observable.

There is yet another approach to the formulation of quantum transport based on the construction of the Wigner distribution function [100]. As we shall show, this distribution function has no simple interpretation in the sense of probability theory but, in lieu of its special properties, can be used directly for calculating expectation values of observables in a manner quite analogous to that of classical theory, i.e., by integrating the product of the observable and the Wigner distribution function over all phase space.

In this section we shall review the salient features of the Wigner distribution function. Although the Wigner function is generally defined in terms of all the generalized coordinates and momenta of the system in question as

$$\begin{aligned}P_W(x_1 \cdots x_n, p_1 \cdots p_n) &= \frac{1}{2\pi\hbar}\int_{-\infty}^{\infty} dy_1 \cdots dy_n\, \psi^*\left(x_1 + \frac{y_1}{2}, \ldots, x_n + \frac{y_n}{2}\right) \\ &\times \psi\left(x_1 - \frac{y_1}{2}, \ldots, x_n - \frac{y_n}{2}\right) \exp\left[\frac{i(p_1 y_1 + \cdots + p_n y_n)}{\hbar}\right],\end{aligned} \tag{B-1}$$

we shall discuss the properties of the Wigner function in terms of a single coordinate and momentum. In this case, we let

$$P_W(x,p) = \frac{1}{2\pi\hbar}\int_{-\infty}^{\infty} dy\, \psi^*\left(x + \frac{y}{2}\right)\psi\left(x - \frac{y}{2}\right)\exp\left(\frac{ipy}{\hbar}\right), \tag{B-2}$$

where $\psi(x)$ refers to the state of the system in the coordinate representation.

The distribution function of Eq. (B-2) has interesting properties in that the integration of this function over all momenta leads to the probability density in real space; conversely, the integration of this function over all coordinates leads to the probability density in momentum space. In mathematical terms,

$$\int_{-\infty}^{\infty} P_W(x,p)\, dp = \psi^*(x)\psi(x) \tag{B-3a}$$

and

$$\int_{-\infty}^{\infty} P_W(x,p)\, dx = \phi^*(p)\phi(p), \tag{B-3b}$$

where

$$\phi(p) = (2\pi\hbar)^{1/2} \int_{-\infty}^{\infty} \exp\left(\frac{-ipx}{\hbar}\right) \psi(x)\, dx.$$

It follows immediately from Eq. (B-3) that, for an observable $\hat{F}(\hat{x},\hat{p})$, which is either a function of momentum operator alone, of position operator alone, or of any additive combination therein, the expectation value of the observable is given by

$$\langle F \rangle = \iint \hat{F} P_W(x,p)\, dx\, dp, \tag{B-4}$$

which is analogous to the classical expression for the average value. Herein lies the interesting aspect of the Wigner distribution function; the result of Eq. (B-4) suggests that it is possible to transfer many of the results of classical transport theory into quantum transport theory by simply replacing the classical distribution function by the Wigner distribution function. However, unlike the density matrix, the Wigner distribution function itself cannot be viewed as the quantum analog of the classical distribution function because it is generally nonpositive definite and nonunique [$P_W(x,p)$ of Eq. (B-2) is not the only bilinear expression in ψ that satisfies Eq. (B-3)].

Further resemblance of the Wigner distribution function to the classical distribution function is apparent by examining the equation of time evolution for $P_W(x,p)$. Upon assuming that $\psi(x)$ in Eq. (B-2) satisfies the Schrödinger equation for a system with Hamiltonian $H = (p^2/2m) + V(x)$, it can be readily shown that $P_W(x,p)$ satisfies the equation

$$\frac{\partial P_W}{\partial t} + \frac{p}{m}\frac{\partial P_W}{\partial x} + \theta \cdot P_W = 0, \tag{B-5}$$

where

$$\theta \cdot P_W = -\frac{2}{\hbar}\sum_{n=0}^{\infty} (-1)^n \frac{(\hbar/2)^{2n+1}}{(2n+1)!}\frac{\partial^{2n+1}V(x)}{\partial x^{2n+1}}\frac{\partial^{2n+1}P_W(x,p)}{\partial p^{2n+1}}. \tag{B-6a}$$

Alternately, $\theta \cdot P_W$ can be expressed as

$$\theta \cdot P_W = -\frac{2}{\hbar}\left[\sin\frac{\hbar}{2}\left\{\frac{\partial}{\partial x}\frac{\partial}{\partial p}\right\}\right] V(x)P_W(x,p), \qquad \text{(B-6b)}$$

where it is understood that the position gradient operates only on the potential energy $V(x)$. It is evident that in the limit $h \to 0$, $\theta \cdot P_W$ in Eqs. (B-6) becomes

$$\theta \cdot P_W = -\frac{\partial V}{\partial x}\frac{\partial P_W}{\partial p}, \qquad \text{(B-7)}$$

so that Eq. (B-5) reduces to the classical continuity equation; it is also clear that the dominant quantum correction to $\theta \cdot P_W$ is of order h^2, with this term also having dependences on the third derivative of potential energy with respect to position and the third derivative of the Wigner distribution function with respect to momentum.

The Wigner distribution function is derivable [109] from the Fourier inversion of the expectation value with respect to state $\psi(x)$ of the operator $\exp[i(\tau\hat{p} + \theta\hat{x})]$ (here, x and p satisfy the commutation relation $[\hat{x},\hat{p}] = i\hbar$). As such,

$$P_W(x,p) = \frac{1}{4\pi^2}\int\int C_W(\tau,\theta)\exp[-i(\tau p + \theta x)]\,d\tau\,d\theta, \qquad \text{(B-8a)}$$

where

$$C_W(\tau,\theta) = \int \psi^*(x)\exp[i(\tau\hat{p} + \theta\hat{x})]\,\psi(x)\,dx \qquad \text{(B-8b)}$$

and the interval of integration is $[-\infty,\infty]$ unless otherwise specified. In order to show that the right side of Eq. (B-8a) is indeed the Wigner distribution function as defined in Eq. (B-2), note from the Baker–Hausdorff theorem [110] that $\exp[i(\tau\hat{p} + \theta\hat{x})$ can be rewritten as

$$\exp[i(\tau\hat{p} + \theta\hat{x})] = \exp(\tfrac{1}{2}i\tau\hat{p})\exp(i\theta\hat{x})\exp(\tfrac{1}{2}i\tau\hat{p}) \qquad \text{(B-9)}$$

in which case $C_W(\tau,\theta)$ of Eq. (B-8b) becomes

$$C_W(\tau,\theta) = \int_{-\infty}^{\infty}\exp[-\tfrac{1}{2}i\tau\hat{p}]\psi(x)\exp(i\theta\hat{x})\exp[\tfrac{1}{2}i\tau\hat{p}]\psi(x)\,dx, \qquad \text{(B-10)}$$

which further reduces to

$$C_W(\tau,\theta) = \int_{-\infty}^{\infty}\psi^*(x - \tfrac{1}{2}\tau\hbar)e^{i\theta x}\psi(x + \tfrac{1}{2}\tau\hbar)\,dx. \qquad \text{(B-11)}$$

Then, by inserting $C_W(\tau,\theta)$ of Eq. (B-11) into the right-hand side of Eq.

(B-8a), integrating over the variable θ by using the relation

$$\int_{\infty}^{\infty} \exp i\theta(x' - x'')\, d\theta = 2\pi\delta(x' - x''),$$

and letting $\tau = y/\hbar$, the desired result is obtained.

The method just outlined for arriving at the Wigner distribution function is based on the notion of a characteristic function. The characteristic function of an observable $\hat{A}$ with respect to state $|\psi\rangle$ (here, the Dirac notation is utilized for purposes of generality) is defined as

$$C_A(\xi) = \langle\psi|\exp(i\xi\hat{A})|\psi\rangle, \tag{B-12}$$

where ξ is a real parameter. Assuming $\hat{A}$ to possess an eigenvalue spectrum given $\hat{A}|A'\rangle = A'|A'\rangle$, $C_A(\xi)$ can be evaluated in the A' representation as

$$C_A(\xi) = \int dA' \int dA'' \langle\psi|A'\rangle\langle A'|\exp(i\xi\hat{A})|A''\rangle\langle A''|\psi\rangle. \tag{B-13}$$

Since $\langle A'|\exp(i\xi\hat{A}|A''\rangle = \exp(i\xi A')\ \delta(A' - A'')$ in the A' representation, $C_A(\xi)$ in Eq. (B-13) reduces to

$$C_A(\xi) = \int dA' \exp(i\xi A')|\psi_{A'}|^2, \tag{B-14}$$

where $|\psi_{A'}|^2 = |\langle\psi\rangle|^2 \equiv P(A')$, the probability distribution function for measuring A' in state $|\psi\rangle$. Hence, the characteristic function for $\hat{A}$ is the Fourier transform of the probability distribution function $P(A')$. Subsequent inversion of Eq. (B-14) leads to

$$P(A') = \frac{1}{2\pi} \int C_A(\xi) \exp(-i\xi A')\, d\xi. \tag{B-15}$$

The Wigner distribution function was derived by taking the Fourier transform of the characteristic function for $\exp[i(\tau\hat{p} + \theta\hat{x})]$. In view of the connection between the probability distribution function and the characteristic function for a given observable, this approach seems to be a natural way of obtaining a distribution function for momentum and position. Unfortunately, the noncommutative nature of the two observables destroys the convenient probability interpretation of the characteristic function implicit in Eq. (B-15).

In order to demonstrate this point, assume the characteristic function of two noncommuting observables, $\hat{A}$ and $\hat{B}$, to be

$$C_{AB}(\xi_1, \xi_2) = \langle\psi|\exp[i(\xi_1\hat{A} + \xi_2\hat{B})]|\psi\rangle. \tag{B-16}$$

Observables $\hat{A}$ and $\hat{B}$ are assumed to have eigenvalue spectra

$$\hat{A}|A'\rangle = A'|A'\rangle, \tag{B-17a}$$

$$\hat{B}|B'\rangle = B'|B'\rangle \tag{B-17b}$$

and are chosen so that $[A,[A,B]] = [B,[A,B]] = 0$. This assumption is imposed so that the identity

$$\exp[i(\xi_1\hat{A} + \xi_2\hat{B})] = \exp(i\xi_1\hat{A})\exp(i\xi_2\hat{B})\exp(-\tfrac{1}{2}\xi_1\xi_2[\hat{A},\hat{B}]) \tag{B-18}$$

may be used.

Inserting Eq. (B-18) into Eq. (B-16), obtaining the matrix elements of $\exp(i\xi_1\hat{A})$ in the A' representation and $\exp(i\xi_2\hat{B})$ in the B' representation, and assuming $[\hat{A},\hat{B}]$ is a C number, we obtain

$$C_{AB}(\xi_1,\xi_2) = \exp(-\tfrac{1}{2}\xi_1\xi_2[\hat{A},\hat{B}]) \int dA' \int dB' \exp[i(\xi_1 A' + \xi_2 B')] \times \langle\psi|A'\rangle\langle A'|B'\rangle\langle B'|\psi\rangle. \tag{B-19}$$

We define $F(A',B')$, a generalized Wigner distribution function, to be

$$F(A',B') = \langle\psi|A'\rangle\langle A'|B'\rangle\langle B'|\psi\rangle \tag{B-20}$$

so that

$$F(A'B') = \frac{1}{(2\pi)^2}\int d\xi_1 \int d\xi_2 \exp(\tfrac{1}{2}\xi_1\xi_2[\hat{A},\hat{B}])C_{AB}(\xi_1,\xi_2) \times \exp[-i(\xi_1 A' + \xi_2 B')]. \tag{B-21}$$

It is evident from Eqs. (B-20)–(B-21) that

$$\int F(A',B')\,dA' \equiv |\langle B'|\psi\rangle|^2 = \frac{1}{2\pi}\int d\xi_2\, C_{AB}(0,\xi_2)\exp(-i\xi_2 B') \tag{B-22a}$$

and

$$\int F(A',B')\,dB' \equiv |\langle A'|\psi\rangle|^2 = \frac{1}{2\pi}\int d\xi_1\, C_{AB}(\xi_1,0)\exp(-i\xi_1 A'). \tag{B-22b}$$

Thus, Eq. (B-21) establishes the relationship between the characteristic function for two arbitrary noncommuting observables and a generalized Wigner distribution function. The generalized distribution function has the essential properties of the conventional Wigner function in that an integration of the generalized function over the eigenvalue spectrum of one observable leads to the probability density in the canonically conjugate observable [Eqs. (B-22)]. However, there is no simple probability interpretation of $F(A',B')$ in Eq. (B-21) because of the necessary overlap between the states of the noncommuting observables. If $\hat{A}$ and $\hat{B}$ are made to commute so that $|\hat{A}'\rangle$ and $|B'\rangle$ have a common set of eigenvectors, then $F(A',B')$ reduces to the classical distribution function for $\hat{A}$ and $\hat{B}$.

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